

SILENT STRATEGIES: INNER SPEECH AND PROBLEM SOLVING IN APHASIA

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SILENT STRATEGIES: INNER SPEECH AND PROBLEM SOLVING IN APHASIA

Inner speech, the experience of “talking to yourself in your head”, plays a crucial role in cognition, communication, and self-regulation. While inner speech has been studied for nearly two centuries, its significance in clinical populations, particularly individuals with aphasia, is an emerging area of research. This dissertation explores the multifaceted nature of inner speech in both healthy aging and aphasia, examining its role in language processing, problem-solving, and psychosocial health through interdisciplinary methods.

Aphasia, a language disorder most often caused by stroke, affects over two million people in the U.S., disrupting various aspects of language production and comprehension. Some individuals with aphasia report experiencing disruptions in their inner speech. This research employs multiple methodologies, including inner rhyme judgments, articulatory suppression, rating scales, questionnaires, and experience sampling, to assess inner speech at the word level, in daily life, during problem-solving tasks, and in relation to psychosocial well-being.

Findings reveal that while many individuals with aphasia continue to use inner speech frequently, their inner speech is less varied in content and function compared to their neurologically healthy counterparts. In daily life, people with aphasia most often use inner speech to make decisions about food, plan activities, solve problems, and self-motivate. Experimentally, disrupting inner speech hinders improvement on complex reasoning tasks, underscoring its role in cognitive processing. Objective measures, such as inner rhyme judgment, are associated with aphasia severity and cognitive abilities like inhibition, reasoning, and problem solving, whereas subjective reports of inner speech are not. Additionally, inner speech use is linked to psychosocial health, with certain patterns, such as replaying past conversations, being associated with lower quality of life in individuals with aphasia.

Overall, this dissertation provides a comprehensive perspective on inner speech, demonstrating its significance beyond language production. By integrating insights from psychology, neuroscience,

linguistics, and speech-language pathology, it advances our understanding of inner speech as a critical component of cognition, communication, and well-being.

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Introduction to the Dissertation

Inner Speech

As explained to the participants in the following studies, inner speech is the experience of speaking to oneself silently in the head, without speaking out loud or moving the mouth or lips. It can be frequent or infrequent, a few words or full sentences, a monologue or dialogue, meaningful or nonsensical, and spontaneous or voluntary. Inner speech often mirrors the words or expressions used in social interactions, typically occurring in one's own voice. However, not all internal experiences qualify as inner speech; for instance, mental images, sensations, pure emotions, and other nonverbal, wordless thoughts do not fall under the category of inner speech. Although there has been a resurgence of interest in inner speech recently, it has been studied in fields such as psychology and philosophy for almost 200 years, with prominent researchers such as Vygotsky (Alderson-Day & Fernyhough, 2015; Vygotsky, 1934, 1962). For further explanation of the terminology and history related to the study of inner speech across disciplines, see Chapter 1. The work presented in the following pages will examine inner speech and its role in the fields of communication sciences and disorders and neuroscience, specifically the clinical implications of inner speech in aphasia and aging.

Inner Speech and Aphasia

Aphasia is an acquired disorder of language, most commonly caused by a stroke. In the United States, more than two million people are living with aphasia, which impacts all aspects of language – reading, writing, speaking, and understanding (The National Aphasia Association, 2023). Recently, researchers have been investigating the way that aphasia may impact self-directed, inner language – inner speech. This research is inspired by the accounts of people with aphasia, as seen in the quotes below:

“I was aware I couldn’t pronounce it in my mind.” (Broussard, 2016) (p.56)

“Those little voices inside your head, reminding you of who you are and where you live, become silent.” (Taylor, 2008) (p. 79)

“...my inner monologue, my self-directed speech, had also gone almost completely mute.”
(Marks, 2018) (p.17)

“...her name erases itself from my brain almost as soon as she says it.” (Schultz, 2010) (p.50)

These quotes are highlighted in a recent metaphor analysis of prose by persons with aphasia, an acquired language disorder (Tichborne et al., 2024). They describe each author’s experience of inner speech, or the little voice in their head. When discussing inner speech, people often jump at the chance to share more about their personal experience – whether they talk to themselves in their heads frequently or infrequently, what sorts of tasks they complete using inner speech, how they cannot imagine a world without inner speech. For some people with aphasia, especially immediately after their brain injury, this inner silence is the new world they must navigate (Marks, 2018; Taylor, 2008). Still other people with aphasia report that their inner speech is much better than their speech out loud, a subjective experience backed by evidence of a dichotomy in abilities to complete tasks in inner speech or find words inside the head but difficulty saying words out loud (Alexander et al., 2023; Fama et al., 2019; Stark et al., 2017).

This intersection of topics features research from a variety of disciplines, and future researchers are encouraged to lean into the natural interdisciplinarity of the study of inner speech in aphasia in the viewpoint that begins this dissertation. This is fitting considering that the dissertation is a culmination of a very interdisciplinary approach to a doctoral degree: coursework from gerontology, medicine, psychology, neuroscience, communication sciences and disorders, cognitive science, data science, and intelligent systems engineering; collaboration with researchers in philosophy, neuroscience, and psychology; and mentorship in STEM education.

Inner Speech and Aging

Inner speech changes throughout the lifespan. Rather than using inner speech, children often speak to themselves out loud with what is called “private speech.” As a result of the development of the ventral and dorsal language streams in the brain, private speech is internalized into inner speech around the

age of seven (Geva & Fernyhough, 2019). Then as young adults, people use inner speech in a variety of ways, with some uses related specifically to the student experience – using inner speech about education or while studying (Racy et al., 2020). Throughout childhood and adulthood, inner speech has been shown to support cognitive tasks, especially involving executive functioning (Alderson-Day & Fernyhough, 2015). It has not yet been investigated, but older adults may also have unique experiences with inner speech as they age, related to life experiences such as retirement, changes in family roles, and health.

Some of the content of this dissertation is reproduced from previously published, peer-reviewed journal articles. Namely, Chapter 1 is adapted from Alexander, J. M., & Stark, B. C. (2024).

Interdisciplinary approaches to understanding the inner speech, with emphasis on the role of incorporating clinical data. *European Journal of Neuroscience*, 60(5). <https://doi.org/10.1111/ejn.16470>.

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Chapter 1: An interdisciplinary overview of the examination of inner speech

It is necessary to understand how interdisciplinary study of inner speech has shaped its understanding, and why evaluating inner speech through the lens of aphasia aids continued learning about inner speech (Alexander & Stark, *European Journal of Neuroscience*, 2024). Chapter 1, a viewpoint article, urges neuroscientists to expand their understanding and methodology of inner speech, informed by the robust theories, techniques, and approaches from other fields. When reconsidering what may “count” as inner speech to include inner experiences beyond impoverished motor speech, researchers explore the multiple dimensions of inner speech and its uses in daily life. The focus then shifts to the

importance of clinical data, especially from aphasia, posing questions which the rest of the dissertation attempts to address.

Chapter 2: Measuring inner speech

Alexander, Stark, and Langland-Hassan (*Aphasiology*, 2023) investigate the challenges and recommendations for measuring inner speech in aphasia, with an emphasis on comparing objective and subjective methods from prior experimental work. The relationship between inner speech and aphasia severity is examined, with special attention paid to the differences in “in-the-moment” inner speech (including objective judgments of rhyme and subjective reports of inner speech during naming) and general impressions of inner speech (e.g., how frequently they experience successful inner speech when struggling to name something, how it has changed post-stroke, how it compares to their speech out loud). This first study addresses one of the first questions that people ask about studying inner speech – how can you measure a phenomenon that takes place in someone’s head, and what information do different types of measurements give?

Chapter 3: Subjective experience of inner speech in daily life by persons with aphasia

An examination of subjectively perceived inner speech in the daily lives of people with aphasia is presented in Chapter 3 (**Alexander**, Hedrick & Stark, *Frontiers in Psychology*, 2024). I examine the experience of inner speech in the daily lives of people with aphasia, showcasing additional methods of measuring inner speech which allow researchers a depth of information about the frequency, contents, functions, and activities of inner speech use. Most prior work on inner speech in aphasia examines inner speech at the word level, looking at tasks such as naming or rhyme judgment to understand the ability to say single words in one’s head. Work outside of aphasiology considers additional perspectives on the idea of inner speech, extending beyond the word level to the inner narrative that people use when they go about their days – planning, solving problems, motivating themselves. According to research with young adults, many activities of daily living are supported by inner speech. Evaluating inner speech through experience sampling and questionnaires provides a better understanding of the

experience of living with aphasia and the role of inner speech in a person's daily life after stroke. This second study extends the work from the first study, addressing a broader question – how do people with aphasia use inner speech in their daily lives, beyond the word level?

Chapters 4 and 5: Study Procedure

Chapters 4 and 5 were collected as part of the same study on inner speech in adults with and without aphasia. The study procedure is shown in Figure 1 to provide context to those chapters as to the organization of the sections and the counterbalancing/randomization that was done to combat order effects.

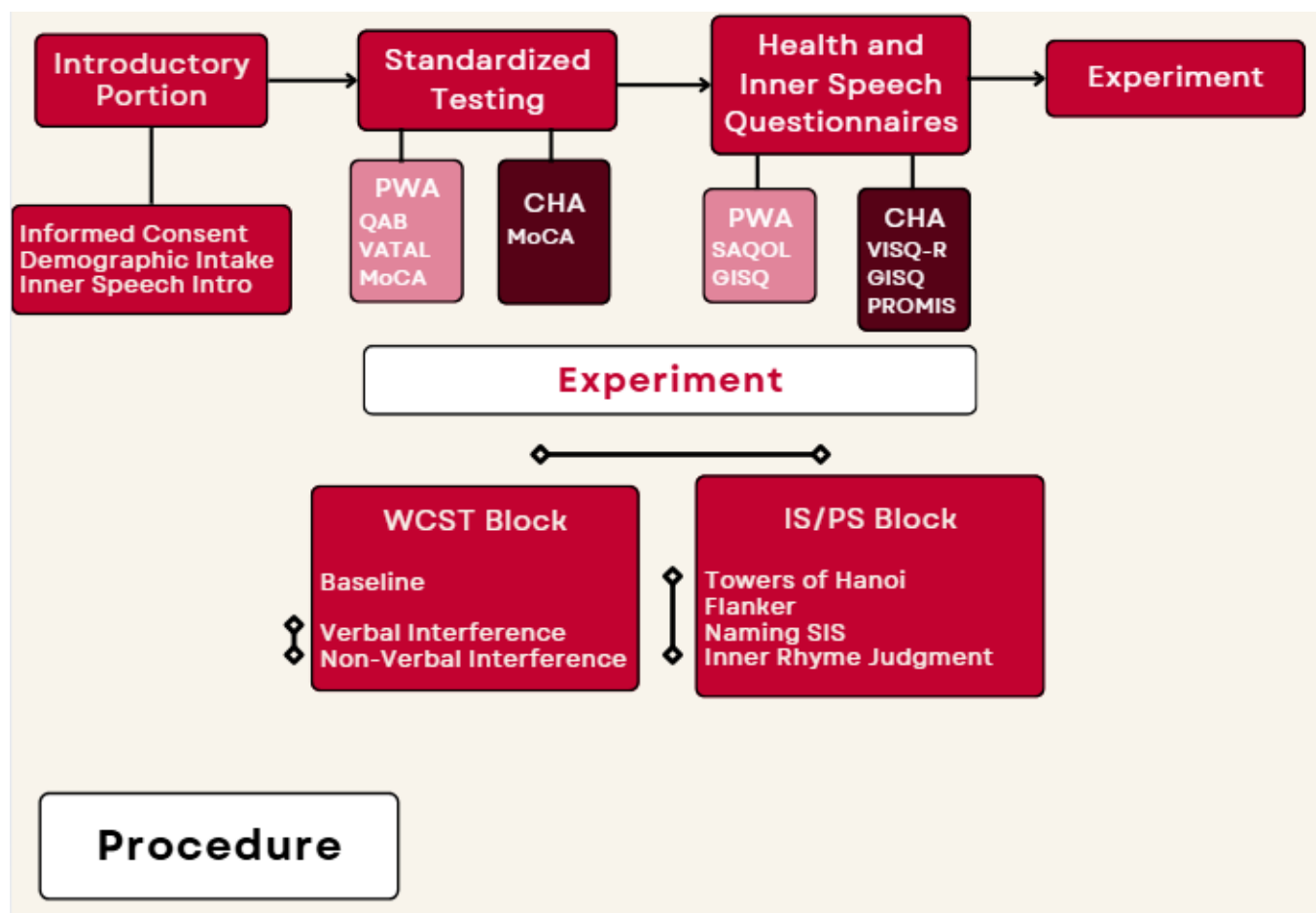


Figure 1: Diagram of the experimental procedure. PWA = people with aphasia. CHA = cognitively healthy adults. QAB = Quick Aphasia Battery. VATA-L = Visual Analogue Test Assessing Anosognosia – Language. MoCA = Montreal Cognitive Assessment. SAQOL = Stroke and Aphasia Quality of Life

Scale. GISQ = General Inner Speech Questionnaire. VISQ-R = Varieties of Inner Speech Questionnaire. PROMIS = Patient Reported Outcome Measurement Information System. WCST = Wisconsin Card Sorting Test. IS/PS = inner speech and problem solving. SIS = successful inner speech. Lines with diamonds at the end indicate randomization.

Chapter 4: The relationship between inner speech and problem solving in people with and without aphasia

Complementing Chapter 3, in Chapter 4, I take a more experimental approach, manipulating the process of solving problems by interfering with the ability to use inner speech during complex reasoning tasks. Participants completed a variety of tasks to assess inner speech and problem solving, and this study explores the connections between these abilities along with the consequences of making inner speech more difficult. Through these approaches, this study addresses two questions – what happens to people with and without aphasia when we disrupt the use of inner speech while they solve non-verbal problems (i.e., how is language related to non-verbal reasoning skills) and how are different ways of measuring inner speech related to different types of cognitive tasks?

Chapter 5: Subjective experience of inner speech, emphasizing problem solving and psychosocial health

Finally, in Chapter 5, participants with and without aphasia are compared on their general use of inner speech, with a focus on problem solving and psychosocial factors. This final study expands on the findings from the study in Chapter 3 to allow comparison with age-similar groups and investigate specific questions related to the general experience of inner speech: How are people with and without aphasia using inner speech – for problem solving, emotional control, motivation, rehearsal? Do more people with aphasia talk about health and identity in their inner speech than adults without aphasia? How does inner speech relate to psychosocial health?

Chapter 6: Discussion, conclusions, and future directions

After presenting these distinct yet related ways of examining and characterizing inner speech, there will be an overall discussion of the findings, highlighting the connections between the studies and the information that can be gained from them when examined in depth. This discussion will also address the clinical implications of this line of research and propose future directions in the field of inner speech and aphasia. Finally, the dissertation will conclude with a summary of recommendations and takeaway messages regarding this line of research, emphasizing the broader picture of inner speech in the fields of speech-language pathology and neuroscience.

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Chapter 1: Interdisciplinary Approaches to Understanding the Inner Speech, with Emphasis on the Role of Incorporating Clinical Data

As reproduced from: Alexander, J. M., & Stark, B. C. (2024). Interdisciplinary approaches to understanding the inner speech, with emphasis on the role of incorporating clinical data. *European Journal of Neuroscience*, 60(5). <https://doi.org/10.1111/ejn.16470>

Abstract

Neuroscience has largely conceptualized inner speech, sometimes called covert speech, as being a part of the language system, namely a precursor to overt speech and/or speech without the motor component (impoverished motor speech). Yet, interdisciplinary work has strongly suggested that inner speech is multidimensional and situated within the language system as well as in more domain general systems. By leveraging evidence from philosophy, linguistics, neuroscience, and cognitive science, we argue that neuroscience can gain a more comprehensive understanding of inner speech processes. We will summarize the existing knowledge on the traditional approach to understanding the neuroscience of inner speech, which is squarely through the language system, before discussing interdisciplinary approaches to understanding the cognitive, linguistic, and neural substrates/mechanisms that may be involved in inner speech. Given our own interests in inner speech after brain injury, we finish by discussing the theoretical and clinical benefits of researching inner speech in aphasia through an interdisciplinary lens.

1. Inner Speech across Disciplines

Inner speech, the conscious process of “talking to oneself in one’s head,” is a concept that has been situated in philosophy and psychology since Plato and Aristotle (Duncombe, 2016). Inner speech is an important aspect of human experience, with Heavey & Hurlburt (2008) estimating that approximately one quarter of people’s conscious (awake) life consists of inner speech. Across disciplines, inner speech has been called covert self-talk, self-talk, internal verbalization, covert verbalization, inner voice, inner ear, and verbal thought. It has also become clear from interdisciplinary work that not only do terminology and operational definitions differ pragmatically – they may also describe different concepts. The simplest definitions describe the experience of a “little voice in the head” (from aphasiology and psychology, Geva, Bennett, et al., 2011) or “silent self-directed speech,” (from psychology and philosophy, Vygotsky, 1962). However, some researchers (from across disciplines) have teased apart differences within the inner speech system, with inner voice being a generative language process and inner ear being more highly related to the phonological/verbal working memory system (Perrone-Bertolotti et al., 2014). In philosophy literature, researchers disagree as to which components of inner speech are required, theorizing about matters such as how conscious people are of their inner speech, how much phonetic and articulatory information is involved, and how inner speech interacts with other cognitive processes (Langland-Hassan et al., 2018). The neuroscience literature has most often conceptualized inner speech as a part of the language system, typically a precursor to overt speech. Table 1 highlights disciplines in which inner speech has been researched, the disciplines’ primary interest in inner speech, and methods of study within that discipline.

There are overlaps and distinctions across disciplines in a few key areas: goals, perspectives, and methodology. As seen in Table 1, neuroscientific research on inner speech is distinct due to its narrow focus on brain-behavior mapping, with perspectives centered on the connections with the language network. Through imaging studies, such as functional magnetic resonance imaging (fMRI), researchers have pinpointed brain areas crucial in the generation of inner speech. Neuroscience helps

elucidate the pathways that underlie the transition from external to internal speech and the brain's role in this process. Philosophy addresses the more abstract and theoretical aspects of inner speech, pondering its implications for understanding consciousness, self, and identity. Philosophers debate the nature of inner speech, its phenomenology, and its role in self-conscious thought and introspection. They also examine the ethical and existential dimensions of inner speech, exploring how it affects our sense of autonomy and moral reasoning. Methodology may overlap across disciplines, however, with neuroscientists combining the use of neuroimaging with inner speech tasks commonly used in other disciplines, such as covert rhyme judgment, and use of questionnaires being common in linguistics, philosophy, and psychology. In aphasiology, methods and perspectives organically converge to reflect the influence of neuroscience (due to brain injury) and linguistics/psycholinguistics (due to disordered language). Goals, however, may be distinctly tailored to understanding the uses of inner speech after brain injury and could extend to questions of quality of life and a deeper comprehension of living with aphasia. Together, these disciplines contribute to a comprehensive understanding of inner speech, revealing its multifaceted nature and its significance in human cognition and social behavior. Each field not only deepens our knowledge of how inner speech works but also how it serves as a bridge between the neural and subjective experiences of thought.

Table 1: Study of inner speech across disciplines.

Discipline	Goals/Interests of study	Perspectives on inner speech	Validity of methods	Papers for further reading
Cognitive Neuroscience	Brain-behavior mapping, e.g., networks underlying inner speech	Evaluating the neural underpinnings of inner speech using neuroimaging methodology, typically with an emphasis on its language origins, as well as brain structure and function.	Internal validity is supported by inner speech mapping to 'known' brain areas involved in language and self-referential processing. External validity is based on replication in new, different samples (e.g., different sites; different ages).	(Grandchamp et al., 2019; Løevenbruck, 2018; Løevenbruck et al., 2018; Martin et al., 2018; Perrone-Bertolotti et al., 2014; Pratts et al., 2023; Stephan et al., 2020)
Philosophy	Phenomenology and uses of inner speech	Theory and examination of inner speech's nature, use, functions, and purpose	Internal validity is confirmed by logical consistency and internal coherence of arguments across researchers. External validity based on self-report methodologies from participants, in which similar information (e.g., inner speech use) is found.	(Carruthers, 2018; Hurlburt & Heavey, 2015; Langland-Hassan, 2018; Langland-Hassan et al., 2018; Morin, 2005, 2011b, 2019; Munroe, 2022)
Linguistics	The linguistic underpinning of inner speech, including cultural-linguistic individual differences	Inner speech as a mechanism in speech production, lexical-semantic-phonological access, and self-monitoring of speech.	Internal validity is supported by the close correlation of language ability and inner speech measurement (e.g., greater accuracy and faster reaction time to covert rhyming tasks suggestive of greater verbal working memory ability). External validity is based on replication in new, different samples (e.g., different languages, dialects).	(Dewaele, 2015; Levelt, 1989; Netsell et al., 2016; Oppenheim & Dell, 2008, 2010)

Psychology and Psycholinguistics	The role of inner speech within the greater 'mind' and in the experimental manipulation of inner speech, its relationship with other cognitive functions, and its role/presence in clinical populations	Behavioral relationships with inner speech and language, as well as higher cognitive functions, such as executive function, problem solving, planning, emotional control, and Theory-of-Mind. Also, relationships of inner speech to psychological states (e.g., positive or negative emotion), or psychiatric conditions such as schizophrenia.	Internal validity supported by the close correlation of related cognitive abilities and inner speech measurement (e.g., inner speech disruption impacting problem solving; greater problem-solving skills relating to greater inner speech awareness and use) and demonstrating that inner speech can be disrupted, or magnified, dependent on clinical population (e.g., auditory hallucinations in schizophrenia). External validity is based on replication in new, different samples (e.g., different sites; different ages; clinical populations) and with closely related cognitive concepts (e.g., planning instead of problem solving may also relate to inner speech).	(Akhutina, 2003; Alderson-Day & Fernyhough, 2015; Fernyhough, 2004; G. L. Wallace et al., 2017)
Aphasiology	The preservation and uses of inner speech after brain injury affecting language	Evaluate the role of inner speech in describing aspects of living with aphasia, such as language abilities but also an individual's everyday life and experiences. Evaluate scenarios when inner speech is, and is not, present in those with left hemisphere brain damage.	Internal validity confirmed by demonstrating of a relationship between inner and overt speech, as well as demonstration that inner speech can be preserved despite impaired overt speech. This is likely due to different, and preserved, brain mechanisms underlying inner speech and overt speech. External validity is based on replication in new samples of persons with aphasia who vary in their aphasia presentation (e.g., overall severity; auditory comprehension severity) and in other neuropsychological characteristics (e.g., preserved/impaired executive function; presence of anosognosia, which is a lack of insight about language impairment).	(Alexander et al., 2023, 2024; Fama, Snider, et al., 2019; Fama & Turkeltaub, 2020; Geva, Jones, et al., 2011; Stark et al., 2017)

Internal validity examines whether the study theory, logic, design, methodology, and analysis answer the research questions. External validity examines whether the study findings can be generalized to other contexts. These are the authors' examples of internal and external validity considerations and should not be considered exhaustive

2. Multidimensionality of inner speech

The result of interdisciplinary evaluations of inner speech is integrated models that enable evaluation of a multifaceted inner speech. The ConDialInt Model (Grandchamp et al., 2019) and models that extend and refine it (Pratts et al., 2023) conceptualize inner speech as multidimensional, comprising three dimensions/varieties: condensation, dialogality, and intentionality (Figure 1). The neuroscientific investigation of inner speech has largely conceptualized inner speech through this 'condensation' dimension, situating inner speech within the larger language network. The other two dimensions have had less focus in neuroscience research, more so the result of investigations in philosophy and psychology. In this viewpoint, we argue for the importance of their evaluation to extend our understanding of the neural mechanisms and correlates of inner speech.

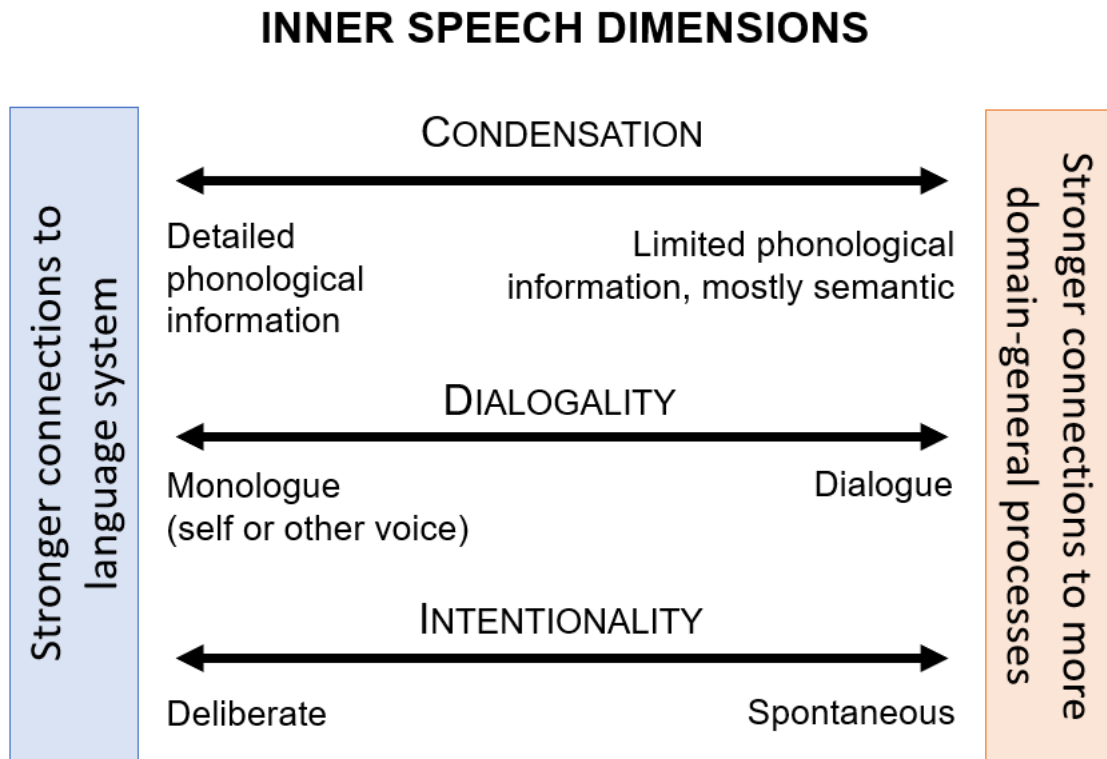


Figure 1: The authors' interpretation of inner speech multidimensionality and its relationship to neural and cognitive processes (blue; orange). Derived largely from ConDialInt model from Grandchamp et al., 2019, with integration of interdisciplinary evidence.

2.1 Condensation

The CONDENSATION dimension—closely tied to the language system—reflects inner speech's representation as a purely condensed form (without all acoustic, phonological, and syntactic information) to an expanded form (including articulatory and auditory properties). Indeed, “thinking in pure meaning” (Vygotsky, 1962) would be the extreme of the condensed form of inner speech. The neuroscientific investigation of inner speech has largely conceptualized inner speech through this ‘condensation’ dimension, situating inner speech within the larger language network. The other two dimensions have had less focus in neuroscience research, more so the result of investigations in philosophy and psychology. In this viewpoint, we argue for the

importance of their evaluation to extend our understanding of the neural mechanisms / correlates of inner speech.

Empirical study of inner speech often conceptualizes it as 'impoverished' speech, i.e., speech lacking the motor component. For example, inner speech is represented as a pre-articulatory code sent to the motor system (Ackermann et al., 2004, 2007). Evaluating inner speech through this lens enables one to differentiate motor planning and motor execution stages of speech production. It should be noted, however, that not all believe that inner speech is just impoverished overt speech. For example, psycholinguistic research has shown that inner speech slips exhibit lexical bias but not the phonemic similarity effect (Oppenheim & Dell, 2008), both of which are evident in overt speech. Authors suggest that inner speech is impoverished at lower (featural) levels, but robust at higher (phonemic) levels, therefore sharing some but not all language characteristics with overt speech. Further, psycholinguistic research suggests that tongue twisters can occur in both inner and overt speech (Corley et al., 2011). Regardless of the relationship of inner to overt speech, inner speech clearly draws upon the language network. The language system comprises a dorsal 'sound-to-speech' stream (comprising left lateralized frontoparietal cortex) and a ventral 'sound-to-meaning' stream (comprising bilateral temporoparietal cortex) (Hickok & Poeppel, 2004). Unsurprisingly, neuroscientific studies of inner speech have found that inner speech involves the activation of brain areas related to language processing. A few select studies are discussed below to demonstrate how inner speech has been associated with these brain areas.

A perspective article by Geva and Fernyhough (2019) reviewed the evidence from fMRI and diffusion tensor imaging (DTI) for the neural correlates of the development of inner speech, arguing that the ventral language stream (temporoparietal cortex) is detectable at birth and matures quickly compared to the dorsal language stream (left lateralized frontoparietal cortex), which is similar anatomically to adult structure by late childhood but does not fully mature until late teenage years or early 20s. This information, in conjunction with Vygotsky's (1962) theory of

the internalization of inner speech, provides evidence that at least some aspects of inner speech developmentally emerge as the dorsal language stream matures. This is corroborated by evidence from a variety of imaging methods, spanning back to the mid-1990s. An early study from McGuire and colleagues used positron emission tomography (PET) to evaluate neural mechanisms of inner speech and auditory verbal imagery (McGuire et al., 1996). Participants were shown single words, which they were then asked to use to generate short, stereotyped sentences without speaking (inner speech) or to imagine the sentences being spoken to them in another person's voice (auditory verbal imagery). The study found that inner speech was associated with increased activity in the left inferior frontal gyrus whilst auditory verbal imagery was associated with increases in the same region, but also in the left premotor cortex, the supplementary motor area and the left temporal cortex. This evaluation of auditory verbal imagery was a precursor to studying the dialogality aspect of inner speech (discussed more in the next section), though 'auditory verbal imagery' should not be conflated with auditory verbal hallucinations often present in schizophrenia. Geva and colleagues utilized voxelwise lesion symptom mapping, which associates behavior with damage voxels in the brain, to investigate the neural correlates of inner speech by examining patients with focal brain lesions and aphasia (Geva, Jones, et al., 2011). Participants were assessed on tasks requiring inner speech, including covert written rhyme judgment. Lesions in the left inferior frontal gyrus were significantly associated with impairments in inner speech tasks, highlighting the critical role of this region in the generation of inner speech. Numerous studies in the neuroscience literature have demonstrated tight functional connectivity between the language dominant frontal lobe and the contralateral cerebellar hemisphere (for an overview, see Ackermann et al., 2007), and it is therefore unsurprising that the right cerebellum has also been implicated in inner speech (Ackermann et al., 1998, 2004).

Temporoparietal cortex has also been associated with inner speech. In a more recent study using electrocorticography to measure electrical signals from the surface of the brain (an

invasive method), participants repeated continuous words or sentences aloud or silently (Brumberg et al., 2016). Motor cortical responses were observed in the overt condition but were expectedly absent in the covert condition. Interestingly, the authors identified left hemisphere superior temporal gyrus activation in both overt and covert conditions. This suggests that auditory processes are recruited even in the absence of planned motor execution, though this finding has been challenged in other studies (e.g., Towle et al., 2008). Tian et al. (2016) compared inner speaking (articulation imagery) and imagining someone else speaking (hearing imagery) by having participants repeat single syllables during fMRI. Participants were trained to mentally articulate during inner speaking and to rely on auditory memory processes during hearing imagery (Tian et al., 2016). Inner speech was associated with greater activity in frontoparietal sensorimotor systems, including sensorimotor cortex, as well as posterior superior temporal sulcus and anterior superior temporal gyrus. Hearing imagery, on the other hand, showed stronger activation in regions implicated in memory retrieval, including middle frontal, inferior parietal cortex and intraparietal sulcus.

2.2 Next generation neuroscience: evaluating a multidimensional inner speech

2.2.1 Dialogality / Egocentricity.

The DIALOGALITY dimension (sometimes referred to as egocentric dimension, from Pratts et al., 2023) reflects inner speech's continuum from monologic (an inner soliloquy) to dialogic (reflecting a back/forth conversation). Combined with research on auditory hallucinations (Ferryhough, 2004), one might speculate that this dimension of inner speech is more apt to be associated with domain general systems of self-awareness and self-monitoring rather than the language system. Grandchamp et al (2019) also suggest that this variety of inner speech involves a perspective switching mechanism, as well as further monitoring and executive control processes. Whilst limited empirical, neuroscientific evaluations on the dialogality dimension exist, there is burgeoning evidence to support this claim. Grandchamp et al., 2019, found

greater activation in the right inferior frontal gyrus, bilateral inferior parietal lobules, precuneus, and cingulate cortex during three controlled inner speech conditions: a monological self-voice condition (participants mentally generated definitions of visually presented objects), a monological other-voice condition (participants mentally imitated a high-pitched voice of an avatar whilst providing a definition of a visually presented object), and a dialogal other-voice condition (participants imagined an avatar was addressing them with a sentence ending with the name of the object). They also had a verbal mind wandering condition. In the monological self-voice condition (contrasted with a baseline), the most highly implicated regions included the left frontal inferior gyrus pars opercularis and frontal superior medial gyrus. This lined up with evidence presented earlier, implicating left frontal regions closely associated with the language network. The monological other-voice condition tended to associate highly with the left supplementary motor area and frontal inferior gyrus pars orbitalis, again implicating language network regions. These results are corroborated by a recent meta-analysis suggesting that high egocentricity (monologic) speech was associated with a large cluster was identified in the left medial frontal gyrus and supplementary motor area, as well as the left precentral gyrus and left inferior frontal gyrus (Pratts et al., 2023). The right insula and right culmen were also implicated. Interestingly, the dialogal other-voice condition tended to associate with right occipital cortex, left supplementary motor area and left frontal inferior gyrus pars triangularis. This result doesn't necessarily align with the meta-analysis by Pratts et al., 2023, which demonstrated that, in the low egocentricity (dialogic) conditions, the right insula and right inferior frontal gyrus were most implicated.

In a more ecologically valid method, Alderson-Day et al. (2016) used fMRI to study monologic and dialogic inner speech. They compared inner speech tasks manipulating scenarios so that they would likely be set in similar places, changing only the number of talkers (monologic/dialogic). For example, the participants were asked to imagine visiting their old school and either making a speech to current students (monologue) or having a conversation

with an old teacher (dialogue). For both dialogic and monologic inner speech, there was increased left hemisphere activation in the inferior frontal gyrus, medial frontal gyrus, caudate, and insula. This is to be expected given the linguistic aspect of inner speech. Additionally, the left precuneus (a core node in the default mode network), right superior temporal gyrus, right cingulate gyrus, left medial frontal gyrus, left inferior frontal gyrus, and right posterior cingulate (and other areas, though to a lesser extent), were all more significantly activated during dialogic inner speech compared with monologic inner speech. This finding aligned with behavioral evidence from a subjective survey (Varieties of Inner Speech Questionnaire), where self-report scores for the dialogic inner speech subscale were associated with activation in the right middle temporal gyrus and right precuneus. Authors postulated this was because these areas may house the phonological representation of other voices and/or Theory-of-Mind. Given that dialogic inner speech tended to particularly activate default mode network nodes (e.g., precuneus, posterior cingulate), there may also be an aspect of mind-wandering, or contemplation, at play.

2.2.2 Intentionality.

The INTENTIONALITY dimension ranges from intentional, as when purposefully rehearsing information or manipulating it silently for later use, to unintentional, as in daydreaming or mind wandering. The intentionality dimension is critical for understanding how inner speech differentially presents in laboratory settings (with highly conscious and intentional tasks, such as covert rhyming) and in real life situations (where inner speech tends to be spontaneously generated). This dimension appears to comprise both the language system (e.g., in rehearsing language information) and domain general systems (e.g., during mind wandering). This dimension becomes more difficult to study because of the phenomenon that the awareness of inner speech as it occurs in daily life (the more spontaneous variety) is a heterogeneous

experience, similar to other phenomena which are not universally experienced, such as aphantasia (Lupyan et al., 2023).

Grandchamp et al., 2019 found greater activation in right inferior parietal gyrus, right superior frontal gyrus, left middle and superior frontal gyrus, right insula, and right anterior cingulate (among other areas) when contrasting verbal mind wandering with monologic self-directed speech. Authors suggested that this finding may be a connection with the default mode network, which has often been associated with mind-wandering and passive contemplation. Few studies have attempted to study inner speech as it presents in real life, which tends to be spontaneous. This is because spontaneous inner speech is very hard to empirically constrain for experimental studies, and thus the majority of lab studies have studied highly intentional inner speech, where the participant completes only the task at hand (e.g., “tell me if these two words rhyme”; “say these words in your head”). Hurlburt et al. (2016) set to compare intentional and more spontaneous inner speech. They leveraged constrained, experimental tasks that many neuroscience researchers use to investigate inner speech, such as silent articulation, and compared them to the neural correlates of spontaneous inner speech, identified through descriptive experience sampling, which requires the person to examine their inner speech in the specific moment before a beep. The participants were trained in descriptive experience sampling in their natural environments and afterwards used the same beeps with expositional interviewing in the scanner, which allowed for the comparison of elicited versus spontaneous inner speech. The researchers found that during elicited inner speech tasks, there was significant deactivation in bilateral Heschl’s gyrus (Brodmann area 41, part of the primary auditory cortex) and significant activation in the left inferior frontal gyrus. Interestingly, during spontaneous inner speech only, the findings were of a significant increase in activation of Heschl’s gyrus and no change in activation in the left inferior frontal gyrus (Hurlburt et al., 2016). This suggests that the predominant finding about inner speech being related to the left inferior frontal gyrus may be an artifact of only investigating intentional inner speech, which is what the

neuroscientific literature of inner speech is predominantly based upon. A recent meta-analysis suggested highly intentional inner speech was associated with left medial frontal gyrus / supplementary motor area and left precentral gyrus, whilst the low intentionality studies tended to implicate left middle and superior temporal gyri (Pratts et al., 2023).

2.3 Function of inner speech

In addition to multidimensionality, inner speech has unique functionality. Through the three dimensions discussed above (condensation, dialogality, intentionality), inner speech satisfies cognitive, emotional, and behavioral needs. For example, Racy et al. (2020), using the General Inner Speech Questionnaire, asked college-aged adults to consider inner speech in three ways: how/why it is used (“I talk to myself in order to”), what it is used for (“I talk to myself about”), and when it is used (“I talk to myself when”). A large portion of respondents reported using inner speech for negative emotions, problem solving, and planning, indicating intentional use of inner speech. Complementary research from psychology has implicated inner speech’s role in problem solving (Alderson-Day & Fernyhough, 2015; Fernyhough, 2004; Vygotsky, 1962; Wallace et al., 2017). Questionnaires, such as the Varieties of Inner Speech Questionnaire-Revised (Alderson-Day et al., 2018) and the Self-Talk Scale (Brinthaup et al., 2009), capture several functions of inner speech, including its evaluative nature (e.g., negative self-talk, motivation to change), the representation of inner speech as another person’s voice inside our own head, the positive and regulatory function of inner speech (e.g., supportive self-talk, emotion regulation, self-management), and inner speech as related to social interaction (e.g., imagining or reviewing past conversations) (Alderson-Day et al., 2018; Brinthaup et al., 2009). Additionally, inner speech has been associated with executive function/cognitive control (Alderson-Day & Fernyhough, 2015; Winsler et al., 2009), Theory-of-Mind (Alderson-Day et al., 2016), and self-awareness (Morin, 2005, 2009, 2011b). Self-awareness, including self-regulation, observation, and introspection, are associated with increased activation in extra-

linguistic areas, including the medial prefrontal cortex, inferior parietal lobules, posterior cingulate/precuneus, and regions of the medial and lateral temporal lobes (Morin, 2011a, 2011b). The interdisciplinary nature of the study of inner speech function extends to work on motivation and endurance in sports sciences, showing that interfering with inner speech decreased performance during sustained physical activity (Nedergaard et al., 2023). A clear next step in the neuroscience field is to evaluate not only multiple dimensions of inner speech, but also how neural mechanisms may differ according to inner speech's intended function. For example, given that a large number of individuals use inner speech to solve problems, it would be fruitful to evaluate the extent to which the neural mechanisms of the two processes overlap, thus shedding more light on the interconnectedness of inner speech with higher cognitive systems.

3. Effectively understanding inner speech in post-stroke aphasia requires the use of interdisciplinary theory/evidence about multidimensionality

Aphasia is a language disorder most commonly occurring after left hemisphere stroke and affecting roughly two million adults in the USA (Simmons-Mackie & Cherney, 2018). It is most often caused by stroke and can affect all aspects of language (both expressive and receptive). Recently, the topic of inner speech has gained interest in aphasiology research (see Fama & Turkeltaub, 2020 for a review), due to the common report from people with aphasia that they “know what it is but can’t say it” and inspired by autobiographical accounts of acute loss of inner speech, followed by a return throughout recovery (Marks, 2018; Taylor, 2009). Dr. Jill Bolte Taylor, a neuroanatomist, suffered a left-hemisphere stroke which caused a temporary but nearly complete loss of inner speech immediately post-stroke. She highlights the importance of inner speech in her book *My Stroke of Insight*, describing the lack of inner speech as “the dramatic silence that had taken residency inside my head” (Taylor, 2009, pp. 75-76). Dr. Taylor stated that the silence affected her self-awareness and sense of self (p. 67), but it is unknown

whether this is common to most people living with aphasia. But, Brumfitt (1993) also stated that one must understand that the “impact on the aphasic person’s experience may be the lost ability to talk to oneself” (Brumfitt, 1993), which leads us to believe that impacted inner speech may be a shared experience across people with aphasia. Interestingly, it has been observed that, in brain damaged patients who eventually recover from their trauma, self-awareness often returns in parallel with inner speech (Ojemann et al., 1996). Given Dr. Bolte Taylor’s many appearances and discussion of her aphasia, it appears that she may likewise have regained her inner speech with time, similar to the observation made by Ojemann et al. (1996). This anecdotal evidence highlights the resiliency of inner speech, as well as a need to carefully evaluate all dimensions of inner speech. These autobiographical accounts denote a change in inner speech due to aphasia, which is understandable given that the brain regions involved in inner speech also tend to be damaged in aphasia (e.g., left inferior frontal gyrus). What is particularly interesting in this population, however, is that inner speech and overt speech abilities can dissociate – there are individuals with better inner speech than speech out loud and vice versa (Fama, Henderson, et al., 2019; Geva, Bennett, et al., 2011; Stark et al., 2017), although the interpretation of this is dependent on the definition of inner speech.

In aphasiology research, a prominent definition of inner speech is “the ability to create an internal representation of the auditory word form, without producing overt sounds or articulation, and to apply computations or manipulations to this representation” (Geva & Warburton, 2019). In this case, inner speech has the phenomenological property of accessing and then manipulating phonological processes, which is not a definition necessarily shared across other disciplines. Alternatively, Fama and colleagues define successful inner speech as “the feeling of being able to say a word correctly in one’s head” (Fama et al., 2019, p. 107). These two definitions differ in whether inner speech requires just access to the word (including semantics and phonology), or if it also requires manipulation. Both definitions are used primarily at the word level when researching word-finding and naming abilities, thus overlooking some

dimensions of inner speech discussed above (spontaneity, dialogality), especially if inner speech becomes more condensed or if there are more linguistically complex thoughts in inner speech. We argue that aphasia is an opportune clinical population to address *all* components of the inner speech phenomenon in an interdisciplinary way: capturing inner speech as it is used in everyday life (its function, its intentionality, its dialogality, its condensation).

The majority of studies cited in the brief overview of inner speech neuroscience, above, tend to use functional imaging (e.g., PET, fMRI), which are indirect ways of measuring the involvement of areas to a behavioral process. Studying inner speech in post-stroke aphasia enables lesion-symptom mapping, which associates brain damage with behavior performance. This facilitates understanding the *necessary* regions underlying function by linking behavioral performance to lesion sites. Geva and colleagues found that difficulties with silent rhyme judgment (a type of highly controlled, intentional, monologic inner speech), over and above impairments of overt speech (reading aloud), were associated with lesions to the left inferior frontal gyrus *pars opercularis* and supramarginal gyrus (Geva, Jones, et al., 2011). Fama and colleagues related the subjective experience of successful inner speech during naming difficulty to lesions in the left hemisphere inferior frontal regions and sensorimotor cortices, which are involved in later stages of speech output processing (Fama et al., 2017). Indeed, Fama and colleagues found that brain damage associated with successful inner speech was different from brain damage associated with the subjective perception of having a word on the tip of the tongue or knowing a word without having all the right sounds for the word. We can also learn more about neuroplasticity and brain reorganization after stroke by studying how people with aphasia use inner speech over the course of their recovery, and how this relates to brain changes (e.g., connectivity) across the recovery period.

Additionally, the clinical population of aphasia enables us to directly evaluate the cognitive and neural components necessary for inner speech by comparing overt and inner speech. For example, if the participant reports successful inner speech during naming with “all

of the right sounds in the right order” and performs well on inner rhyme judgment tasks but has difficulty saying words out loud, we can presume that phonological and lexical access was successful and gain more insight into the mechanisms of inner speech (Fama, Henderson, et al., 2019). Because of the disruption of language abilities, aphasiology research on inner speech is well situated to answer the question of how language, thought, and consciousness interact and may even help to address controversies concerning the modularity of language, which have been posed in philosophy (Langland-Hassan et al., 2018). When language is disrupted, what happens to thought, and how complex can our inner narratives be with impoverished overt speech?

One can also relate inner speech abilities and perceptions to aphasia severity, aphasia type, and specific deficits (e.g., auditory comprehension, motor planning/programming/execution, naming, awareness of deficits), giving a better profile of who experiences inner speech and what that experience is like. This may help to tease apart which brain areas and aspects of cognition are *required* for inner speech versus those that contribute to the process. Our lab has collaborated with philosophers and looked in detail at aphasia severity and anosognosia (lack of awareness of language deficits) and found encouraging results concerning aphasia severity – that our participants did not show differences in inner speech performance or reports of inner speech based on aphasia severity (Alexander et al., 2023, 2024), but that potential anosognosia could be more telling of differences in inner speech frequency (Alexander et al., 2024). Indeed, this finding that those with anosognosia may perceive inner speech less often correlates with findings from psychology and philosophy about the close tie that inner speech has with self-reflection and monitoring (Morin, 2011a, 2011b). Anosognosia has been associated more with a lack of awareness at the domain-general level, again implicating default mode network nodes (Therriault et al., 2018). Therefore, the preservation of inner speech in individuals with post-stroke aphasia may be at least partially due to non-linguistic, domain general areas such as nodes in the default mode network (e.g.,

precuneus, cingulate cortex), which are not typically damaged in aphasia-causing events (i.e., left middle cerebral artery strokes/injuries).

Studying inner speech in aphasia also enables a more in-depth investigation of the role of inner speech in identity, self-esteem, and self-awareness. Inner speech has been implicated in these processes, likely due to its self-reflective functions such as self-perception, self-evaluation, monitoring, introspection, and self-knowledge, but this research has centered on young adults (Alderson-Day & Fernyhough, 2015; Morin, 2005, 2009; Morin & Michaud, 2007). We will define self-awareness as Morin does in his work: that self-awareness is when one attends to oneself and understands their own emotions, attitudes, and attributes, and that this often takes place with inner speech (Morin, 2011a). Due to the hypothesized relationship of inner speech with these factors, it is possible to learn more about personal and psychosocial aspects of aphasia by holistically evaluating inner speech and, on the other hand, to learn more about the self-related functions that inner speech assumes by evaluating it in persons with aphasia. For example, if inner speech is used for positive and negative self-evaluation in individuals with aphasia as it is in neurotypical adults (Alderson-Day et al., 2018; Racy et al., 2020), therapy manipulating inner speech (e.g., cognitive behavioral therapy) may lead to better treatment outcomes and psychosocial health. We are also considering how inner speech may relate to use and effectiveness of metacognitive treatment, which researchers are beginning to explore in people with aphasia to increase self-awareness and independence (Wadams et al., 2022).

Further, identification of the impact of a significant life event, such as stroke and aphasia, and subsequent longitudinal evaluation of inner speech, will enable a comprehensive understanding of the extent to which inner speech is dynamic (e.g., changes over time in response to situations) and resilient after brain injury. We can, for example, evaluate the extent to which inner speech is subjectively experienced by people with aphasia. For example, is it common that people experience *more* inner speech after their stroke? Does this extend to inner

speech across all dimensions, for example, is inner speech used more *intentionally* post-stroke? It may be the case that impairments in overt speech lead to a heightened awareness of inner speech, as we have found that persons with more impaired overt language tend to subjectively experience inner speech more (Alexander et al., 2023). Aphasia may be a unique model by which to understand cognitive (e.g., language) and brain mechanisms associated with inner speech because of the heterogeneity of language impairments (and brain damage) experienced by members of this population. Within the population of people with aphasia, we can further tease apart individual differences, exploring the extent to which condensation, dialogality, and intentionality differ across and within individuals (e.g., over time, situationally). Researchers can collect longitudinal data in neurotypical populations to see how inner speech changes throughout the life span and in people with aphasia to gain empirical evidence of the gradual recovery of inner speech from acute to chronic phases. If certain skills or linguistic forms appear in inner speech prior to returning in overt speech, that could further the possibility of inner speech as a predictor of recovery and therapeutic performance.

Just as aphasia enables a better understanding of the role of inner speech in self-processes, aphasia also enables a better understanding of the function of inner speech: when, how, and for what purpose inner speech is used. Indeed, studying spontaneous inner speech in persons with aphasia has suggested that these individuals, like individuals without brain damage, rely on inner speech when remembering, to problem solve, and to rehearse ahead of time (Alexander et al., 2024). We can also characterize inner speech in unique ways. For example, some scholars separate “imagined interactions” from “inner speech,” where imagined interactions are recalling past conversations or predicting future ones (Morin, 2019). As people with aphasia may feel the need to practice interactions ahead of time or review them afterwards (in their head), it would be interesting to establish the extent to which imagined interactions make up a person with aphasia’s experience of inner speech (and individual differences), and how this makeup compares to neurotypical adults’ inner speech. An exciting avenue is to

explore narrative generation in inner speech to help reframe identity after stroke, as many people with aphasia report a change in their self-concept (Meyerson & Zuckerman, 2019; Shadden, 2005; Taubner et al., 2020), and biographic-narrative intervention can improve reports of health-related quality of life (Corsten et al., 2015). We are also considering how inner speech may relate to use of metacognitive treatment, which researchers are beginning to explore in people with aphasia to increase self-awareness and independence (Wadams et al., 2022).

One can learn about the daily lives of people with aphasia through inner speech. Indeed, aphasia recovery includes recovery across several domains, not limited to recovery of language. After a devastating event, such as a stroke, and with stroke-related sequelae such as aphasia, recovery also occurs across personal, psychosocial, participatory, and activity-related domains (Simmons-Mackie & Kagan, 2007). This holistic view of aphasia recovery is well accepted in the field and endorsed by people living with aphasia (S. J. Wallace et al., 2017). It is therefore highly relevant to identify predictors of, and contributors to, recovery in people living with aphasia. There is preliminary evidence that successful inner speech in naming is related to therapeutic outcomes in people with aphasia (Hayward et al., 2016), and we are currently investigating this with a larger group of participants.

Finally, work in aphasiology could inform research in other populations of interest when considering inner speech, such as autism spectrum disorders (Williams et al., 2016) and schizophrenia (Barber et al., 2021).

4. Conclusions and next steps

Interdisciplinary research on inner speech – and we argue, incorporation of these interdisciplinary methods in populations with brain injury and aphasia – holds great potential for advancing our understanding of the cognitive and neuroscientific bases of this phenomenon. By integrating evidence across disciplines, methodologies, and clinical populations, we can enhance our knowledge of inner speech processes. With a careful combination of theory and

methods from multiple disciplines, researchers can create an experimental design which captures multidimensional inner speech, focusing on the specific dimensions of interest for their research questions. This knowledge can guide future research directions, broaden our conceptual toolbox, deepen our understanding of cognitive and neuroscientific mechanisms, and improve clinical understanding and outcomes for individuals with aphasia and other brain injuries.

Abbreviations

ACL: Administration for Community Living

ConDialInt: Condensation, Dialogality, Intentionality

DTI: diffusion tensor imaging

fMRI: functional magnetic resonance imaging

HHS: Health and Human Services

NIDILRR: National Institute on Disability, Independent Living, and Rehabilitation Research

PET: positron emission tomography

USA: United States of America

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Data Availability Statement

Data presented in this article are discussed in full in a Frontiers in Psychology publication (Alexander et al., 2024). The preprint and data are available on the Open Science Framework (<https://doi.org/10.31219/osf.io/gbw3t>; data available: <https://osf.io/b6vxk/>).

Conflict of Interest Statement

Authors have no financial conflicts of interest. Authors have no non-financial conflicts of interest.

Author Contributions Statement

JMA and BCS equally contributed to the conceptualization, writing, and editing of this manuscript.

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Chapter 2: Measuring Inner Speech Subjectively and Objectively in Aphasia

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Background

Many people with aphasia and people without brain injury talk to themselves in their heads, i.e., have “inner speech.” Inner speech may be more preserved compared with spoken speech for some people with aphasia and may serve a variety of functions (e.g., emotion regulation), which motivates us to provide a high-fidelity characterization of it. Researchers have used multiple methods to measure this internal phenomenon in the past, which we combine here for the first time in a single study.

Aims

We compare performance between individuals with and without aphasia on inner speech tasks that assess inner speech “in-the-moment” to general subjective impressions of inner speech to tease apart the relationship of aphasia severity to inner speech.

Methods and Procedures

Twenty people with mild-moderate aphasia and twenty neurotypical controls completed several inner speech tasks, including objective silent rhyme judgements (picture, written, and auditory), subjective reports of inner speech during naming, and subjective rating scales about inner speech experience more generally.

Outcomes and Results

In-the-moment inner speech during silent rhyming tasks was associated with aphasia severity only for picture and written rhyming but not auditory rhyming. In-the-moment inner speech

reports during silent naming were not associated with aphasia severity, nor were the subjective ratings about general inner speech experience. Individuals with and without aphasia demonstrated a variety of subjective general inner speech experiences, demonstrating heterogeneity of this phenomenon more broadly.

Conclusions

Methods of measuring inner speech complement each other and speak to different facets of the inner speech phenomenon, and clinicians and researchers must carefully choose the method(s) that will provide the information about inner speech that they desire.

Inner Speech and Its Measurement

Inner speech is the familiar experience of a “little voice in the head” (Langland-Hassan, 2021; Perrone-Bertolotti et al., 2014). Research in neurotypical individuals and individuals with aphasia suggests that inner speech retains lexical and phonological properties (Oppenheim & Dell, 2008), and is associated with brain areas closely associated with language production (Fama et al., 2017; Geva & Fernyhough, 2019) and with cognitive processes such as phonological working memory (Fama, Henderson, et al., 2019; Stark et al., 2017). Inner speech is not experienced by all, however; some estimates suggest that people experience inner speech with a wide range (0-75%) of frequency (Heavey et al., 2019; Hurlburt et al., 2021). Compellingly, we and others have shown that inner speech, compared to overt speech, is relatively preserved in a majority of individuals with aphasia (Fama et al., 2017; Geva, Bennett, et al., 2011; Stark et al., 2017), which has led to the hypothesis that inner speech may have a consequential role in language recovery in aphasia (Fama & Turkeltaub, 2020; Stark et al., 2017).

Inner speech is complex. One method of conceptualizing inner speech’s multifaceted nature is the ConDialInt model, which describes three dimensions of inner speech: condensation, dialogality, and intentionality (Grandchamp et al., 2019). On the condensation dimension, inner speech ranges from condensed (without all acoustic, phonological, and syntactic information) to expanded (including articulatory and auditory properties). Dialogality dimension refers to whether the inner speech captures just one voice/perspective, like a monologue or inner soliloquy, or whether it is dialogical (e.g., an inner verbal conversation in one’s own voice or with the addition of other voices). For the intentionality dimension, inner speech can be purposeful, as when intentionally rehearsing information or manipulating it silently, or it can be unintentional/spontaneous, as in daydreaming or mind wandering

(Grandchamp et al., 2019). Therefore, different methods for measuring inner speech likely also target different facets of inner speech.

Within the aphasia literature, there are two definitions of inner speech, differing based on whether inner speech involves active manipulation of the word form in the head (as in Geva, Bennett, et al., 2011; Stark et al., 2017), or access to the word form without the need for manipulation (as in, e.g., Fama, Snider, et al., 2019). Notably, both definitions evaluate what we call ‘in-the-moment’ inner speech, meaning that they are aiming to tap into inner speech generated during isolated moments of time.

It is not surprising, given the differing definitions of inner speech from the aphasia literature, that researchers have elected to measure inner speech using different, though complementary, methods: objective and subjective tasks. Inner speech involving manipulation has been most often measured objectively using silent rhyming decisions, where individuals are shown either two pictures, two written words, or listen to two aurally presented words and then make a decision via button press about whether or not they rhyme. Therefore, within objective tasks measuring inner speech, cognitive processes differ. Auditory judgement of rhymes relies on phonological access and the phonological loop (sometimes termed phonological working memory; Baddeley & Hitch, 2019). Written judgement of rhymes relies on orthographic-to-phonologic conversion, especially in instances where two words rhyme but have dissimilar orthography (e.g., bear, chair), in addition to the phonological loop. Rhyme judgement using pictures requires access to lexical-semantic and phonological information via the visual route, and also the phonological loop. A benefit to objective measurement is that these methods assure that the participants are completing the tasks asked of them (e.g., by responding via button press to a rhyme judgement) and because the experimenters largely control the situation (e.g., by providing stimuli that range in orthographic match and by monitoring the subject to ensure that they are not moving their lips or tongue) (Geva, Jones, et al., 2011). Objective methods like rhyme judgment require the extra step of manipulation beyond access to the word,

making it more difficult to separate inner speech access from cognitive-linguistic abilities such as phonological working memory. Yet, manipulation is not exactly equivalent to phonological working memory. While Geva and colleagues (2011) identified that rhyme judgement requires working memory, they did not find that homophone judgement relied on phonological working memory to the same extent. Their definition of inner speech, however, requires manipulation.

Subjective assessment of inner speech in aphasia has likewise relied on measuring in-the-moment experience of inner speech, most commonly during confrontation naming. For example, when presented with a picture and asked to name it silently in one's head, a participant would say 'yes' to having subjective inner speech if they could access the word and it sounded right (Fama, Henderson, et al., 2019). Silent picture naming has been related to phonological access skills, but not necessarily phonological working memory given that no manipulation is necessary (in contrast to the objective tasks, described above) (Fama, Snider, et al., 2019). As with all subjective tasks, whether in the moment or generalized, there is a lack of experimental control inherent to measuring internal, phenomenological experience (Lupyan et al., 2023).

The other way in which inner speech has been measured subjectively is by evaluating the function and uses of inner speech more generally. This is the method most commonly used to measure inner speech in neurotypical adults by employing questionnaires/surveys (Alderson-Day et al., 2018; Heavey et al., 2019; Racy et al., 2020), thought listing (Racy et al., 2020), interviews and other dynamic sampling methods (e.g., experience sampling methodology, Hurlburt et al., 2021). The goal of these methodologies is to evaluate how often inner speech occurs across some timeframe (e.g., the last week), for what reasons, and in what contexts. Research in non-brain-damaged adults suggests inner speech frequency ranges from 0-75% of sampled instances (Hurlburt et al., 2016); is used for a variety of functions, including problem-solving (Wallace et al., 2017), self-awareness (Morin, 2011b, 2011a) and emotion regulation (Morin & Michaud, 2007), and varies by context (e.g., may be more present during cognitively

difficult scenarios) (Morin et al., 2018; Racy et al., 2020). A benefit to this way of measuring inner speech is these methods have high face validity for how inner speech is perceived in everyday life.

The choice to measure inner speech objectively and subjectively is experimenter-specific and based on the goals of the study, for example, whether the goal is to evaluate inner speech in the moment (using rhyme [objective] or inner naming [subjective]) or more generally (e.g., via survey). The tasks tell us slightly different information about inner speech because they theoretically invoke different cognitive processes. However, there is also, theoretically, considerable overlap in the cognitive processes employed by some objective and subjective tasks. For example, objective picture rhyme judgment and subjective inner naming are both thought to require phonological access via visual routes and have both been shown to implicate the dorsal stream of language, thereby reflecting some shared neural components (e.g., Fama, Snider, et al., 2019; Geva, Jones, et al., 2011).

To our knowledge, our study is the first to employ both objective and subjective means of measuring inner speech in aphasia. In the current study, we collect data in adults with and without aphasia for the three in-the-moment objective rhyming tasks described above (picture, written word, auditory word; Section 1.1 of Methods) as well as a subjective in-the-moment task (silent naming; Section 1.2 of Methods) and subjective Likert-style questions about inner speech experience more generally (Section 2 of Methods). Few studies have contrasted performance between neurotypical adults and individuals with aphasia across objective and subjective means of measuring inner speech, and thus we will examine this in Research Question 1. To flesh out the differences and similarities between tasks all meant to evaluate some facet of inner speech, we will examine the relationship between performance on objective tasks, on tasks evaluating inner speech in-the-moment, and between subjective tasks (Research Question 2). We will compare objective and subjective performance whenever possible.

An interesting phenomenon that has been observed in at least two studies using objective rhyming tasks to measure inner speech is that inner speech is relatively preserved compared to overt speech, even in individuals with more severe aphasia (Geva, Bennett, et al., 2011; Stark et al., 2017). Note, though, that the individuals with more severe aphasia, whilst having relatively preserved inner speech compared with their overt speech, still demonstrated worse performance on both inner and overt speech tasks compared to individuals with milder aphasia. There is a theoretical basis for explaining the impact of aphasia severity on inner speech performance, and it is related to the cognitive processes used during these objective inner speech tasks. Two of the objective tasks – picture and written rhyme decisions – are thought to engage the “inner voice” whilst auditory rhyme decisions are thought to engage the “inner ear” (Hubbard, 2010; Smith et al., 1995). The “inner ear” is understood to be a more passive process, involving just auditory information, with the “inner voice” being a more active process related to subvocalization, thus including pre-articulatory processes and, often, rehearsal. Another group makes a similar distinction between “inner speaking” and “inner hearing” (Hurlburt et al., 2013), emphasizing the difference between whether a person is truly saying something in their head or if they are simply hearing it, as if played back on a recording. There are likely different underlying processes involved in inner voice/speaking versus inner ear/hearing (Hubbard, 2010). The “inner voice” requires lexical access and retrieval and other cognitive processes, like phonological working memory, compared to the “inner ear,” which does not. Because decision making based on inner voice tasks likely requires access to more cognitive-linguistic processes than decision making based on inner ear tasks, inner voice tasks might be more affected in people with more severe aphasia. Tree and Playfoot (2019), in an individual with conduction aphasia and impaired phonological short-term memory, and Langland-Hassan et al. (2015) in a large group of individuals with aphasia, both showed that inner voice tasks are typically more difficult for individuals with aphasia, though they did not explicitly look at aphasia severity as further moderating performance on inner voice tasks.

Studies evaluating inner speech using subjective methods like silent naming have likewise demonstrated that individuals with varying aphasia severities demonstrate varying inner speech abilities (Fama, 2019; Fama et al., 2017; Fama, Snider, et al., 2019), though the impact of aphasia severity on subjective inner speech has had limited evaluation. Therefore, we empirically evaluate the relationship of aphasia severity with inner speech performance across objective and subjective tasks in Research Question 3.

Research Questions

Below is a summary of our three research questions, with hypotheses generated from the literature discussed above:

Research Question 1: *How do neurotypical controls differ from persons with aphasia regarding inner speech?* This is largely a confirmatory research question, as prior evidence has demonstrated that individuals with aphasia, overall, do worse than their neurotypical peers on objective measures of inner speech, like rhyming tasks. Therefore, we expect to confirm these results for objective inner speech tasks. However, comparatively less literature has evaluated differences between individuals with and without aphasia using subjective measurements of inner speech. Our hypothesis regarding group differences during subjective inner speech tasks is therefore exploratory, but we also anticipate finding that the aphasia group demonstrates different performance (likely, more variability) than the neurotypical group. Given that some of our subjective measurements do not have a “better” or “worse,” as the objective measurements do, we cannot make a directional hypothesis in the same way. Therefore, this exploratory hypothesis about subjective inner speech is derived from the fact that inner speech is typically variable in neurotypical adults (e.g., Hurlburt et al., 2021; Roebuck & Lupyan, 2020) and inner speech in aphasia may be more variable (Fama, Snider, et al., 2019; Fama & Turkeltaub, 2020; Geva, Bennett, et al., 2011; Langeland-Hassan et al., 2015; Stark et al., 2017).

Research Question 2: *To what extent does performance on each inner speech task relate to performance on other inner speech tasks within similar genres (i.e., objective, subjective)? Relatedly, to what extent do subjective measures and objective measures relate to one another?* Inner speech tasks tap into different cognitive processes but share underlying processes. We therefore expected to find positive correlations between inner speech tasks within the same genre (objective, subjective) and which share underlying cognitive processes (i.e., picture rhyme decision [objective] and silent naming of pictures [subjective]).

Research Question 3: *Is performance on inner speech tasks mediated by aphasia severity?* We expected individuals with aphasia to do worse, overall, on inner voice tasks (measured objectively using picture and written word rhyme judgments) compared to inner ear tasks (measured objectively using auditory word rhyme judgments). We also expected there to be an impact of severity, where individuals with most severe aphasia also demonstrate the lowest accuracy on inner voice tasks. For hypotheses regarding the impact of aphasia severity on subjective measurements of inner speech, there is sparse evidence, so these are largely exploratory. We hypothesized that individuals with more severe aphasia would show more impairments with silently naming objects (described in Section 1.2 of Methods, “Naming SIS”), and that aphasia severity would impact general interpretation of inner speech in daily life (described in Section 2 of Methods).

Method

Participants

Twenty-three participants with chronic post-stroke aphasia were recruited by author PLH for a related study (details can be found in Langland-Hassan et al. (2021)), which was approved by the University of Cincinnati’s ethical review board (IRB #2012-4185). The original study collected the Western Aphasia Battery-Revised (WAB-R) (Kertesz, 2007) to establish aphasia severity and type. To ensure adequate comprehension of experimental tasks, the original study

included participants only if their auditory-verbal comprehension score on the WAB-R was ≥ 4 (out of 10). To provide a general overview of cognitive functioning, the original study also had individuals with aphasia complete the non-linguistic subtests of the Cognitive Linguistic Quick Task (CLQT) (Helm, 2003), and the participants were subsequently classified as within normal limits, mildly impaired, moderately impaired, or severely impaired. If a participant obtained ratings of “within normal limits” and/or “mild” impairment on the non-linguistic subtests of the CLQT, this was taken as evidence of auditory and visual perception adequate to complete the experimental tasks and participants were included in the study.

N = 20 individuals with aphasia were included in the following analyses, excluding three of the original 23 individuals with aphasia due to Attention and Visuospatial Skill scores on the CLQT indicating more than mild impairment. They were aged 35-72 years ($M=56.4$, $SD=9.09$), and all had chronic aphasia (1.5-31 years with aphasia, with a mean of 8.85 ± 8.09). Four types of aphasia were represented: anomic ($n=7$), Broca's ($n=6$), conduction ($n=6$), and transcortical motor ($n=1$). Only one participant was classified as having severe aphasia according to the WAB-R Aphasia Quotient (AQ) (an AQ score of <50), while all others were classified as mild-moderate ($M=73.1 \pm 13.9$). The average level of education was approximately equivalent to a bachelor's degree ($M=15.9 \pm 1.79$ years).

A sample of prospectively matched non-brain damaged individuals served as a control group. This group was at least 18 years of age, had no history of language impairments, had no history of major psychotic or neurologic episodes or intractable substance abuse, had at least a high school education, and were native speakers of American English. Twenty adults were included (Table 1).

Between the aphasia and control groups, there were no significant differences in age, gender/sex, or education. Table 1 includes demographic information and statistical comparisons.

Table 1: Demographic information of all included participants

	Aphasia (N=20)	Control (N=20)	Statistics
Age			
Mean (SD)	56.4 (9.09)	55.2 (9.66)	Independent t-test t(38) = -0.42, p = 0.68
Median [Min, Max]	55.5 [35.0, 72.0]	55.0 [41.0, 70.0]	
Sex			
F	11 (55.0%)	13 (65.0%)	Chi-square test $\chi^2 = 0.38$, p = 0.54
M	9 (45.0%)	7 (35.0%)	
Years Education			
Mean (SD)	15.9 (1.79)	16.5 (1.82)	Wilcoxon Rank Sum Test W = 239.5, p = 0.27
Median [Min, Max]	16.0 [12.0, 18.0]	18.0 [14.0, 18.0]	
Years with Aphasia			
Mean (SD)	8.85 (8.09)	NA	NA
Median [Min, Max]	5.00 [1.50, 31.0]	NA	
Aphasia Type			
Anomic	7 (35.0%)	NA	NA
Broca's	6 (30.0%)	NA	
Conduction	6 (30.0%)	NA	
Transcortical Motor	1 (5.0%)	NA	
WAB-R Aphasia Quotient			
Mean (SD)	73.1 (13.9)	NA	NA
Median [Min, Max]	70.9 [47.8, 94.0]	NA	

WAB-R = Western Aphasia Battery – Revised.

Materials and Procedure

An inner speech battery, described below, was presented on a 21" touchscreen using a JavaScript application stored on a secure external web server. There are two major sections to the battery. The first section evaluates “in-the-moment inner speech,” which is further divided into subjective and objective methods, whilst the second section evaluates how participants generally, subjectively interpret their inner speech.

Section 1: In-the-moment inner speech

Section 1 was further subdivided into Objective inner speech tasks (S1.1) and Subjective inner speech tasks (S1.2).

Section 1.1: Objective tasks. The objective portion of the inner speech battery (see Figure 1 for a visual representation) included the following three subtests:

1. Picture Rhyme Judgement, in which the participants saw two pictures of objects and were asked if the words for them rhyme (as in Langland-Hassan et al., 2015). This task requires access to semantic, lexical and phonological information via visual routes, as well as phonological working memory. It is commonly referred to as an “inner voice” task;
2. Written Rhyme Judgement, in which the participants saw two written words, some of which varied in orthography (e.g., Bear / Chair), and were asked if they rhyme. This task requires access to orthographic and phonological information as well as phonological working memory. It is commonly referred to as an “inner voice” task;
3. Auditory Rhyme Judgement, in which the participants heard two words and were asked if they rhyme. Participants could hear the words an unlimited number of times. This task requires access to phonological representations and phonological working memory. It is commonly referred to as an “inner ear” task.

The materials for the picture rhyming task were adapted from the materials of Geva and Warburton (2019). However, the sets used were slightly different from the published version of

Geva and Warburton (2019)'s materials (those materials were normed on British English-speaking adults) and so cannot be considered strictly equivalent.

The words used in each of the three inner rhyming tasks did not differ significantly in terms of their frequency, familiarity, or imageability, based on t-tests ($p > 0.3$ for all comparisons) of ratings from the online MRC Psycholinguistic Database (Coltheart, 1981). For all tasks, participants were instructed to give a yes/no response via button press on the touchscreen and to complete the task in silence and without moving their mouth or tongue. A research assistant monitored performance. There were 38 trials involving picture rhyming, 40 trials for written word rhyming, and 40 trials for auditory rhyme judgements with no time limits for each trial and no instructions about speed of answering.

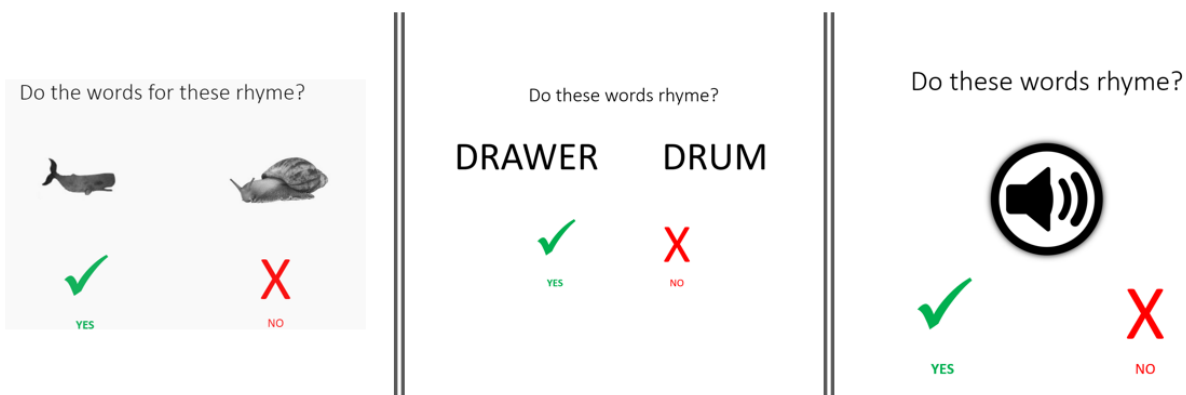


Figure 2: Examples of displays for picture rhyme judgement, written rhyme judgement, and auditory rhyme judgement. Participants responded by touching the green check mark or the red x.

Section 1.2: Subjective task: Naming Successful Inner Speech (SIS). Participants also completed subjective reports of inner speech during a silent naming task in which they were shown one picture at a time and were asked to indicate whether they were able to say the word for the object silently in their head and have it sound right (“naming SIS”). Again, they responded with a yes/no button press to the prompt. The subjective report of single word inner speech requires only access to phonological information but not manipulation of phonological information.

In order to compare inner speech to overt speech ability, participants completed overt (out loud) naming of the same items. They were shown one picture at a time and asked to name it out loud. The experimenter recorded their response in English orthography.

Section 2: General, subjective interpretation of inner speech

This section included four subjective rating questions, which were presented on the screen and read aloud by the experimenter. In contrast to the in-the-moment objective and subjective tasks above, these ratings were about participant’s experience in general, not just during the task. The subjective rating questions used a categorical visual rating scale (continuous arrow with numbers and descriptions) from 1-5 with verbal descriptions and are as follows:

1. Idea without Word (IwW), “How often do you see something and know what it is but can’t say the word for it out loud?” with 1 being almost never and 5 being almost always;
2. Post-Stroke, given to aphasia group only, “Do you think you talk to yourself in your head less now, more now, or about the same as before your stroke?” with 1 being much less now and 5 being much more now;
3. Successful Inner Speech (SIS), “A few minutes ago we talked about how we

sometimes see something and know what it is but can't say the word for it out loud. When that happens to you, how often can you say the right word for it in your head, and have it sound right?" with 1 being almost never and 5 being almost always.

4. Inner vs. Overt Speech, "In general, how do you think your abilities to speak in your head compare to your abilities to speak out loud?" with 1 being much better in head and 5 being much better out loud.

Two of the subjective rating questions (IWW and SIS) were adapted from Fama's work (e.g., Fama et al., 2019, Fama & Turkeltaub, 2020). As noted above, the subjective responses were on a 5-point scale, which changed based on the question, and were about general abilities rather than subjective reports about specific single words. Therefore, they served as a complement to the subjective single word inner speech data also collected. The questions were read aloud and presented visually. Participants selected their answer on a touchscreen. Individuals with aphasia answered all four subjective questions, while the control group answered all but the post-stroke question.

To summarize, we distinguish the inner speech tasks in two ways: whether the method is objective or subjective, and whether the measurement is happening "in-the-moment" of inner speech or is a general interpretation of the overall inner speech experience.

Experimental Procedure

Participants with aphasia attended two sessions: a screening session in which they completed neuropsychological assessments (e.g., WAB-R), and a second session in which they completed the inner speech tasks and a categorization task relevant to a larger abstract thought study (Langland-Hassan et al., 2021). Sessions were scheduled one to three weeks apart, depending on participant availability. Neurotypical controls attended only one session and completed the inner speech tasks and categorization task in that session.

Regarding the inner speech task ordering, the subjective ratings described in Section 2 were always presented to participants first, and the order of the subjective questions was not randomized (they were presented in the order described above). Naming SIS (Section 1.2) came after the general subjective ratings from Section 2. Next, participants completed the rhyme judgement tasks (Section 1.1). The questions from Section 1.1 were presented in pseudo-randomized order by participant, alternating among six different orderings of the tasks. Overt naming (also described in Section 1.2) was always the last task.

Steps were taken to ensure that individuals with aphasia understood the questions. In all cases, the question prompts were both displayed visually and slowly read aloud by the experimenter. Participants had unlimited time to consider and answer the prompts. The important concept of “knowing what something is without knowing the word for it,” was explained in a training trial that preceded the experimental trials. That training trial showed a picture of a watch and read, in part, “For example, a person might look at the object pictured below and know that it is something used for keeping track of the time, and that you wear on your wrist, but not be able to remember that it is called a **watch**. This would be a case of knowing what something is without remembering the word for it.” Additionally, we want to highlight that the mean Auditory Verbal Comprehension score from the WAB-R was 8.9 out of 10, indicating comprehension suitable for understanding the questions.

Data Analysis

All data analysis was completed in R version 4.1.1 with RStudio Desktop 2022.07.1+554.

The data analysis was designed to examine the research questions, below.

Research Question 1: Neurotypical Controls compared to Participants with Aphasia

To ensure that the objective inner speech tasks were valid (i.e., that controls performed near ceiling / with high accuracy on all tasks) and to replicate findings from Geva and Warburton (2019), we computed descriptive statistics for the objective inner speech tasks. We also statistically compared the neurotypical controls to the individuals with aphasia on the inner speech tasks that both groups completed (i.e., excluding the subjective post-stroke question because only the stroke group completed this question). Note that two members of the control group were missing data for the general, subjective rating about SIS, and the missing values were ignored in analysis. For continuous metrics, we used either independent t-tests or Wilcoxon rank sum tests dependent on data normality, and for categorical metrics (the subjective questions), we used Wilcoxon rank sum tests.

Research Question 2: Inner Speech Task Relationship with one another

We estimated the magnitude of the relationship between performance on each of the Section 1.1 (objective “in-the-moment” inner speech) tasks using Spearman correlations. Additionally, we investigated the relationship between tasks from Section 1.2 (subjective inner speech “in-the-moment” and overt naming) using the same methods. We used Spearman’s correlation to evaluate the relationship between select Section 1.1 and Section 1.2 tasks, specifically the successful inner speech in-the-moment (Section 1.2) and the objective picture rhyming task (Section 1.1). We chose these two tasks because they both require inner speech achieved through picture-based visual semantic access and thus should partially share cognitive mechanisms. Finally, we used Spearman correlations to explore the relationship between Section 2 tasks (general, subjective interpretation of inner speech using Likert scales).

For Research Question 2, we only evaluated the aphasia data, given that the data from neurotypical controls approached ceiling for most Section 1.1 and 1.2 measures.

Research Question 3: Relationship between Aphasia Severity and Inner Speech

We first computed Spearman correlations between severity, as measured by the WAB-R AQ, and each of the inner speech in-the-moment tasks (objective, subjective). In addition to computing correlations with this continuous metric of aphasia severity, we separated the aphasia group into two severity groups based on *a priori* severity groups identified by the WAB-R: a “mild” group (AQ scores of ≥ 76 , but note that one person included in this group scored a 94, which could be considered “latent”, i.e., non-clinical aphasia) and “moderate” (AQ scores of 51-75, but note that we included one person who scored 47.8 in this group). There were $n = 9$ in the mild group and $n = 11$ in the moderate group. Mild and moderate aphasia are commonly used distinctions from the WAB-R, relevant to clinical practice. We then analysed the extent to which inner speech scores differed by severity group using the Wilcoxon rank sum test with continuity correction, given that the data did not satisfy normality assumptions (Meek et al., 2007). To analyse aphasia severity impact on Section 2 tasks (subjective, general interpretation of inner speech), we computed the Wilcoxon rank sum test with continuity correction for each question to compare aphasia severity groups.

We had originally planned to also evaluate the impact of aphasia type (fluent vs. non-fluent) on inner speech, but we ultimately decided that due to the scoring criteria on the WAB-R (in which severity/AQ has a large bearing on type of aphasia), evaluating by type would not add sufficient additional information to our study and would be conflated with aphasia severity.

Results

Research Question 1: Descriptive statistics and group comparisons

Section 1 Tasks: In-the-moment inner speech

Controls’ data on Section 1 assessments confirmed that the tasks were valid (i.e., performance was high/ at ceiling for all tasks) (see Table 2). To assess the differences between individuals with aphasia and controls on all Section 1 in-the-moment inner speech tasks,

Wilcoxon rank sum tests were conducted. As expected, the control group significantly outperformed the individuals with aphasia group on all Section 1 tasks except for Naming SIS (Section 1.2), as both groups were near ceiling.

Section 2 Tasks: General, subjective interpretation of inner speech

We conducted Wilcoxon rank sum tests to assess the differences between individuals with aphasia and controls on the general, subjective questions. As shown in Table 2, for the subjective questions that asked more generally about inner speech, we saw similar answers for the controls and individuals with aphasia. An exception was that individuals with aphasia more commonly reported that their inner speech was better or much better than their speech out loud, while the controls sometimes reported better speech out loud than in their heads. Figure 2 represents the subjective answers from the individuals with aphasia group and the control group, visually demonstrating that individuals with aphasia and control groups only differ on their answers of whether they speak better in their head vs. out loud. Additionally, the individuals with aphasia and control groups' answers vary considerably within groups.

Table 2: Descriptive Statistics for individuals with aphasia and Controls on Objective and Subjective Inner Speech Tasks, with Section 1 as proportions and Section 2 as ratings from 1-5.

Section	Task	Aphasia (N=20)	Control (N=20)	Statistics
Section 1.1	Inner Picture Rhyming			
	Mean (SD)	0.643 (0.166)	0.946 (0.0646)	W = 18,
	Median [Min, Max]	0.592 [0.368, 0.947]	0.970 [0.790, 1.00]	p < .001
	Inner Written Rhyming			
	Mean (SD)	0.634 (0.169)	0.948 (0.0409)	W = 22,
	Median [Min, Max]	0.625 [0.350, 0.950]	0.950 [0.880, 1.00]	p < .001
Section 1.2	Inner Auditory Rhyming			
	Mean (SD)	0.755 (0.148)	0.961 (0.0341)	W = 28.5,
	Median [Min, Max]	0.750 [0.450, 1.00]	0.950 [0.880, 1.00]	p < .001
	Naming SIS			
	Mean (SD)	0.972 (0.0397)	0.993 (0.0133)	W = 142.5,
	Median [Min, Max]	0.973 [0.838, 1.00]	1.00 [0.970, 1.00]	p = .08
Section 2	Overt Naming			
	Mean (SD)	0.786 (0.217)	0.992 (0.0170)	W = 28,
	Median [Min, Max]	0.865 [0.135, 1.00]	1.00 [0.950, 1.00]	p < .001
	Idea without Word			
	Mean (SD)	2.70 (1.08)	2.35 (0.933)	W = 236.5,
	Median [Min, Max]	3.00 [1.00, 5.00]	2.00 [1.00, 4.00]	p = .30
Section 2	Successful Inner Speech			
	Mean (SD)	3.55 (1.32)	3.39 (1.61)	W = 184,
	Median [Min, Max]	4.00 [1.00, 5.00]	4.00 [1.00, 5.00]	p = .91
	Inner vs. Outer			
	Mean (SD)	1.90 (0.718)	2.95 (0.826)	W = 74,
	Median [Min, Max]	2.00 [1.00, 3.00]	3.00 [2.00, 5.00]	p < .001
Section 2	Post-Stroke			
	Mean (SD)	3.25 (1.02)	NA	NA
	Median [Min, Max]	3.00 [1.00, 5.00]	NA	

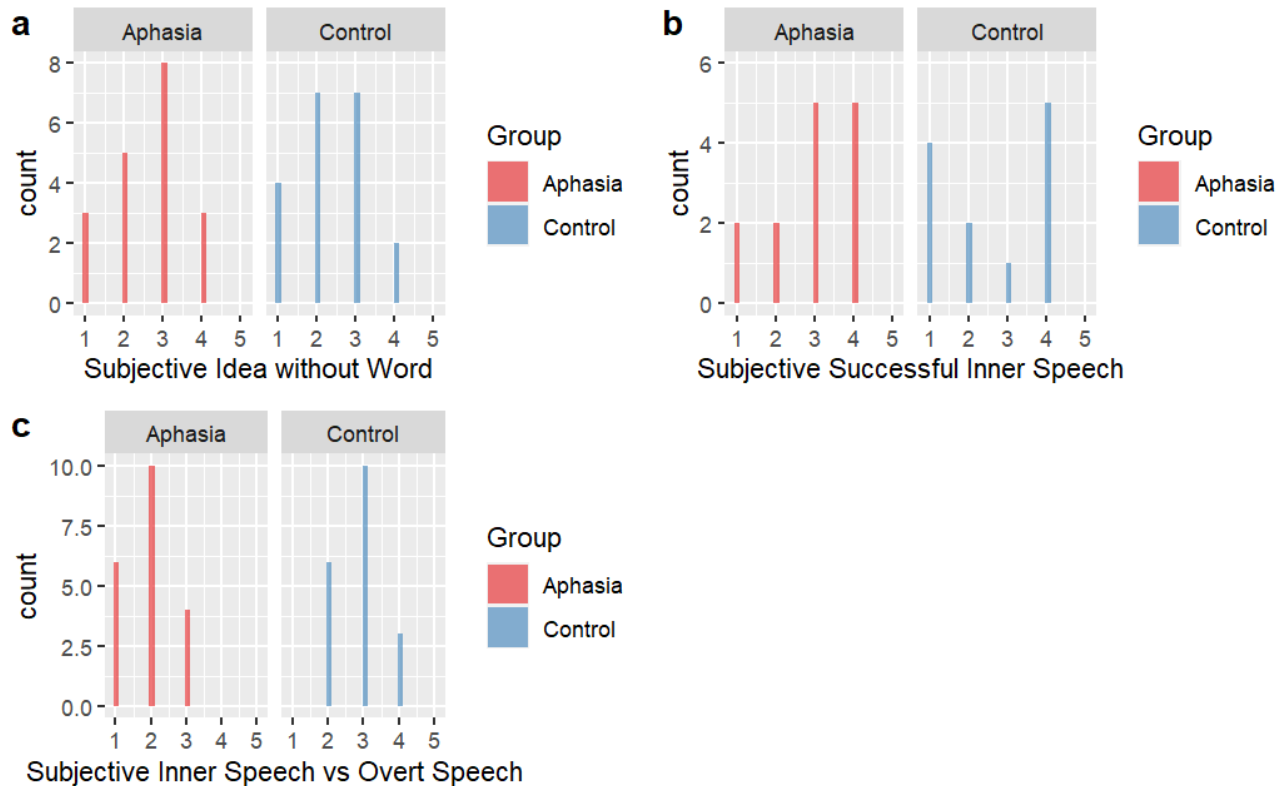


Figure 2: Subjective Ratings with persons with aphasia (individuals with aphasia) and neurotypical controls comparison.

Idea without Word (IwW), “How often do you see something and know what it is but can’t say the word for it out loud?” with 1 being almost never and 5 being almost always; Successful Inner Speech (SIS), “A few minutes ago we talked about how we sometimes see something and know what it is but can’t say the word for it out loud. When that happens to you, how often can you say the right word for it in your head, and have it sound right?” with 1 being almost never and 5 being almost always; Inner vs. Overt Speech, “In general, how do you think your abilities to speak in your head compare to your abilities to speak out loud?” with 1 being much better in head and 5 being much better out loud

Research Question 2: Inner speech task relationship with one another

Figure 3 shows a comprehensive representation of Spearman correlations between general subjective inner speech questions (Section 2) and inner speech in-the-moment tasks (Section 1) as well as the demographic variables of age, years with aphasia, education (in years), and aphasia severity (WAB-R AQ).

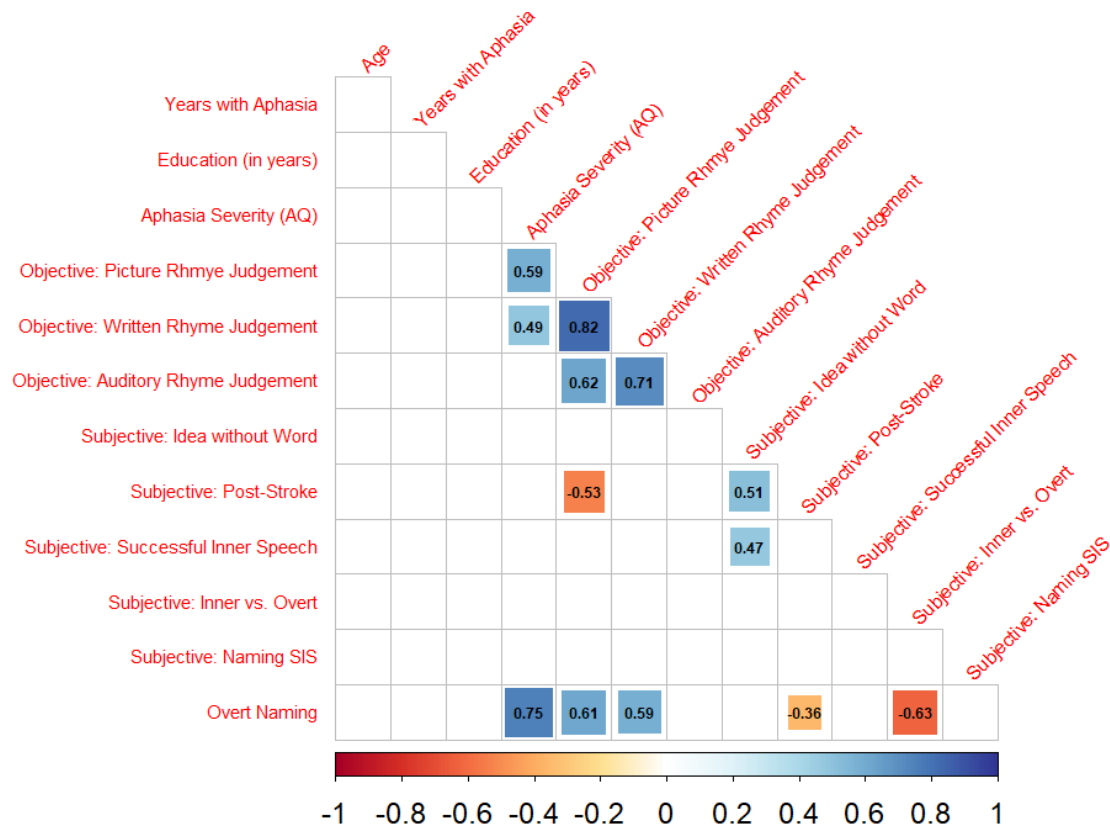


Figure 3: Spearman's correlations between objective and subjective measures, as well as demographic variables. All correlations shown are significant with a 95% confidence interval. Larger squares and darker hues indicate stronger correlations, also shown by the numbers in each square.

Section 1 Tasks: In-the-moment inner speech

As hypothesized, the objective rhyme tasks from Section 1.1 were significantly correlated with one another (picture and written, $r_s = 0.82$, $p < .001$; picture and auditory, $r_s = 0.64$, $p = 0.004$; written and auditory, $r_s = 0.71$, $p = 0.0004$). Overt naming was also related to the two objective rhyming judgements which required the "inner voice," picture ($r_s = 0.61$, $p = 0.004$) and written ($r_s = 0.59$, $p = 0.007$), but not the "inner ear" task, auditory ($r_s = 0.29$, $p = 0.22$).

We only had one in-the-moment, subjective inner speech task (Section 1.2), Naming SIS. Therefore, we could not truly compare this to the more general subjective tasks from Section 2. However, we did evaluate the extent to which silent naming (Naming SIS) was

related to overt naming in the aphasia group and found a not-significant relationship ($r_s = -0.17$, $p = 0.47$, Figure 4a). This finding shows that overt naming abilities were not related to reports of silent naming in individuals with aphasia, with few reporting impoverished silent naming.

We evaluated the relationship between an objective in-the-moment method (picture rhyme judgment) and a subjective in-the-moment method (Naming SIS) that we hypothesized involved substantially overlapping cognitive processes, namely picture-based visual access to lexical and phonological information. Using Spearman's correlations, there was no significant relationship between the picture rhyme task and Naming SIS ($r_s = 0.14$, $p = 0.55$, Figure 4b).

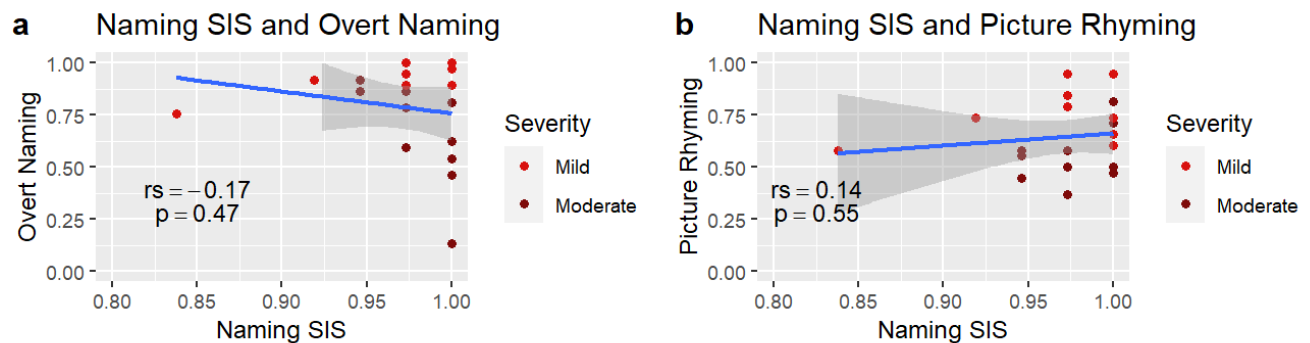


Figure 4: Correlations between reports of successful inner speech (SIS) in-the-moment of silent naming (“Naming SIS”) with a) overt (out loud) naming performance and b) performance on the objective picture rhyme judgement task for individuals with aphasia. Spearman’s correlation coefficient (r_s) and significance values are provided. Additionally, dots represent each participant, with light red dots representing those with mild aphasia severity, and dark red dots representing those with moderate aphasia severity. The trend line (in blue) includes all participants with aphasia.

Section 2 Tasks: General, subjective interpretation of inner speech

We found only a single significant correlation between subjective, general tasks, which was that lwW was significantly related to Post-Stroke ($r_s = 0.51$, $p = 0.02$). This suggested that people with aphasia who tended to have the idea without word (lwW) also tended to experience more inner speech post-stroke.

Research Question 3: Evaluating the impact of aphasia severity on inner speech

Section 1 Tasks: in-the-moment inner speech

We first compared performance on the objective rhyme judgement tasks using Wilcoxon rank sum tests, because we hypothesized that individuals with aphasia would do worse on inner voice compared with inner ear tasks. As expected, individuals with aphasia performed better on the auditory task than the written and picture tasks (Table 2 for means), and there was no difference in performance between the written and picture tasks (auditory vs picture: $W = 279.5$, $p = 0.03$; auditory vs written: $W = 286$, $p = 0.02$; picture vs written: $W = 202.5$, $p = 0.96$). It was also of interest how individuals with aphasia did when we parsed apart the residuals from each inner speech task, i.e., evaluated performance on picture rhyming when considering variance from written and auditory rhyming performances, and so on. For example, given that the picture rhyming task is thought to involve different cognitive processes, with some overlapping with the cognitive processes required in the written and auditory rhyming tasks, considering variance from the other objective tasks enabled us to evaluate performance uniquely per task. When accounting for variance from each of the other objective tasks, a one-way ANOVA showed that individuals with aphasia did not perform significantly differently on any task (three paired t-tests, all $p > 0.99$). This suggests that, when evaluating each objective task as comprised of unique cognitive components, individuals with aphasia performed similarly on auditory, written, and picture inner speech tasks. The similar performance when accounting for variance suggested that the poor performance on picture and written rhyme tasks by the individuals with aphasia group as a whole (no regression of other tasks) may be due to the specific cognitive processes engaged by these tasks.

Next, we explicitly assessed the role of aphasia severity. To assess how the continuous variable of aphasia severity was related to the objective inner speech tasks, we conducted Spearman's correlations. People with milder aphasia (based on the WAB-R AQ) tended to

perform better on the objective inner speech tasks, especially picture ($r_s = .31$, $p = .01$) and written ($r_s = .23$, $p = .03$) tasks, but aphasia severity was not related to performance on the auditory task ($r_s = .17$, $p = .06$). As WAB-R AQ increases (indicating milder aphasia severity), performance on the picture and written tasks also increases, but more of the participants at all aphasia severity levels are at or near ceiling for the auditory task (Figure 5).

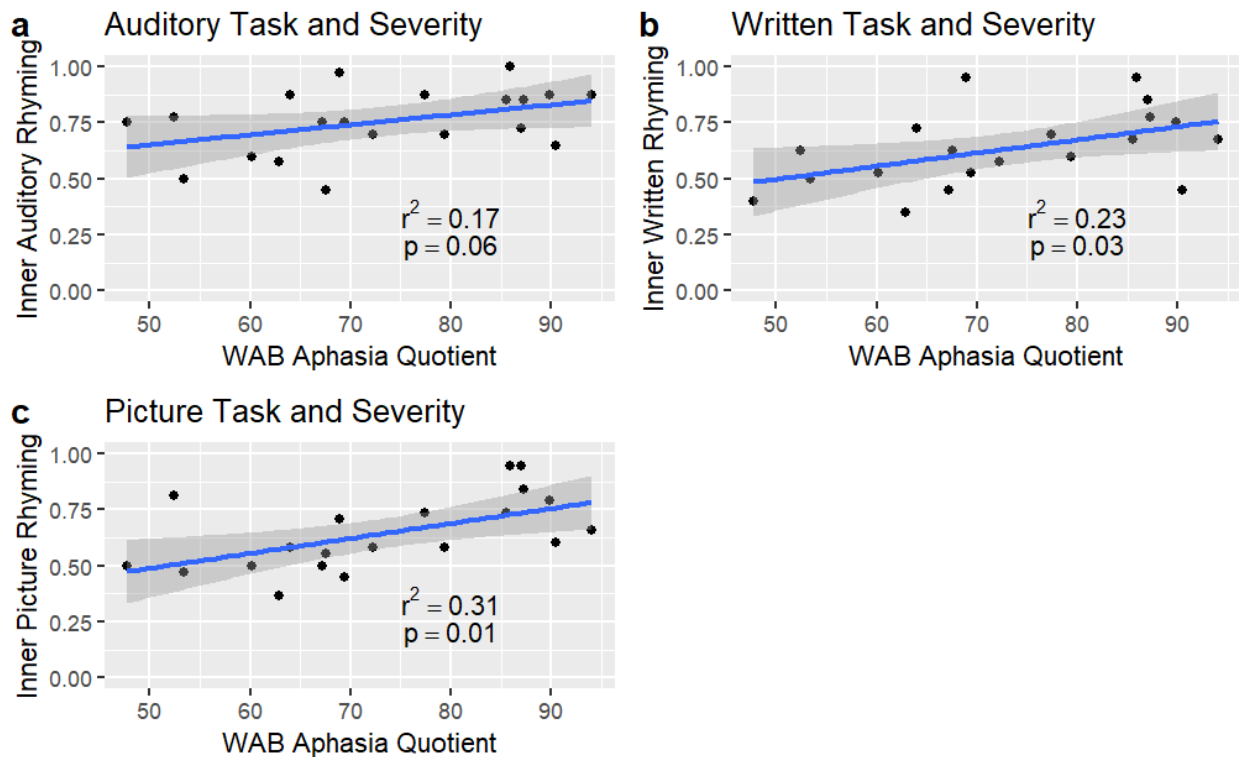


Figure 5: Objective Inner Rhyming Judgement Tasks - a) Auditory: participants hear two words; b) Written: participants see two words; c) Picture: participants see drawings of two objects. x-axis is WAB-R AQ, where a higher score is milder aphasia severity.

When we grouped the individuals with aphasia into mild and moderate aphasia severity groups based on their WAB-R AQ, we found similar results using the Wilcoxon rank sum tests, with the mild group performing more accurately than the moderate group on the picture ($W = 89$, $p = 0.003$) and written ($W = 77$, $p = 0.04$) inner speech rhyme judgement tasks, but not on the auditory rhyme judgement task ($W = 71$, $p = 0.11$). As seen in Figure 6, the severity distinction appears related to inner speech abilities, as the mild group outperformed the moderate group on the tasks which required inner “voice” (picture, written), but not inner “ear” (auditory).

When evaluating subjective in-the-moment inner speech (Naming SIS), we did not find a significant association with aphasia severity, as measured by AQ ($r_s = 0.02$, $p = 0.60$) or using severity groups ($W = 47.5$, $p = 0.90$).

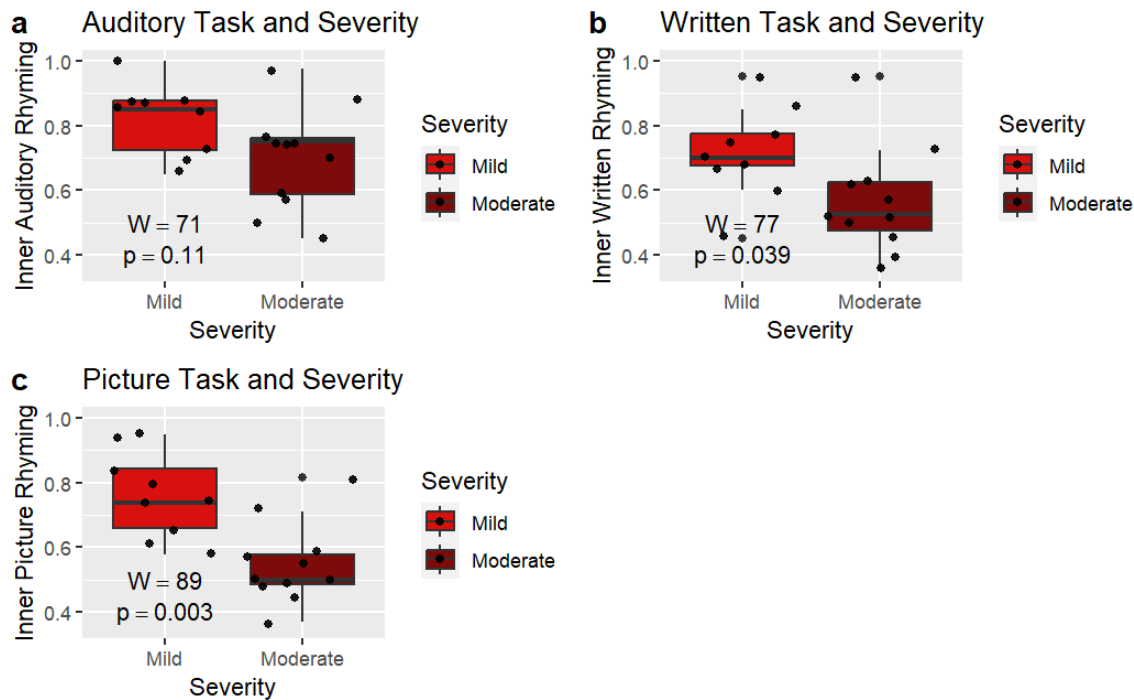


Figure 6: Objective Inner Rhyme Judgement and Categorical Severity - a) Auditory; b) Written; c) Picture. The groups were split into mild severity and moderate severity based on the cut-offs from the WAB-R. Jittered dots represent individual performance.

Section 2 Tasks: General, subjective interpretation of inner speech

Aphasia severity groups answered similarly on all the general subjective questions about inner speech (Section 2) based on the Wilcoxon rank sum test with continuity correction.

Individuals from the different severity groups reported similar frequency of successful inner speech ($W = 48.5$, $p = 0.97$) and idea without word ($W = 37.5$, $p = 0.36$). Both participant groups reported using inner speech more often now or at the same frequency as they did before their stroke, with no significant differences between the groups ($W = 38$, $p = 0.36$). Both groups reported equal or better inner speech than overt speech, with no significant differences between

the groups ($W = 33.5, p = 0.20$). As seen in Figure 7, subjective ratings were similar between the two severity groups, with intra- and inter-group variability.

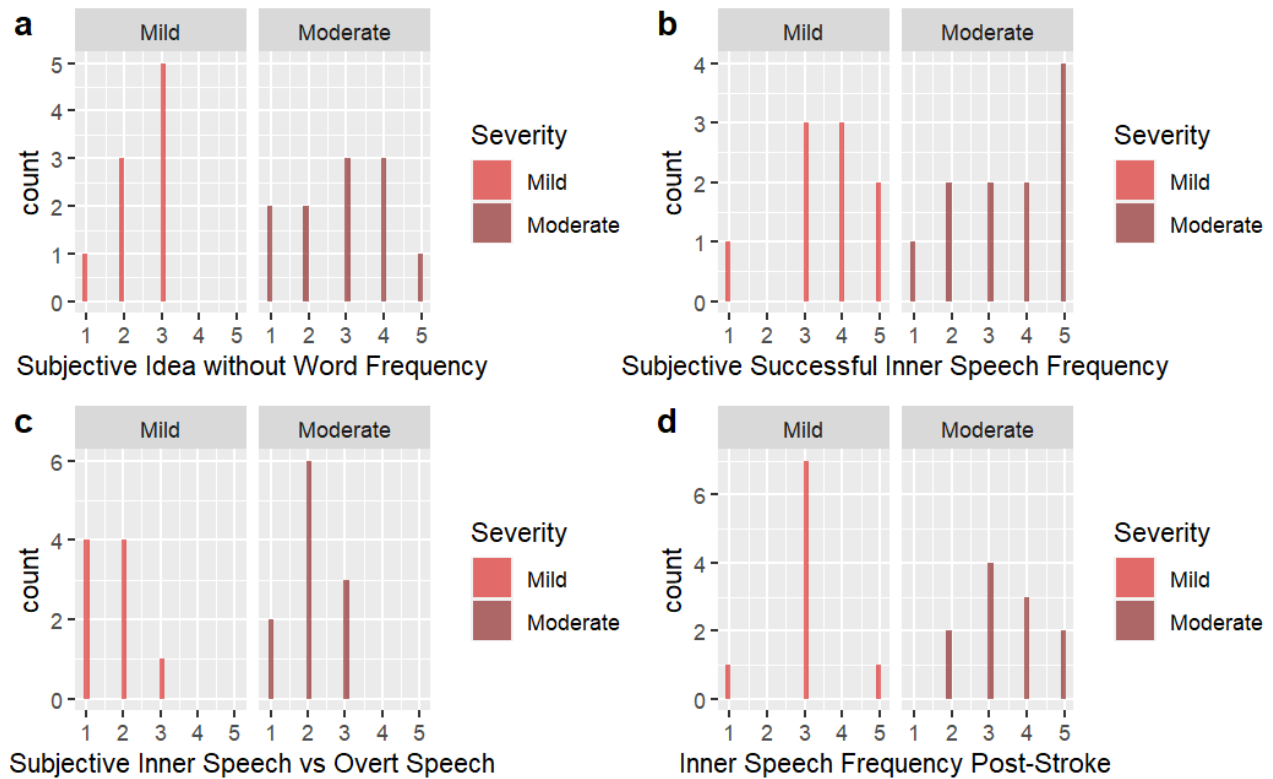


Figure 7: Subjective Ratings, comparisons between mild and moderate aphasia.
Idea without Word (IwW), “How often do you see something and know what it is but can’t say the word for it out loud?” with 1 being almost never and 5 being almost always; Post-Stroke, “Do you think you talk to yourself in your head less now, more now, or about the same as before your stroke?” with 1 being much less now and 5 being much more now; Successful Inner Speech (SIS), “A few minutes ago we talked about how we sometimes see something and know what it is but can’t say the word for it out loud. When that happens to you, how often can you say the right word for it in your head, and have it sound right?” with 1 being almost never and 5 being almost always; Inner vs. Overt Speech, “In general, how do you think your abilities to speak in your head compare to your abilities to speak out loud?” with 1 being much better in head and 5 being much better out loud

Cases examining individual differences in inner speech performance

Looking at individual participants illuminates the differences in abilities and reports of inner speech. Hence, we wanted to take time in the Results to discuss specific examples. For example, in participants with moderate aphasia, there are varying strengths and weaknesses on tasks requiring lexical-semantic access with pictures (Table 3). Three of these participants

(1003, 1008, and 1011) performed similarly on picture rhyme judgement but differed on the other tasks, while the fourth participant (1016) performed better on picture rhyme judgement, allowing for another contrast. First consider participant 1008, who made more phonemic naming errors (i.e., similar speech sounds, e.g., “bowl” for “bow”) rather than semantic errors (i.e., similar meaning, e.g., “bowl” for “cup”). Other errors were either unrelated to the target or no response. Participant 1008 endorsed having 100% naming SIS, but overt naming score for the same items was 46%. They also scored poorly on the objective picture task (50%). On the rating scales for general subjective inner speech questions (Section 2), participant 1008 marked a “5” for how often they experience SIS, suggesting agreement between Naming SIS and general SIS experiences. Participant 1011 scored 50% on picture rhyme judgement, 13.5% on overt naming (with more semantic than phonemic errors), but 100% on reports of successful inner speech for the same pictures and rated the frequency of successful inner speech as 5 (almost always). In contrast, participant 1003, who also has moderate aphasia, scored 55% on picture rhyme judgement but 92% on overt naming (with no phonemic errors and some semantic) and 95% on reports of successful inner speech and rated the frequency of successful inner speech as lower (3, sometimes). It is possible that participants have difficulty with different cognitive processes required for picture rhyme judgement (semantic and phonological access, phonological awareness, working memory) even though they performed similarly on that task (~50%), and aphasia severity does not completely explain performance on these tasks. Those differences appear in overt naming scores and their perceptions of inner speech.

Another contrast is from participant 1016, who performed well on the picture rhyme judgement task (82%) but poorly on overt naming (54%, with more semantic than phonemic errors), while rating the frequency of SIS in cases of anomia out loud as 2 (rarely). We postulate that participant 1016’s primary impairments are in articulatory and motor components, but if they cannot find a word, it is because they usually do not have the semantic or phonetic access. Participant 1008 may have a different issue because they make more phonemic errors, with

difficulty in phonetic access which makes both the picture rhyming and overt naming challenging.

Table 3: Scores on tasks and ratings requiring lexical-semantic access, highlighting the individual differences in participants with moderate aphasia.

Participant	Aphasia Severity (AQ)	Overt Naming Errors (%)	Overt Naming Score (%)	Naming SIS (%)	Rating-SIS (Scale 1-5)	Objective-Picture Rhyming (%)
1003	Moderate (67.6)	Phonemic: 0 Semantic: 50 Other: 50	92	95	3 - sometimes	55
1008	Moderate (47.8)	Phonemic: 28 Semantic: 0 Other: 72	46	100	5 - almost always	50
1011	Moderate (60.2)	Phonemic: 3 Semantic: 22 Other: 75	14	100	5 - almost always	50
1016	Moderate (52.4)	Phonemic: 0 Semantic: 24 Other: 76	54	100	2 - rarely	82

Discussion

Inner speech is difficult to measure by nature, as it occurs entirely in one's head. Nevertheless, it has been measured using both subjective and objective paradigms, in-the-moment and more generally, with individuals demonstrating its validity by showing that perception of inner speech relates to expected brain areas (Fama et al., 2017). Understanding how inner speech relates to severity when it is measured differently has downstream implications for more comprehensive study of inner speech in aphasia. It is important to evaluate the relationship of aphasia severity and inner speech, as aphasia severity is related to quality of life and participation (Williamson et al., 2011), and inner speech is likely related to quality-of-life factors, such as self-awareness (Morin, 2009, 2011; Morin & Michaud, 2007) and psychological distress (Heavey & Hurlburt, 2008). By evaluating how aphasia severity is related to inner speech, we can gain insight into the mechanisms behind changes in self-awareness and psychological health with aphasia. With the many methods of measuring inner speech that

have been utilized in prior research, it is important to reveal what information we can gain from each method and how the methods complement each other.

How do neurotypical controls differ from persons with aphasia on inner speech tasks?

Regarding objective tasks, we expected and found that individuals with aphasia did worse on objective measures of inner speech than individuals without aphasia. This makes sense, given that inner speech measured using silent rhyme judgement has been closely tied to phonological awareness and manipulation and most individuals with aphasia have impairments with these language functions (Stark et al., 2017; Fama et al., 2019). Regarding group differences for subjective inner speech tasks, our exploratory hypothesis was that the aphasia group would demonstrate different performance (likely, more variability) than the neurotypical group. For the subjective, general inner speech questions, we found that both the aphasia and the control group had high within-group variability with no significant group differences. This suggests that people have a range of experiences of inner speech, regardless of history of brain injury, which has been shown before in neurotypical adults (Alderson-Day et al., 2018; Racy et al., 2020). We likewise did not identify a significant difference on the in-the-moment subjective measure of inner speech, Naming SIS, between the two groups. In the case of Naming SIS, both groups reported accuracy near ceiling (see Table 2 for means). Therefore, our hypothesis regarding wider variability for the aphasia group on subjective questions did not come to fruition. A null finding related to group differences during subjective tasks may be because the subjective tasks did not employ the phonological working memory component that the objective rhyming tasks did, resulting in a higher performance from the aphasia group.

Inner speech measured across tasks

We first expected to find positive correlations between inner speech tasks within the same genre (objective, subjective) and modality (in-the-moment) for the aphasia group. We found this to be the case for the objective rhyming tasks, which showed positive correlations with one

another. We also found that, within the more general subjective questions, lwW (idea without word) was significantly, positively related to using their inner speech post-stroke. It may have been the case that, more generally, people are thinking more about concepts related to speaking (i.e., lwW) after the stroke.

We also expected to find positive correlations between tasks that shared underlying cognitive processes despite being collected using different methods (i.e., picture rhyme decision [objective] and silent naming of pictures [subjective]). Surprisingly, we did not find a significant relationship between picture rhyme judgment and silent picture naming, which suggests that, even whilst at face value these tasks appear to substantially overlap in cognitive demands, this may not be the case in practice. Indeed, we found that individuals with aphasia had a much higher rate of Naming SIS (all > 80%, refer to Table 2) than they did accuracy on picture rhyming (mean of 64%, refer to Table 2), suggesting that this null relationship may be due to the difficulty of the picture rhyming task. This difficulty could be the result of the added cognitive process of phonological working memory, the fact that they were asked to come up with two names (e.g., Bear vs. Chair) versus one name (Naming SIS), or inflated reporting of accuracy in Naming SIS.

We did not identify a significant relationship between Naming SIS and overt naming performance in the aphasia group. At face value, this seems strange, but this finding jives with prior literature (e.g., Geva et al., 2011). To be related in a positive manner, this would entail that everyone with very good naming also has very good inner speech (not rare, and we do find this) but also that everyone with poor naming has poor inner speech (rare). Our sample did not report poor inner speech during naming, as shown in Figure 4A. That is, the median for Naming SIS was 97% (range ~84-100%) whilst overt naming had a median of 86.5% (range ~13.5 – 100%). The markedly preserved inner speech alongside relatively impoverished overt naming has been shown in a variety of papers evaluating inner speech in aphasia (e.g., Stark et al., 2017; Geva et al., 2011), where the majority of participants had more preserved inner speech compared to

overt speech. Thus, we suspect that the non-significant relationship between Naming SIS and overt naming is driven by a lack of people reporting poor inner speech.

Is performance on inner speech tasks mediated by aphasia severity?

We expected individuals with aphasia to do worse, overall, on inner voice tasks (measured objectively using picture and written word rhyme judgments) compared to inner ear tasks (measured objectively using auditory word rhyme judgments). Confirming our hypothesis, we found that individuals with aphasia performed better on the auditory rhyme task (inner ear) than the written or picture rhyme tasks (inner voice tasks). We also expected there to be an impact of severity, where more severe individuals would show the worst performance on inner voice tasks, in particular. Individuals with milder aphasia performed better on the objective inner speech tasks involving the inner voice (picture, written). We did not identify a significant relationship between severity and performance on the auditory rhyme task, likely because the performance was near ceiling for most participants. These findings confirm prior research (Fama, Henderson, et al., 2019; Langland-Hassan et al., 2015; Tree & Playfoot, 2019) and, coupled with complementary findings from other labs, lead us to conclude that objective methods that require several cognitive-linguistic steps for accessing inner speech (e.g., picture rhyme, written rhyme) are more liable to be testing the language production-like component of inner voice and more likely to be related to aphasia severity.

Our exploratory hypothesis was that individuals with more severe aphasia would show more impairments with silently naming objects and that aphasia severity would impact general interpretation of inner speech in daily life. In contrast to what we identified for the inner voice tasks, above, we did not find that the subjective task of Naming SIS was related to aphasia severity for our sample, even though it is also an in-the-moment inner speech task. One likely explanation is that this is due to the difference in cognitive resources required for access alone (Naming SIS task) vs. access and manipulation (inner voice objective inner speech tasks). That

is, aphasia severity may become an important factor when internal manipulation is required. Note that overt naming was related to aphasia severity (as shown in Figure 3), even though overt naming does not require manipulation. This correlation makes sense given that the aphasia quotient from the WAB-R is partially derived from performance on word-finding and object naming. Further, overt naming requires post-phonological processes (articulation, motor) that inner speech does not, and these may be particularly related to the way that the WAB-R quantifies aphasia severity. Alternatively, Naming SIS reports may have been inflated if participants were inaccurate with reporting inner speech.

We likewise explored the extent to which aphasia severity mediated the more general subjective inner speech tasks. We found that aphasia severity groups did not demonstrate significantly different responses for any of the subjective questions about how they generally experience inner speech. There was high within-group variability for most subjective questions, and we did not find that ratings were related to severity. We identified some trends suggesting that severity of aphasia may play a role in general subjective perception of inner speech if evaluated in a larger sample: for example, participants with moderate aphasia reported using inner speech more often now than they did before their stroke, while participants with milder aphasia tended to report having the same frequency of inner speech use now as they did prior to their stroke. This could be because people with moderate aphasia are intentionally using inner speech more often due to their more impaired overt speech. Both groups reported equal or better inner speech than overt speech. This finding strongly encourages further investigation into the nature and uses of inner speech across a diverse sampling of individuals with aphasia.

Figure 7 is an example of heterogeneity on the general subjective perceptions about inner speech in the mild and moderate aphasia groups. When asked about having an Idea Without Word (“IwW”), there was wide variation within the moderate aphasia group (all scale ratings used), whereas the mild group used only three of five of the scale ratings. This suggests greater homogeneity of experience in the mild group as it relates to Idea Without Word. For

Successful Inner Speech (“SIS”) frequency, both the mild and moderate groups showed high intra-group heterogeneity, using all scale options (moderate) or four out of five scale options (mild). Both the mild and moderate groups trended in feeling that their inner speech was better than their overt speech, demonstrating intra-group homogeneity. For the frequency of inner speech pre-stroke, the mild group tended to be most homogeneous, with nearly all members selecting option three (“about the same as before my stroke”) whereas the moderate group tended to show higher intra-group heterogeneity but with many participants choosing a higher scale option (“experience inner speech more now”). While these subjective scales did not significantly relate to aphasia severity or demonstrate significant differences between participant groups (control, individuals with aphasia), they serve to clearly illustrate individual differences in the experience of inner speech. Some of these individual differences were likely present pre-stroke, because as we saw, the control group also had heterogeneity in their answers (Figure 2). Individual differences in inner speech have also been shown in individuals with aphasia in past literature (Fama, Henderson, et al., 2019; Geva, Bennett, et al., 2011; Langland-Hassan et al., 2015; Stark et al., 2017), and often in individuals without aphasia (e.g., Hurlburt et al., 2013).

Contextualizing the differences between inner speech tasks

We want to conclude this Discussion by mapping back how these tasks relate to a multidimensional framework used to understand inner speech, ConDialInt (Grandchamp et al., 2019). If the relationships between aphasia severity and the ways of measuring inner speech are dissimilar, it is possible that they are measuring different facets or experiences of inner speech, as described in the ConDialInt model (Grandchamp et al., 2019). Picture and written rhyme judgements likely use expanded, monologue, and intentional inner speech, based on the ConDialInt model. These rhyme judgements require phonetic information, are not conversational, and are purposeful, and the auditory task may share these qualities, although it may be less “intentional.” While the general subjective rating questions were designed to test

perception of inner speech, it is possible that participants were thinking beyond inner speech used for isolated tasks (e.g., rhyming, naming/word-finding) and more about their experience of an inner narrative, which may be more condensed and less intentional but likely still monologue. Inner speech throughout daily life, measured by some of our subjective general questions, relates to work by Morin and colleagues on the relationship between inner narrative and extra-linguistic processes such as self-regulation and problem-solving (Morin, 2011a; Morin & Michaud, 2007), which are likely less related to aphasia severity. The differences in the condensation and intentionality dimensions may contribute to the results which indicate that there are not strong relationships between the subjective, general methods of measuring inner speech and how they relate to aphasia severity. For example, condensed inner speech would not require total phonetic access, so participants with difficulty accessing the phonemes would not necessarily report worse inner speech more generally.

Conclusion and Clinical Implications

Aphasia severity was related to in-the-moment objective inner speech tasks that involve the inner voice, including picture and written rhyme judgments, but not to other in-the-moment tasks, including subjective inner speech during naming and objective auditory rhyme judgment. Aphasia severity was also not related to subjective interpretation of the general frequency/use of inner speech. We therefore suggest that the difference may have been due to the cognitive-linguistic processes involved and alternative interpretations of the questions on the subjective rating scales.

Individuals with aphasia may retain inner speech despite impoverished lexical or phonological abilities (Fama et al., 2017; Fama, 2019; Fama, Snider, et al., 2019; Fama & Turkeltaub, 2020; Geva, Bennett, et al., 2011; Stark et al., 2017). It has long been understood that inner speech plays a role in aphasia, e.g., Brumfitt (1993) states that one must understand that the ‘impact on the aphasic person’s experience may be the lost ability to talk to oneself’.

Moreover, in brain damaged patients who eventually recover from their trauma, self-awareness often returns in parallel with inner speech (Ojemann et al., 1996). The importance of inner speech in aphasia recovery is emphasized by Dr. Jill Bolte Taylor in her book, *My Stroke of Insight*, in which she describes her experience recovering from a left hemispheric stroke (Bolte Taylor, 2006). Dr. Taylor recalls a total lack of inner speech: “The dramatic silence that had taken residency inside my head” (pp. 75–76) and emphasizes that this silence greatly impacted her self-awareness and sense of self (p. 67).

Various methods of measuring inner speech provide meaningful information about the experience of inner speech in both neurotypical controls and people with aphasia. These methods are likely capturing unique facets of inner speech. Because there is the possibility of relatively strong inner speech abilities in comparison to overt speech in individuals with aphasia (Geva, Bennett, et al., 2011; Stark et al., 2017; Fama et al., 2019; Fama & Turkeltaub, 2020), characterizing inner speech in aphasia has clinical implications. It is important, however, to consider the specific facets of inner speech that are relevant to the clinical situation. For example, if a clinician is interested in the underlying naming abilities of their client, they may be more likely to use in-the-moment methods of measuring inner speech to assess the accuracy of access to the word (Fama, Snider, et al., 2019). The clinician could then gain valuable information about where their client was having difficulty in the process of speech production (i.e., difficulty with access alone vs. difficulty with access and manipulation of language information). If a clinician wanted to know more about the frequency of inner speech in daily life, they may use subjective rating scales and adapt them to their desired questions. Finally, if the clinician were interested in the contents and functions of their client’s inner speech, they could opt for open-ended subjective questions or adapt pre-existing surveys (Racy et al., 2020) for individuals with aphasia (our lab is presently working on this). As always, clinicians and researchers must carefully select their methods of measuring a phenomenon based on the desired outcome.

Limitations

Not all aphasia severity levels were represented in the aphasia group, so relationships with severity may be stronger or appear differently if people with more severe aphasia as well as more mild aphasia (e.g., latent aphasia) participate in future studies. At a case study level, we explored the relationship between phonemic and semantic errors in overt naming and inner speech abilities, but this may benefit from a more detailed analysis with a larger set of naming data.

While we did match the stimuli in the objective tasks for frequency, familiarity, and imageability, we did not calculate Levenshtein's distance, which may have informed us further on task difficulty.

Some of the subjective questions about general inner speech may not have been answered in the same way by the two groups, as some members of the control group answered that they sometimes or often cannot say a word out loud but have successful inner speech (SIS), meaning that they had access to the word and its phonemes in their head. This seems unlikely for people who do not have a speech or language disorder, except possibly if someone were extremely exhausted or recently had oral surgery. Because the experience of anomia is likely uncommon for the control group, they may have assumed that the question was addressing a different experience, such as having partial phonetic access (tip-of-the-tongue phenomenon).

When using subjective reports of an internal phenomenon, there is always the possibility that the reports are inaccurate. We recognize that self-report may be harder to interpret in our population with aphasia, which may affect comprehension and self-monitoring/error detection, as discussed in Fama and Turkeltaub (2020). Readers can be encouraged that with similar methods, subjective reports of inner speech were meaningful in individuals with aphasia (Fama,

Snider, et al., 2019). We acknowledge that some of the subjective questions may also have had difficult wording, making them linguistically difficult for individuals with aphasia to interpret.

Future Directions

Future work should focus not only on inner speech in the moment for isolated tasks (e.g., rhyming, naming), but as a part of the daily experience of people with aphasia. The focus may also shift to include more clinical applications of relatively strong inner speech compared to overt speech, such as how successful inner speech may predict therapy outcomes. We are presently evaluating both of these directions in our lab as part of a National Institute on Disability, Independent Living and Rehabilitation Research (NIDILRR)-funded grant to Stark.

Data Availability Statement

The data that support the findings of this study are openly available in The Open Science Framework at <https://osf.io/8sjny/>, reference number DOI 10.17605/OSF.IO/8SJNY.

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Chapter 3: Inner Speech in the Daily Lives of People with Aphasia

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Abstract

Introduction: This exploratory, preliminary, feasibility study evaluated the extent to which adults with chronic aphasia (N = 23) report experiencing inner speech in their daily lives by leveraging experience sampling and survey methodology.

Methods: The presence of inner speech was assessed at 30 time-points and themes of inner speech at three time-points, over the course of three weeks. The relationship of inner speech to aphasia severity, demographic information (age, sex, years post-stroke), and insight into language impairment was evaluated.

Results: There was low attrition (<8%) and high compliance (>94%) for the study procedures, and inner speech was experienced in most sampled instances (>78%). The most common themes of inner speech experience across the weeks were ‘when remembering’, ‘to plan’, and ‘to motivate oneself’. There was no significant relationship identified between inner speech and aphasia severity, insight into language impairment, or demographic information. In conclusion, adults with aphasia tend to report experiencing inner speech often, with some shared themes (e.g., remembering, planning), and use inner speech to explore themes that are uncommon in young adults in other studies (e.g., to talk to themselves about health).

Discussion: High compliance and low attrition suggest design feasibility, and results emphasize the importance of collecting data in age-similar, non-brain-damaged peers as well as in adults with other neurogenic communication disorders to fully understand the experience and use of inner speech in daily life. Clinical implications and future directions are discussed.

The Inner Speech Phenomenon

Inner speech's definition varies by discipline, having been investigated under the contexts of psychology, neuroscience, philosophy, and cognitive science. The simplest definitions describe the experience of a "little voice in the head" (Geva, Bennett, et al., 2011) or "silent self-directed speech" (Vygotsky, 1962). The ConDialInt Model (Grandchamp et al., 2019) envisions inner speech as multidimensional, comprising dimensions of condensation, dialogality, and intentionality. The condensation dimension reflects inner speech's representation as a purely condensed form (without all acoustic, phonological, and syntactic information) to an expanded form (including articulatory and auditory properties). Indeed, "thinking in pure meaning" (Vygotsky, 1962) would be the extreme of the condensed form of inner speech. The dialogality dimension reflects inner speech's continuum from monologic (an inner soliloquy) to dialogic (reflecting a back/forth conversation). The intentionality dimension ranges from intentional, as when purposefully rehearsing information or manipulating it silently for later use, to unintentional, as in daydreaming or mind wandering. Recent research suggests that inner speech may be better conceptualized (and empirically studied) by re-envisioning the ConDialInt model (Pratts et al., 2023). Pratts et al. (2023) eliminate "condensation," as they argue that this dimension is not empirically manipulable or measurable, and replace "dialogality" with egocentricity, measuring the extent to which inner speech is a recreation of one's own voice (high egocentricity), or the voice of another individual (low egocentricity). They argue that this more clearly differentiates the distinct types of inner speech (own-voice vs. other-voice) to enable empirical investigation. Pratts et al. (2023) also replace "intentionality" with spontaneity, largely to improve the definition of this aspect of inner speech to represent the fact that spontaneous inner speech may or may not have intent. The spontaneity continuum ranges from highly deliberate inner speech, elicited by explicit task demands (e.g., rhyme judgements), to

highly spontaneous inner speech emerging in the absence of cues or clear task demands (e.g., mind wandering).

In philosophy literature, researchers disagree as to which components of inner speech are required, theorizing about matters such as how conscious people are of their inner speech, how much phonetic and articulatory information is involved, and how inner speech interacts with other cognitive processes (Langland-Hassan et al., 2018). To summarize, the definitions of inner speech, the rationale behind measuring inner speech, and the methods used to measure inner speech (discussed in more detail later in the Introduction) likely have some bearing on the types of data gathered and their interpretation. Despite the variability of studies, the accumulation of evidence on inner speech from a variety of disciplines, methodologies, and subject groups makes more complete and clear the role of inner speech in cognition and, more broadly, in daily life.

Frequency and Content of Inner Speech

Inner speech experiences happen variably in young adults (in 0-75% of cases, estimated using experience sampling methods; Heavey & Hurlburt, 2008). Studies using surveys rather than experience sampling, asking more generally about inner speech experience, have identified a majority of young adults as experiencing inner speech (e.g., 68% using Nevada Inner Experience Questionnaire and 67% using the Self-Talk Scale; Brinthaup et al., 2009; Heavey et al., 2019).

Through surveys such as the General Inner Speech Questionnaire (Racy et al., 2020), inner speech has been shown to vary by context (e.g., using inner speech when driving, alone, or working) and have many different uses (e.g., problem-solving, controlling emotions, or self-evaluating). Through various methodologies in neurotypical populations, inner speech has been closely linked with self-awareness and monitoring (Morin, 2011a, 2011b; Morin & Everett, 1990; Nooteboom, 2004; Perrone-Bertolotti et al., 2014), emotion regulation and assessment (Kross et

al., 2014; Morin & Racy, 2022; Racy et al., 2020), and cognitive processes such as problem-solving (Alderson-Day & Fernyhough, 2015; Wallace et al., 2017), planning (Morin & Racy, 2022; Racy et al., 2020), and memory (Racy et al., 2020). In fact, interrupting inner speech disrupted problem-solving abilities in young adults (Wallace et al., 2017), suggesting that inner speech may be critical for solving certain kinds of problems.

Inner Speech in Aphasia

Aphasia is a language disorder that affects two million adults in the USA and is most commonly caused by a stroke to the dominant (left) hemisphere (Simmons-Mackie & Cherney, 2018). Typically, aphasia persists into the chronic stage (usually defined as more than six months post-injury) in 20-40% of individuals with acute aphasia (Flowers et al., 2016). In adults with aphasia who have impairments of verbal language, inner speech can still be relatively intact (Alexander et al., 2023; Fama et al., 2017; Geva, Bennett, et al., 2011; Stark et al., 2017). This is supported by findings from cognitive neuroscience research, demonstrating that inner speech has distinct neural correlates from overt speech (Fama et al., 2017; Geva, Jones, et al., 2011; Pillay et al., 2014), and that subjective experiences of inner speech dissociate from experiences of ‘tip of the tongue’ phenomena and ‘having an idea without a word’ (Fama, Snider, et al., 2019). That is, inner speech is not simply ‘impoverished’ overt speech. To date, the majority of empirical studies evaluating inner speech in aphasia have leveraged experimental, non-spontaneous, and highly constrained tasks, such as silent rhyme judgment (Geva, Jones, et al., 2011; Geva & Warburton, 2018; Stark et al., 2017), as a proxy for inner speech presence. These investigations have highlighted inner speech’s relationship with language functions like phonological working memory and phonological retrieval (Fama, Henderson, et al., 2019; Stark et al., 2017).

Inner speech is interesting in aphasia not only because of its ability to help identify the source of anomia (word finding impairments, for an overview see Fama & Turkeltaub, 2020), but because many individuals with aphasia anecdotally speak about being ‘able to say a word in

their head' and not aloud, and who anecdotally speak about inner speech during daily life.

Stroke survivor Shai Anbar, in *Goddess Aphasia: A Stroke Survivor and his Dual Muse*, wrote that all forms of language are affected by aphasia, including inner speech (Anbar, 2022). Dr. Jill Bolte Taylor, a neuroanatomist, suffered a left-hemisphere stroke which caused a temporary but nearly complete loss of inner speech immediately post-stroke. She highlights the importance of inner speech in her book *My Stroke of Insight*, describing the lack of inner speech as “the dramatic silence that had taken residency inside my head” (Taylor, 2008, pp. 75–76). Dr. Bolte Taylor stated that the silence affected her self-awareness and sense of self (p. 67), but it is unknown whether this is common to most people living with aphasia. Both Mr. Anbar and Dr. Bolte Taylor describe the loss of inner speech acutely, with some indication that inner speech has the potential to return over the course of stroke recovery. Indeed, inner life and the experience of inner speech may be more rich post-stroke than it was pre-stroke, with recent evidence suggesting that individuals with moderate aphasia subjectively say that they experience inner speech more frequently after their stroke (Alexander et al., 2023). Because inner speech plays a variety of roles for adults without brain damage, the preservation and uses of inner speech by adults with aphasia may constitute an important factor in their recovery and in living well with aphasia.

Measuring Inner Speech in Daily Life

In neurotypical adult research, there has been progress in formalizing models and measurements of inner speech, with a great emphasis on being able to “objectively” measure this phenomenon with experimental control, such as silent reading, rhyming, and homophone judgments. However, this progress in formalizing models and measurement of inner speech (and inner speech judgments) may have come at a cost of ignoring broader elements of the psychology and philosophy of inner speech, such as uses of inner speech (e.g., to problem

solve), frequency of inner speech experiences in daily life, and the influence of context (e.g., location, mood) on these perceptions of inner speech.

The focus on daily life experiences of inner speech has been comparatively rarer yet has led to the creation of an assortment of tools that enable one to sensitively and, ideally, reliably measure the experiences of inner speech. Methods that describe inner speech in daily life are subjective/introspective. These methods, therefore, come with their own strengths and weaknesses. While experimental control can be ideal, it can also directly impact the nature and content of the inner speech (e.g., priming, requiring non-spontaneous uses). On the other hand, self-report methods are easy to use, yet are also more prone to inter-individual variability (for more detailed reviews of reliability and validity issues with self-talk measures, see Brinthaupt & Morin [2023] and Van Raalte et al. [2019]). To summarize: assessing inner speech *in situ* – as it happens – is inherently difficult yet comes with many benefits. Below are several commonly used methods that have evaluated inner speech in daily life.

Methods of sampling behavior in everyday life go by many names, such as experience sampling, ecological momentary assessment, and real-time data capture (Hektner et al., 2007). This type of assessment is defined by three critical features: people are assessed in their natural environments; researchers attempt to measure events or experiences in real-time; and researchers intensively assess people over time, such as several assessments per day across a week (Hektner et al., 2007). Experience sampling is appealing to measure inner speech because inner speech is known to vary by context and thought to not be present all of the time (Hurlburt et al., 2021). Therefore, using intensive methods in a real environment may most sensitively and accurately reflect experiences of inner speech. As with every other method that aims to measure inner speech, there are benefits and drawbacks to experience sampling. A major benefit is the large amount of data collected and the ability to evaluate within-subject and between-subject variability. For example, the collection of seven assessments per day for seven days in 20 participants elicits 980 datapoints and robust statistical power to detect potentially

small effects. However, a logical barrier with these designs is a lower rate of compliance and a higher rate of attrition than might be expected from a cross-sectional design, and/or a design confined to a laboratory setting (Silvia & Cotter, 2021). Another benefit is its ecological validity (its ability to reflect information from real life), but a drawback is generally considered to be its lower experimental control (and, therefore, increase in statistical noise). Note, though, that experience sampling can still have some experimental control (e.g., development of a valid assessment protocol at each sample collection), though it is different from the control within a lab environment.

There are no typical or best practice standards about number of times, or duration of sampling, for experience sampling. Results of a recent meta-analysis across research fields demonstrated that, on average, experience sampling studies in neurotypical populations scheduled six assessments per day, lasted for seven days, and obtained a compliance of 79% (Wrzus & Neubauer, 2023). Note that most individuals being sampled were young adults with no concomitant clinical conditions. The authors found that studies with more assessments per day scheduled fewer assessment days but did not find that the number of assessments predicted compliance or dropout rates (Wrzus & Neubauer, 2023). A review evaluating mobile experience sampling in clinical populations found that the median duration of studies was 12 days and the median prompt frequency was five per day across clinical and non-clinical samples (Williams et al., 2021). There was no significant relationship between compliance and prompts per day or items per prompt in clinical samples (Williams et al., 2021).

Experience sampling has been largely used to evaluate the experience of inner speech in young adults, typically undergraduate students. In one study, inner speech was measured six times per day at pseudo-random intervals, for three separate days (Heavey & Hurlburt, 2008). Participants received a notification from a beeper and were asked to describe their inner experience in that moment. Within 24 hours, they completed an unstructured interview with the experimenters, detailing their inner experience during the six sampled times (Heavey & Hurlburt,

2008). As previously mentioned, inner speech was found to be a common form of inner experience, with an overall frequency of 26% of sampled instances (with a wide range: 0-75%). In another study by the same group, also with young adults as participants, Hurlburt et al. (2016) combined experience sampling with neuroimaging, with participants reporting if they were experiencing inner speech just before a beep while in an MRI scanner. This study too found an overall frequency of inner speech of 29% across all participants. Authors found that elicited inner speech (rhyme judgments) and spontaneous inner speech (experience sampling) showed differences in activation of Heschl's gyrus (includes primary auditory cortex) and the left inferior frontal gyrus (activated in language production). Elicited, non-spontaneous inner speech was associated with a significant decrease in activity in Heschl's gyrus and a significant increase in left inferior frontal gyrus whereas spontaneously occurring inner speech was associated with a significant increase in activity in Heschl's gyrus but no significant change in left inferior frontal gyrus compared to a baseline. These areas are often part of the injuries which cause aphasia, emphasizing the interest in inner speech in aphasia and the differences in neural patterns of inner speech according to elicitation method.

Questionnaires and surveys are another means of assessing inner speech, and complement other methods by highlighting potentially shared uses of inner speech across a population (Morin & Racy, 2022). Through data obtained with a thought-listing procedure, in which participants wrote about their inner speech over the past six months, Morin and colleagues developed the General Inner Speech Questionnaire (GISQ) (Racy et al., 2020). The General Inner Speech Questionnaire focused on specific themes of when participants talk to themselves (during what activities), what they talk to themselves about (the contents), and why they talk to themselves (the functions) (Racy et al., 2020). In five studies conducted by Morin and colleagues from 2011 to 2019 using thought-listing and the General Inner Speech Questionnaire, they found that the most frequently reported contents and functions of inner speech in neurotypical young adults included problem solving, negative emotions, emotional

control, planning, and self-motivation (Morin & Racy, 2022). There are other questionnaires with slightly different aims; for example, the Varieties of Inner Speech Questionnaire-Revised (Alderson-Day et al., 2018) assesses the phenomenology of a person's inner speech (e.g., whether it is a dialogue or monologue and if it contains all phonetic information) and the functions of inner speech (e.g., self-evaluative, self-regulatory) (Alderson-Day et al., 2018). The Self-Talk Scale (Brinthaup et al., 2009) focuses on aspects of self-regulation, such as self-criticism, -reinforcement, and -management.

Advantages to questionnaires and surveys are that they can be tailored to specific research questions with theoretical bases, and they provide quantitative results about those specific questions, with some reporting preliminary construct validity and psychometric qualities (Brinthaup et al., 2009; Racy et al., 2020). A limitation to questionnaires is that they are retrospective and require long-term memory, and as participants recall what happened in the past, their memories may be inaccurate (inflating or underestimating certain experiences of inner speech). Questionnaires may also be less sensitive to subtle changes in context, and they may be affected by the researcher's preconceptions about the phenomenon of inner speech.

Aims, Hypotheses and Rationale for the Current Study

Most of the research on inner speech in aphasia thus far has been in the context of experimentally controlled, non-spontaneous tasks, such as confrontation naming, word retrieval, and silent rhyme/homophone judgments (for an overview, see Fama & Turkeltaub, 2020). To our knowledge, a single study has used experience sampling methodology to evaluate a cognitive component in adults with acquired, post-stroke aphasia: Sather and colleagues (2017) used experience sampling to investigate "flow state" (intense concentration often resulting in distorted time passage and loss of self-consciousness) in people with aphasia. No study to date has used experience sampling to investigate inner speech in aphasia. The feasibility of this methodology

to sample inner speech in aphasia, and the frequency and themes of inner speech experienced during daily life by adults with aphasia, remain unclear.

The following exploratory, preliminary feasibility study was approved by Indiana University (IRB #10549) and was pre-registered on Open Science Framework (<https://osf.io/qf45v>). There were hypotheses detailed in the pre-registration, described in more detail, below. Registered exploratory analyses also included the exploration of the association of inner speech with language impairment awareness (or an impairment in this, “anosognosia”) and with demographic variables, such as age, years post-stroke, and sex.

Exploratory Research Question 1: Explore the extent to which individuals with aphasia can reliably engage with descriptive experience sampling and multiple surveys.

A compliance of about 80% was expected, given the evidence from prior research (in aphasia: Sather et al., 2017; in adults without brain damage: Wrzus & Neubauer, 2023).

Exploratory Research Question 2: Investigate the frequency of inner speech in individuals with aphasia.

Adults with aphasia have been shown to have varying levels and perceptions of inner speech, measured in a variety of ways though always at the single word level (e.g., rhyme judgments, picture naming) and non-spontaneously. Therefore, it remains unclear how inner speech presents itself spontaneously in daily life in this population. Inner speech in daily life has been shown to be highly heterogeneous in young adults (from 0-75% across different studies) (Brinthaup et al., 2009; Hurlburt et al., 2021). Therefore, it was difficult to motivate a directional hypothesis, and it is therefore posed as an exploratory question: to what extent do people with chronic aphasia have inner speech in their daily lives?

Exploratory Research Question 3: Case analysis.

Because of the novel procedures and topic (inner speech), a detailed case-by-case presentation (see Tables 1, 2, 4 and 5) is also provided to explore potential patterns in the data. Exploratory analyses include an investigation of the relationship between inner speech and selected demographics (age, sex, and years post-stroke), and with a neuropsychological measure, the severity of language awareness impairment (i.e., anosognosia).

Hypothesis 1: Content, functions and activities of inner speech.

We hypothesized that people with aphasia likely have some shared contents of inner speech with each other and with young adults, but they may have unique themes or have certain themes that are more prominent because of their age and/or brain injury. As an exploratory part of this hypothesis, we expected that inner speech would not be consistent over three weeks, as it is a heterogeneous experience in young neurotypical adults, known to be impacted by context (Alderson-Day & Fernyhough, 2015). To our knowledge, no study (in a clinical or non-clinical sample) has systematically evaluated the consistency of inner speech experience in this way, making this a highly exploratory question.

Hypothesis 2: Relationship of inner speech with aphasia severity.

Prior research is divided on the relationship between aphasia severity and inner speech, and the method of measuring inner speech may affect whether it relates to aphasia severity (Alexander et al., 2023). In Alexander et al. (2023), self-reported inner speech (the subjective ratings of inner speech in general and successful inner speech during naming) did not relate to aphasia severity. Experience sampling has not been used to study inner speech in aphasia before, but it is similar in that it requires self-report, so no relationship between aphasia severity and inner speech was expected.

Method

Participants

Participants were recruited over a period of one year. They were proficient English speakers (defined by self-report; speaking another language did not exclude), aged 18-80 years old, and had aphasia due to a stroke occurring at least six months prior. To ensure that auditory comprehension abilities were strong enough to participate in the study, participants were required to score 30 or greater (out of 48) on the auditory sentence comprehension subtest of a short, standardized aphasia battery (Quick Aphasia Battery [QAB]; Wilson et al., 2018). If a participant scored lower than that, study personnel examined their comprehension of verbal instructions, written communication, and practice with procedures to ensure adequate comprehension to include them (this occurred for two participants, who were ultimately included). One exclusion parameter involved the presence of disorders that may make their speech unintelligible, such as severe apraxia of speech or dysarthria. Speech motor impairments were rated using the speech motor programming section of the Quick Aphasia Battery (see Table 2 for Speech Motor Programming scores). No participants were excluded due to a severe motor speech impairment. The final exclusion parameter involved the self-report of any other neurological disorder (e.g., Alzheimer's disease); no exclusions were made based on this.

Verbal consent was acquired for 26 adults with aphasia. Two adults with aphasia dropped out after the introductory session/a few experience sampling responses due to busyness (did not have time to complete all study components and proactively dropped out because of this). It was realized at the end of the study that one adult with aphasia had a concomitant stroke and traumatic brain injury, and their data are excluded from these results. Detailed demographic and neuropsychological data for the final sample of 23 is shown in Tables 1 and 2.

Table 1. Individual participants with demographic information.

ID	Years post-onset aphasia	Age at testing	Sex	Education (years)	Race	Is Hispanic or Latino?	Bilingual?
2	21.50	50	Female	18	White	No	No
3	0.81	75	Male	22	White	No	No
5	3.98	68	Male	14	White	No	No
6	25.00	45	Female	16	Asian	No	Yes (Hindi, L1; English, aged 5)
7	4.00	52	Male	16	White	No	No
8	10.29	62	Male	18	White	No	Yes (Hebrew L1; English, most of life)
10	2.64	45	Female	18	White	No	No
11	7.38	47	Female	16	White	No	No
14	2.87	62	Female	20	Unknown / Not Reported	No	No
16	4.43	62	Male	16	White	No	No
19	5.85	63	Female	14	White	No	No
20	1.88	71	Male	14	White	No	No
22	2.58	52	Female	12	Black or African American	No	No
24	6.79	59	Female	18	White	No	No
25	1.67	56	Male	16	White	No	No
29	7.00	67	Female	16	White	No	No
30	10.00	50	Male	20	Asian	No	Yes (English L1; Cantonese for 30+ years)
35	3.00	63	Male	16	More Than One Race	No	No
46	3.97	38	Female	20	White	No	No

47	16.34	46	Female	17	American Indian/Alaska Native	Yes	Yes (English and Spanish simultaneous)
48	11.00	54	Female	18	White	No	No
63	5.71	58	Female	14	White	No	No
78	7.00	70	Female	18	White	No	Yes (Russian, Czech, Hungarian L1s/simultaneous; English from age 8)
Mean (SD) or percentage	7.20 years (6.24)	57.2 years (9.78)	60.87% Female	16.8 years (2.39)	73.91% White 8.70% Asian 4.35% Black or African American	4.35% Hispanic or Latino	21.74% bi- or multilingual
Min, max	0.81, 25.00	37.8, 75.1		12, 22	4.35% More Than One Race 4.35% American Indian/Alaska Native		
					4.35% not reported		

Table 2. Individual participants with neuropsychological data.

ID	Overall	Severity	Quick Aphasia Battery						VATA-L			
			Word Comp.	Sent. Comp.	Word Finding	Gram. Construct.	Speech Motor Program.	Repetition	Reading	Discrepancy*	Participant	Other
2	8.35	mild	10	3.33	9	9.63	10	8.75	10	-10	15	5
3	8.25	mild	10	4.58	8.25	9	10	8.75	9.58	1	15	16
5	9.01	latent / no aphasia	10	6.67	9	9.5	10	8.33	10	8	16	24
6	7.19	moderate	10	7.08	6.25	3.75	10	6.67	8.75	1.5	17	18.5
7	7.39	moderate	10	3.33	6.75	8.88	10	6.67	8.33	0	19	19
8	8.7	mild	10	5.83	7.75	9.5	10	7.92	10	1	8	9
10	9.6	latent / no aphasia	10	10	9	10	10	10	10	-4	10	6
11	8.85	mild	10	8.33	8	8.38	10	10	9.58	8	5	13
14	8.08	mild	7.92	7.92	8.5	8	7.5	8.33	9.58	0	16	16
16	8.8	mild	8.75	6.67	9	9.38	10	9.58	9.58	-4.5	15.5	11
19	7.9	mild	10	4.58	9.75	9.75		8.33	9.58	-22	37	15
20	7.88	mild	10	10	8.5	5.5	5	6.67	7.5	-7	22	15
22	7.47	moderate	9.58	7.08	5	8.75	7.5	6.67	5.83	-4	24	20
24	8.77	mild	10	9.58	8	7.5	10	8.33	8.33	0	18	18
25	9.36	latent / no aphasia	10	8.75	9	9.5	10	10	10	3	1	4
29	9.41	latent / no aphasia	10	10	10	8	10	10	10	1	4	5

30	9.55	latent / no aphasia	10	9.17	10	9.88	7.5	9.58	10	-16.5	24.5	8
35	7.66	mild	10	8.75	7.25	7	10	7.5	0 [^]	-7.5	30.5	23
46	9.76	latent / no aphasia	10	10	9	10	10	10	10	-4	9	5
47	9.65	latent / no aphasia	10	10	10	9.88	7.5	9.58	10	7	6	13
48	8.49	mild	8.33	8.33	4	10	10	9.17	10	5	8	13
63	8.88	mild	10	8.33	7.5	8.38	10	9.58	10	-1	9	8
78	7.38	moderate	10	9.58	5.5	6	5	6.25	8.33	-8	21	13
Mean (SD) or %	8.54 (0.81)	30.4% latent 52.2%	9.76 (0.59)	7.73 (2.17)	8.04 (1.63)	8.53 (1.64)	9.09 (1.64)	8.55 (1.29)	8.91 (2.21)	-2.30 (7.35)	15.2 (8.80)	12.9 (5.95)
Min, max	7.19, 9.76	mild 17.4% moderate	7.92, 10.0	3.33, 10.0	4.00, 10.00	3.75, 10.0	5.0, 10.0	6.25, 10.0	0, 10.0	-22.0, 8.0	1.0, 37.0	4.0, 24.0

Comp. = Comprehension; Sent. = Sentence; Gram. Construct = Grammatical Construction; Program. = Programming;
VATA-L = Visual-Analogue Test assessing Anosognosia for Language Impairment.

Severity: latent/no aphasia (8.90-10), mild (7.50-8.89), moderate (5-7.49)

* = Discrepancy score of a more negative value indicates a higher probability of anosognosia related to language impairment.

^ = Result due to self-disclosed visual perception issues; all text made larger and read aloud to participant (via software on their own phone and also during experiment)

Procedure and Materials

Participants took part in a three week long mixed methods study, containing four primary components: (1) Introduction to inner speech and the study procedures, (2) Experience sampling, (3) General Inner Speech Questionnaire (Racy et al., 2020), and (4) Interviews (Figure 1). All procedures were conducted remotely and virtually using secure Microsoft Teams platform (Introduction; Interviews), Qualtrics (General Inner Speech Questionnaire, Descriptive Experience Sampling), and Apple iPhone Automations (SMS notification to engage with Qualtrics).

The mixed methods design was an embedded one, with a primary emphasis on quantitative data, which are reported here (Creswell, 2007). The qualitative data (interview and thought listing) was meant to enhance and elaborate on the quantitative at a later date. The qualitative results are not reported in this paper and are currently being analyzed.

Data collection

Introductory session.

There were several purposes for the Introductory session: (1) screening for inclusion and collecting more information about aphasia characteristics, (2) characterizing insight into language impairment, (3) introducing the participant to 'inner speech' and its definitions, and (4) practicing the technology / methods used in the study. The motivation for the multiple purposes of the Introductory session was due to both Morin and Hurlburt's inner speech research emphasizing the need to create a shared understanding of what inner speech is and is not (Brinthaup & Morin, 2023; Heavey et al., 2019; Hurlburt et al., 2021; Hurlburt & Akhter, 2006; Racy et al., 2020); for the purposes of more comprehensively documenting each person's demographic and neuropsychological characteristics; and because of our prior research using virtual testing in aphasia emphasized that troubleshooting technology and methodology together leads to better outcomes and low attrition (Doub et al., 2020; Stark et al., 2023). Note that

Hurlburt's *descriptive* experience sampling methodology was not employed in its entirety in this study, as that methodology is highly iterative and involves interviews after each day, with many instances of discussing what inner speech is and is not after each sampling period.

The typical Introductory session lasted between 90 – 120 minutes and occurred entirely on Microsoft Teams. Several breaks were given to mitigate fatigue.

Study Workflow

1. Introductory Session (90-120 mins)

1A. Screening

Inclusion parameters
Quick Aphasia Battery

1B. Anosognosia Testing

VATA-L: Visual Assessment of Anosognosia
of Language Impairment

1C. Inner Speech Education

Discussion of what inner speech is and is not
Generating ideas of inner speech (e.g., silent reading)
Generating ideas of non-inner speech (e.g., feelings)

1D. Methodology / Tool Practice

Introduction to Descriptive Experience Sampling
Practice receiving SMS and clicking link to Qualtrics
Practice using Qualtrics survey on smartphone/tablet

Introduction to adapted General Inner Speech Questionnaire

Review protocol (identify inner experiences across week)
Review each question and ask comprehension questions
Practice receiving SMS and clicking link to Qualtrics
Practice using Qualtrics survey on smartphone/tablet

*** in four cases, individuals chose to receive paper
copies of General Inner Speech Questionnaire in lieu of
filling in Qualtrics online survey

2. Descriptive Experience Sampling and General Inner Speech Questionnaire

Week 1 (5 days)

2A. Descriptive experience sampling: twice per day

Are you having inner speech right now?
YES or NO or NOT SURE
Thought listing (descriptive) portion
not reported in current study.

2B. General Inner Speech Questionnaire: once

Participants 'check off' experiences, choosing from
32 different experiences that may have
occurred throughout the week

2C. Interview: once

Follow-up interview about week's experiences;
not reported in current study.

Weeks 2 & 3 Repeat Week 1's procedures

Examples of procedures on smartphone

12:29

Are you having inner experiences?

Yes

No

Not sure

Descriptive experience sampling to
measure "in the moment" inner speech:
yes/no/not sure

12:29

Please select the response(s) that you agree with. I talk to myself about:

Positive emotions

Negative emotions

What others think about me

Example from General Inner
Speech Questionnaire to measure inner speech themes

Figure 3: Workflow of study design.

Screening for Inclusion and Characterization of Aphasia

Prior to enrolling in the study, participants were emailed a link to a REDCap survey, where they self-reported demographic information about inclusion and exclusion parameters (Harris et al., 2009, 2019). If found to be potentially eligible, they were invited to the virtual Introductory session. During the Introductory session, the screening for inclusion involved verifying that auditory comprehension was adequately preserved for the purposes of participation, as described under Participants. The Quick Aphasia Battery was also used to further characterize each participant's aphasia (Table 2).

Characterizing Insight into Language Impairment

If eligible to participate in the study, participants were administered the VATA-L: Visual-Analogue Test assessing Anosognosia for Language Impairment (Cocchini et al., 2010) to assess insight into language deficits. Assessing anosognosia (lack of insight into a particular disorder or disease) in aphasia has traditionally been very difficult, with few validated assessments available. The VATA-L is one such way of evaluating anosognosia specific to language impairment, and has the participant, as well as a familiar peer (e.g., significant other, speech therapist), use a visual analogue scale to identify the severity (0 = no difficulty; 1 = some difficulty; 2 = difficult; 3 = very difficult) of communication difficulty during common language production and comprehension situations (e.g., writing sentences, getting the sounds of a word, picking up mistakes in your speech, reading notices, understanding people). The VATA-L also contains several “check” items to ensure that the anosognosia does not extend beyond language (e.g., VATA-L asks about awareness of difficulty waving, hearing a fire engine with sirens on, and jumping over a truck/lorry).

Introduction to Inner Speech

It was very important for the fidelity of the study to ensure that all participants understood what “inner speech experience” was and was not, a clarification that has been made clear by other

researchers using similar methods to study inner speech (Hurlburt et al., 2021; Racy et al., 2020). Therefore, following screening and further characterization of aphasia and insight into language impairment, the Introductory session involved a detailed discussion (also using visual aids) between the participant and author JA about what inner speech is and is not (Table 3). Table 3's exemplar inner speech examples were generated between authors JA, BCS, and inner speech expert, Alain Morin, prior to the study's initiation. After discussing the exemplars in the table, JA had participants generate other examples of what inner speech is and is not to ensure their understanding. Some representative examples from participants that were considered inner speech included: reading to oneself, praying, planning out a conversation and hearing those words, and singing to oneself. Some representative examples from participants that were correctly not considered inner speech included: feeling sad, and "seeing" an item in their head but not having the word for it (like an "Idea Without Word" concept, as described in Fama, Snider, et al., 2019; Fama & Turkeltaub, 2020).

Table 3: Inner speech definitions. This table was generated as a result of discussions with Alain Morin, PhD, and was used in the Introductory session of the study.

Inner speech is:	Inner speech is NOT:
Verbal (thinking)	Non-verbal thinking
A commentary	Silent thinking
In words	Sensations
Self-talk	Images
Inner speaking	Pure emotions
Talking to yourself	Wordless thinking
In your own voice	
Words or sentences	
Mostly silent	
Monologue or dialogue	
Meaningful or nonsensical	
More or less frequent	
Spontaneous or voluntary	

Practicing Methodology for Study

A core part of the study was adherence to procedures and the use of (potentially) new technology. Therefore, the end of each Introductory session focused on practicing the methodology with author JA.

Participants practiced receiving the SMS notification to start the experience sampling procedure (this procedure is described in more detail, below). Each participant was sent an SMS by author JA, and they were tasked with opening the link in the text message and going through the protocol on Qualtrics (see Figure 1 workflow). Author JA verified that the Qualtrics response had been received and probed about the participant's experience, ensuring they understood how to use the technology.

Finally, JA and the participant practiced answering the General Inner Speech Questionnaire, reinforcing the need to accurately represent the participant's experience of inner speech, including that participants should leave items blank if they did not experience them. If participants indicated that they preferred a paper copy rather than doing either the experience sampling or the General Inner Speech Questionnaire survey via Qualtrics, a paper packet was shipped to their home with return postage included. Four participants opted for the paper packets. The paper packets are located on the OSF repository for this study (<https://osf.io/b6vxx/>).

Experience sampling methodology.

The goal of the current study was to acquire 30 experiences per participant, in line with the best practices suggested by Hurlburt & Akhter (2006). Experience sampling methodology, broadly, aims to understand behavior as it occurs in daily life. Each experience sampling occurrence prompted each participant first about whether or not they were experiencing inner speech, and then asked to elaborate on it via a thought listing methodology (discussed below). For the

purposes of this quantitative study, only the experience sampling component (whether or not they were experiencing inner speech) is reported.

In the current study, the experience sampling procedure took place over three weeks, five days per week, twice per day. The time of the sampling was pseudo-random per day (Hektner et al., 2007), where participants received probes between 9:00-10:00 AM and 4:00-5:00 PM. The times differed only if a participant indicated that they were not usually awake at those times (i.e., one participant requested a later hour for the first prompt because he is a late riser). This pseudo-random design was chosen to limit poorly timed disruptions to participants' daily activities and increase their ability to respond to prompts immediately, thereby maximizing construct validity. The choice of using two probes per day was made as a means of compromising on feasibility and compliance alongside collecting enough total data-points per participant.

Participants were sent a prompt via SMS (text message), which included a link to the short survey in Qualtrics. The Qualtrics link began with having participants answer a simple question: were you having inner speech when the alarm/beep was received? The response options were 'yes,' 'no,' and 'not sure' (see Figure 1). This data is referred to as "in the moment" inner speech. Or, if the participant had opted for the paper version, the SMS message indicated that they should fill out their paper copy.

There was also a section of the Qualtrics survey where participants were prompted to make notes about their experience, if they answered 'yes' to the "are you experiencing inner speech" prompt. This constituted the descriptive portion of the study. This 'thought listing' could be done via entering text from the keyboard, recording their own voice (via the Phonic component of Qualtrics), or by uploading a picture of something they had written or drawn. An additional component (an interview) was conducted three times throughout the three-week design, in which inner speech experiences were further probed. During these interviews, author JA also resolved any questions or issues that arose during the week related to the study. The

thought listing and interviews constituted the ‘explore’ qualitative component of this mixed method project, and per the recommendation of Creswell (2007), these qualitative data will be presented in more depth in a following publication. They are presently being analyzed and are not included in this quantitative study.

General Inner Speech Questionnaire (GISQ)

At the end of each of the three weeks, an additional SMS message was sent to individuals which contained the link to the adapted General Inner Speech Questionnaire. Or, if the participant had opted for the paper version, the SMS message indicated that they should fill out their paper copy. Following an “in the moment” question (are you experiencing inner speech right now?), the participants were then asked to think about their inner speech throughout the week. The requirement of reflection over the course of the week was stated in the questionnaire itself and also discussed whilst practicing the General Inner Speech Questionnaire during the Introductory session. The General Inner Speech Questionnaire asks about the presence or absence of inner speech in three sections: “I talk to myself about” (contents), “I talk to myself in order to” (function), and “I talk to myself when” (activities). The original General Inner Speech Questionnaire contains 57 total items (Racy et al., 2020).

For the purposes of this study, an adapted, shortened version of the General Inner Speech Questionnaire was developed between the author team and Alain Morin, PhD, senior developer of the General Inner Speech Questionnaire. First, items that were infrequently chosen by young adults in past administrations of the General Inner Speech Questionnaire were removed, such as thinking about education or studying. Then, items that may be inapplicable to, or insensitive to include, were pre-emptively removed, such as ‘when working’ (in case of forced retirement). For all removals, see Supplementary Table 1. The wording of three items from the original survey was modified: “about how I am perceived by others” was reworded more simply to “what others think about me,” “using language” was clarified to be “when reading,” and

“leisure” was broadened to “what I want to do.” This resulted in 32 items on the adapted version. Finally, black and white drawings (free clipart) accompanied each item in the adapted version to clarify their meaning. There were between five and six drawings/choices on each page of the paper packet.

After each of the three sections—“I talk to myself about” (contents), “I talk to myself in order to” (function), and “I talk to myself when” (activities)—there was a 1-5 Likert-scale that probed the participant’s confidence about how accurately they recalled their inner speech for that section. Above the 1 marker were the words “not confident” and a black/white clipart of a person with thumb pointing down, and above the 5 marker was the word “very” and a black/white clipart of a person with thumb pointing up. In prior work in neurotypical adults, most tended to be fairly confident to very confident (scoring between 3 and 5) (Racy et al., 2020), and thus this was expected to be reflected in the current exploratory study using the adapted General Inner Speech Questionnaire in aphasia.

For an example of one page of the adapted General Inner Speech Questionnaire filled out on the paper form, see Figure 2. For an example of how the General Inner Speech Questionnaire looked on a web browser, tablet and mobile phone as displayed through Qualtrics, see Figure 1. The adapted General Inner Speech Questionnaire used in this study (including pictures) is located in Supplementary File 1. For ease of access, all the materials used in the current study are available on the Open Science Framework (<https://osf.io/b6vxk/>).

Please circle or cross out the response(s) that you agree with.

I talk to myself about:

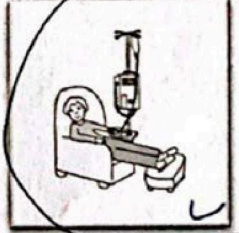
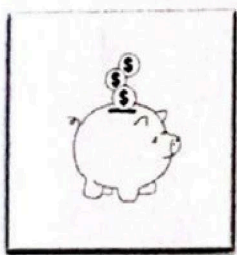
 Food/what to eat	 Current events	 Health
 Financial situation	 Future	 Relationships

Figure 4: Real example of response using paper packet adapted General Inner Speech Questionnaire.

Data Analysis

All analyses were computed using R 4.3.2 and RStudio 2023.12.0 Build 369. Outliers are identified in each section (using boxplots generated in R) but were kept in analyses because of the preliminary nature of the data and overall desire to characterize the whole sample.

Exploratory Question 1: Procedure Compliance

To evaluate the extent to which participants could complete the rigorous procedures, the number of responses received from participants was divided by the total number of notifications sent to each participant to complete the experience sampling (30 times) and the number of General Inner Speech Questionnaire surveys completed (out of three chances). This provided a compliance percentage for each portion of the study.

Exploratory Question 2: Frequency of inner speech

The experience sampling “are you having inner speech right now” data (including the three times this question was also asked on the General Inner Speech Questionnaire) were summed. For analyses, the proportion of “yes” responses for the “in the moment” data were used. There were very few “not sure” answers throughout (~3%). Descriptive analyses were used to quantify the frequency of response types.

Exploratory Question 3: Case Analysis

Exploratory analyses include an investigation of the relationship between inner speech and selected demographics (age, sex, and years post-stroke), and with a neuropsychological measure, the severity of language awareness impairment (i.e., anosognosia). Appropriate correlation tests (Spearman’s or point-biserial) were conducted to evaluate these relationships, and power (β) is also included to aid interpretation because of the known small sample size. Power was calculated using the *pwr* package in R 4.3.2. Boxplots of the “in the moment” data were used to reveal outliers, leading to in-depth analysis of their clinical presentation based on subtests of the QAB and the discrepancy score on the VATA-L.

Hypothesis 1: Content, Functions, and Activities of Inner Speech Use

A pre-registered hypothesis revolved around identifying patterns in the General Inner Speech Questionnaire themes across the participants. Descriptive statistics were used to quantify the number of themes within the three categories (content, function, and activities) summed across the three weeks to identify patterns of inner speech experience. This was done for the entire sample (although two individuals did not complete all surveys, see Compliance in Results). To ascertain what percentage of the sample selected a theme, the total number of individuals selecting the theme was summed and then this was divided by the total number of participants. The average confidence ratings for each section of the General Inner Speech Questionnaire overall are also reported. This hypothesis was a null one: that there would be no difference in

number of inner speech themes/items between sample with aphasia and prior data collected from young adults. As such, Fisher's null hypothesis testing protocol was followed.

Despite the research team's best efforts, it was realized that a question on the General Inner Speech Questionnaire which was on the Qualtrics version was not on the packet version ("Listen to my own voice," from the functions section). Therefore, the four participants who used the packet do not have data for that question, and results should be interpreted as such. In the written packet version, one item was included twice ("Replay past conversations"). If they selected one or both instances, this was counted as being experienced.

The consistency of inner speech themes across the weeks was evaluated using Friedman tests for total themes of inner speech experienced, and then the number of themes selected within each inner speech category: content ("I talk to myself about"), function ("I talk to myself in order to") and abilities ("I talk to myself when").

Hypothesis 2: Relationship of Inner Speech to Aphasia Severity

A pre-registered hypothesis evaluated the relationship of inner speech with aphasia severity.

The "in the moment" inner speech data (proportion of 'yes' responses) was correlated with aphasia severity (total Quick Aphasia Battery score). The total number of inner speech themes derived from the General Inner Speech Questionnaire (summed across three administrations) was correlated to aphasia severity for the $n=21$ with all three administrations of the survey.

Given this is a null hypothesis, the Fisher null hypothesis testing paradigm was employed, following the classic procedure of stating significance level ($\alpha = 0.05$), calculating test statistic and p-value, and then deciding to reject or fail to reject the hypothesis. Correlations were the main statistical test, and power (β) is also included to aid interpretation because of the known small sample size.

Given that prior research has found relationships with overt speech (more inner speech tends to associate with better overt speech), and with phonological processing, exploratory

correlations were run between the inner speech variables (proportion of 'yes' in the moment responses; total General Inner Speech Questionnaire themes) and the QAB Word Finding and QAB Repetition subtests. Correlations were the main statistical test, and power (β) is also included to aid interpretation because of the known small sample size.

Results

Exploratory Question 1: Procedure Compliance

For the 23 individuals included in the study, there was 94.64% (SD=10.33) compliance for the "in the moment" inner speech assessments and 95.65% compliance on the General Inner Speech Questionnaire. There were three outliers identified for the experience sampling portion: one with a compliance of 87%, one with 83%, and one with 57%. There were two outliers identified for the General Inner Speech Questionnaire administrations: one with 67% (completed two of three) and one with 33% (completed one of three). The two dropouts noted in the Methods are considered attrition, and therefore, there was a total participant attrition rate of 7.69% (26 enrolled, two dropping out prior to study completion). The excluded individual (due to late discovery of other non-eligible brain injury) completed all study procedures but is not further reported on and not considered part of attrition or compliance.

There were three participants who answered the General Inner Speech Questionnaire more than three times (although sometimes incomplete), and only the three times that corresponded with when they were sent the notification (weekly on day 5 in the evening) were counted.

As this was an exploratory study for the procedures being implemented in this specific population, the research team wanted to evaluate the type of technology that was used by participants to complete the procedures. No rules were implemented on what kind of technology could be used. In the study, 13.04% used a tablet, 78.26% used a mobile phone, and 4.35%

used a computer. 17.39% opted to use the paper packet at least once over the course of the procedure.

Exploratory Question 2: Frequency of inner speech

Participants responded to the question “are you having inner speech?” on 30 different occasions. Most of the time (>78%), they reported “yes” to having inner speech in that moment (Table 4). As suspected, there was wide variability in the experiences of inner speech across the study (i.e., 23.23% ‘yes’ answers across study for one participant vs. 100% ‘yes’ answers across study for several participants). There were two outliers identified for the ‘yes’ responses, with 30% and 23%. There was one outlier identified for the ‘no’ responses, with 63% (this participant was also the participant with 23% yes responses). There were three outliers identified for the ‘not sure’ responses, with 13%, 13%, and 17%. The subject with 17% not sure was also an outlier for the ‘yes’ responses (with 53% yes).

For the responses recorded on the Qualtrics survey (but not for the paper packets), exact timestamps of responses were extracted. Based on anecdotal reports of people with aphasia experiencing exhaustion and corresponding language difficulties later in the day, the proportion of “yes” responses was compared between the morning and late afternoon/evening sampling times. All responses recorded before noon were considered morning, while responses recorded after noon were considered evening, regardless of when the participants were scheduled to receive a prompt. For this subgroup who answered via the Qualtrics survey, the proportion of “yes” responses in the morning was nearly equivalent to the evening (83.7% vs. 82.0%, respectively).

Table 4. Individual participants with inner speech data: in the moment inner speech.

ID	Yes (%)	No (%)	Not Sure (%)	Compliance (% out of possible 30)
2	84.85	15.15	10	100
3	100	0	0	100
5	78.79	6.06	12.12	96.97
6	72.73	9.09	15.15	96.97
7	100	0	0	100
8	78.79	9.09	9.09	96.97
10	51.51	42.42	6.06	100
11	90.91	9.09	0	100
14	81.82	3.03	0	84.85
16	100	0	0	100
19	57.58	36.36	6.06	100
20	21.21	63.64	0	84.85
22	90.91	3.03	0	93.94
24	69.70	12.12	3.03	84.85
25	93.94	3.03	0	96.97
29	90.91	9.09	0	100
30	36.36	21.21	0	57.58
35	93.94	6.06	0	100
46	96.94	0	0	93.94
47	100	0	0	100
48	81.82	9.09	0	90.91
63	78.79	21.21	0	96.97
78	54.55	30.30	15.15	100
Mean (SD)	78.26 (21.74)	13.33 (15.83)	3.04 (5.31)	94.64 (10.33)
Range (min, max)	23.33, 100	0, 63.33	0, 16.67	57.58, 100

“In the moment” instances constituted 30 possible prompts per participant, resulting in a total of 690 possible responses across the study (30 prompts x 23 participants).

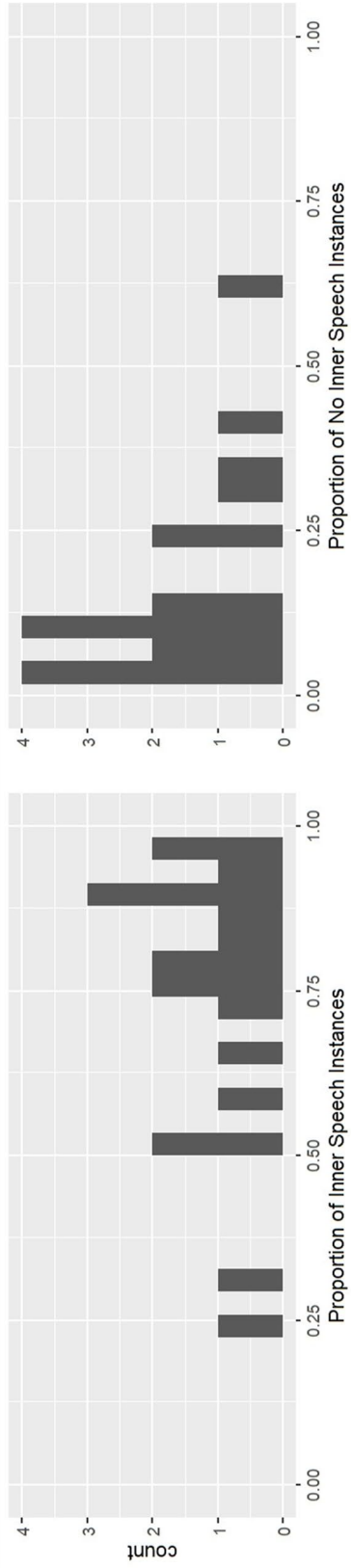


Figure 3: Frequency of “in the moment” inner speech occurrences across the experience sampling paradigm. The y-axis represents a count of participants experiencing inner speech (x-axis computed as a proportion of responses out of 30), for the presence of inner speech (left) and the non-presence of inner speech (right).

Exploratory Question 3: Case Analysis

Given the novelty of this study, detailed participant information is available in Tables 1, 2, 4 and 5, enabling a careful evaluation of each participant's demographic, neuropsychological, and inner speech variables.

Demographic relationships with inner speech

An exploratory Spearman's correlation did not identify a significant relationship between age and proportion of 'yes' "in the moment" responses ($r_s = -0.10$, $p = 0.65$, $\beta = 0.07$), or between total types of inner speech experiences across the week during the experiment (i.e., general inner speech questionnaire checked categories summed across the study) ($r_s = -0.42$, $p = 0.06$, $n = 21$, $\beta = 0.49$). There were no outliers identified for age. Similarly, an exploratory Spearman's correlation did not identify a significant relationship between years post-stroke and inner speech variables ("in the moment" inner speech proportion of 'yes' responses, $r_s = -0.16$, $p = 0.46$, $\beta = 0.07$; total types of inner speech experiences, $r_s = 0.04$, $p = 0.86$, $n = 21$, $\beta = 0.05$). Two subjects were considered outliers, having experienced a stroke many years prior to the study (21.5 years and 25 years). Finally, an exploratory point-biserial correlation did not identify a significant relationship between sex and inner speech variables ("in the moment" inner speech proportion of 'yes' responses, $r_{pb} = -0.02$, $p = 0.93$, $\beta = 0.05$; total types of inner speech experiences, $r_{pb} = -0.29$, $p = 0.20$, $n = 21$, $\beta = 0.25$).

Language impairment awareness relationship with inner speech

Generally, the sample showed great variation in VATA-L discrepancy scores, ranging from -22 (where the participant's estimate of their abilities far exceeded the peer's estimate of their abilities, reflective of anosognosia) to 8 (where the peer's estimate of the participant's abilities slightly exceeded the participant's own estimate) ($M = -2.30 \pm 7.35$). A more negative discrepancy is indicative of anosognosia, or impaired insight into one's own language impairment, with one prior study suggesting that a score of lower than 13.1 (in the negative direction, i.e., a more

negative score than -13.1) is reflective of anosognosia in aphasia. In the current sample, there were two participants who met that criterion for anosognosia (see Table 2). No discrepancy, or a positive discrepancy, suggests self-awareness or insight into one's own language impairment. A very high positive value (which we did not observe in this study) would suggest heightened awareness/insight into one's own language impairment.

Spearman correlation did not identify a significant relationship between “in the moment” inner speech (proportion of ‘yes’ responses) and language impairment insight with inner speech (discrepancy score from VATA-L) ($r_s = 0.33$, $p = 0.13$, $\beta=0.35$), suggesting that those who may have anosognosia (receiving a negative discrepancy score, thus rating themselves higher than the ‘other’ person rated them) are not less likely to say that they have “in the moment” inner speech across the descriptive experience sampling timepoints. The total types (i.e., number) of inner speech themes derived from the General Inner Speech Questionnaire (summed across three administrations) was not significantly related to anosognosia ($r_s = 0.24$, $p = .31$, $n=21$, $\beta=0.18$). Spearman's correlation showed that aphasia severity ($r_s=0.25$, $p=0.25$, $\beta=0.21$) did not demonstrate a significant relationship with language impairment insight, and therefore we did not control for either variable when computing analyses between these variables and the inner speech variables. Note that there were two outliers identified for their VATA Discrepancy scores, both scoring very low (lacking insight), with scores of -16.5 and -22.

Participants with less frequent inner speech

When examining the frequency of inner speech (proportion of “yes” responses for in the moment inner speech) in specific cases, two outliers were identified – Participant 20 and Participant 30. To reflect an accurate characterization of the sample (including heterogeneity) and because of the preliminary nature of the study, these participants are described in a case-wise fashion here. As shown in Table 4, they reported having inner speech less often than the other participants (Participant 20 at 21% and Participant 30 at 36%, compared to the group

average of 78%). Upon exploring the clinical profiles of these two participants (Table 2), it was clear that Participant 20 had a more severe motor speech impairment than most (a 5 out of 10 on the Speech Motor Programming subtest of the QAB). Interestingly, one other participant demonstrated the same score (5 out of 10) for motor speech impairment and also responded with “yes” less often than the group as a whole (participant 78, who responded with “yes” in 55% of instances per Table 4). Participant 78 was not an outlier, however. Participant 30 had very mild aphasia (latent/no aphasia by the QAB), and a 7.5 out of 10 on Speech Motor Programming, showing some mild motor programming impairments. Participant 30 was highly educated, worked in a demanding job during the course of the study, and was the participant with the lowest compliance in responding to prompts in this study (58% compliance on the “in the moment” prompts compared to the group average of 95%). Participant 30 also had a larger negative discrepancy in VATA-L scores (-16), indicating greater likelihood of anosognosia.

Hypothesis 1: Content, Functions, and Activities of Inner Speech Use

The extent to which participants used inner speech to talk about similar things (*content* – “I talk to myself about”), for similar reasons (*functions* – “I talk to myself in order to”) and during similar situations (*activities* – “I talk to myself when”) was characterized by analyzing the percentage of participants who selected a theme across the administrations. For the purposes of characterizing patterns in the data, all participant data (n=23) was used. To streamline Results, a broad overview is presented here and readers are referred to Table 5 and Figure 4 for detailed results.

Across all survey administrations, the highest percentage of participants used inner speech ‘when remembering’, which was followed closely by using inner speech ‘to plan’, to ‘motivate myself’, to ‘think critically’, and about ‘food / what to eat’. Other themes that were endorsed, on average, by more than 70% of the participants included using inner speech ‘to solve problems’, about ‘what I want to do’, about ‘negative emotions’, about preferences, in

order 'to concentrate' and in order 'to rehearse ahead of time.' Generally, themes from the General Inner Speech Questionnaire "I talk to myself in order to" section were the most commonly experienced by participants in this study. As hypothesized, patterns across the group were identified alongside variability, which is evidenced in Figure 4. No outliers were identified when evaluating the total number of General Inner Speech Questionnaire themes identified over the course of the three weeks. When evaluating the total number of themes for each theme area (content, functions, activities) across the three weeks (in the N=21 with 100% compliance), no outliers were identified for either 'content' or 'activities' themes. One outlier was identified for 'functions' theme, who identified only seven inner speech functions across the three weeks (compared to a group mean of 20.76).

Participants' confidence ratings in their answers were very high overall (across all theme areas, max score of 5 [high confidence], Week 1: $M=4.54\pm0.31$, Week 2: $M=4.35\pm0.34$, Week 3: $M=4.00\pm0.57$, Average: $M=4.14\pm0.31$), reflecting what had been previously found in neurotypical young adults (Racy et al., 2020) and giving investigators confidence in interpreting the findings as being reflective of participants' inner experiences. No outliers were identified for confidence across any of the themes.

Table 5. Individual participants with inner speech data: General Inner Speech Questionnaire

ID	Compliance (%completed)	Total “content” (I talk to myself about)			Total “function” (I talk to myself in order to)			Total “activities” (I talk to myself when)			Total items		
<i>Week</i>		1	2	3	1	2	3	1	2	3	1	2	3
<i>Maximum possible</i>		16			11			5			32		
2	100	14	12	10	7	6	5	5	1	3	26	19	18
3	100	13	13	16	10	9	10	5	5	5	28	27	31
5	100	4	3	3	3	2	2	2	1	1	9	6	6
6	100	11	8	7	5	5	7	3	2	4	19	15	18
7	100	8	10	13	7	9	7	5	5	5	20	24	25
8	100	9	7	8	7	2	4	4	2	2	20	11	14
10	100	15	11	14	9	7	7	2	2	4	26	20	25
11	100	16	15	16	10	10	10	5	5	5	31	30	31
14	100	10	12	10	7	6	8	3	3	5	20	21	23
16	100	10	9	7	7	9	9	4	4	4	21	22	20
19	100	3	13	6	6	8	6	2	3	2	11	24	14
20	100	12	4	4	9	1	2	5	1	1	26	6	7
22	100	10	5	12	7	6	5	3	2	2	20	13	19
24	100	11	11	14	8	8	9	3	3	4	22	22	27
25	100	13	10	14	9	4	9	3	2	3	25	16	26
29	100	5	5	5	7	7	7	2	2	2	14	14	14
30	33.33	10	NA	NA	9	NA	NA	4	NA	NA	23	NA	NA
35	100	3	14	2	3	7	3	0	3	0	6	24	5
46	100	11	9	11	8	6	9	3	4	4	22	19	24
47	100	16	16	16	10	10	11	4	4	4	30	30	31
48	66.66	14	15	NA	9	10	NA	3	3	NA	26	28	NA
63	100	5	7	5	5	7	7	4	2	4	14	16	16
78	100	12	7	13	8	9	9	3	5	3	23	21	25

Mean (SD)	95.65 (15.26)	10.22 (3.97)	9.82 (3.76)	9.81 (4.56)	7.39 (1.99)	6.73 (2.64)	6.95 (2.64)	3.35 (1.27)	2.91 (1.34)	3.19 (1.47)	20.96 (6.51)	19.45 (6.84)	19.95 (7.97)
Range (min, max)	33.33, 100	3, 16	3, 16	2, 16	3, 10	1, 10	2, 11	0, 5	1, 5	0, 5	6, 31	6, 30	5, 31

The General Inner Speech Questionnaire (adapted) was filled out on three different occasions by each participant, thus constituting a total of 69 possible surveys across the study (3 surveys x 23 participants).

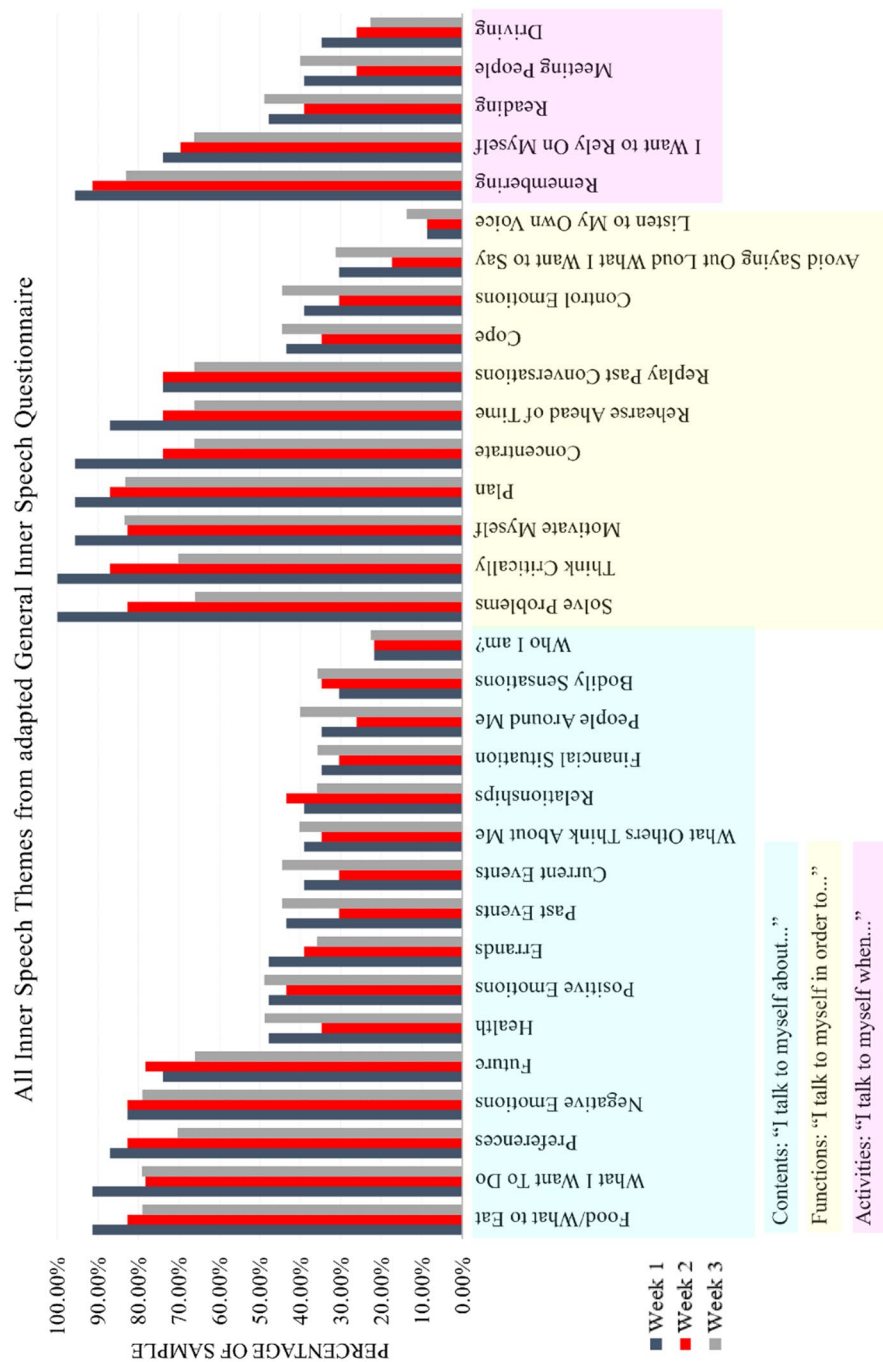


Figure 4: Prevalence of themes of inner speech from the General Inner Speech Questionnaire (n=23 participants). There were three administrations of this survey (Weeks 1, 2, 3). The categories of inner speech are highlighted in light blue ("I talk to myself when"), light yellow ("I talk to myself in order to"), and light pink ("I talk to myself when"), to highlight how the themes of inner speech fit into these categories.

The similarity of total inner speech experiences across the three weeks was evaluated using a Friedman test, finding that there was not a significant difference in the total number of inner speech themes selected across each of the three weeks in the sample ($X^2_r=2.98$, $p=0.23$, $df=2$, $n=21$). For week 1, $M=20.62$ themes or 64.43% of all possible themes were chosen; for week 2, $M=19.05$ themes or 59.52% of all possible themes were chosen; and for week 3, $M=19.95$ themes or 62.35% of all possible themes were chosen. Three further Friedman tests confirmed that this was likewise the case within each category of inner speech experiences: content ("I talk to myself about," $X^2_r=2.57$, $p=0.28$, $df=2$, $n=21$), functions ("I talk to myself in order to," $X^2_r=2.54$, $p=0.28$, $df=2$, $n=21$), and abilities ("I talk to myself when," $X^2_r=1.12$, $p=0.57$, $df=2$, $n=21$). Given no statistical test was significant, multiple comparison correction was not implemented.

Hypothesis 2: Relationship of Inner Speech to Aphasia Severity

The hypothesis was that there would not be a relationship between inner speech and aphasia severity. The statistical tests failed to reject the null hypothesis. There was no significant relationship between the proportion of saying 'yes' during "in the moment" inner speech and aphasia severity ($r_s=0.11$, $p=0.63$, $\beta=0.08$). There was no significant relationship between total types of inner speech themes selected from the General Inner Speech Questionnaire and aphasia severity ($r_s=0.19$, $p=0.41$, $n=21$, $\beta=0.13$).

Given that prior research has found relationships with overt speech (more inner speech tends to associate with better overt speech), an exploratory correlation was run between the QAB Word Finding subtest, the proportion of saying 'yes' during "in the moment" inner speech ($r_s=0.03$, $p=0.90$, $\beta=0.05$), and the total types of inner speech themes selected from the General Inner Speech Questionnaire ($r_s=0.02$, $p=0.94$, $n=21$, $\beta=0.05$). Prior research has also broadly implicated phonological output processes as being related to inner speech, and therefore

exploratory correlations were conducted between the QAB Repetition subtest, the proportion of saying 'yes' during "in the moment" inner speech ($r_s=0.24$, $p=0.27$, $\beta=0.20$), and the total types of inner speech themes selected from the General Inner Speech Questionnaire ($r_s=0.32$, $p=0.16$, $n=21$, $\beta=0.30$).

Discussion

This is a preliminary study reporting on a relatively small and novel sample of the experience of daily life inner speech by persons with chronic, post-stroke aphasia.

Exploratory Question 1: Procedure compliance

A goal of this study was to examine the feasibility of conducting a study including naturalistic and introspective sampling procedures in adults with chronic aphasia. Feasibility was defined as demonstrating high compliance with study procedures. The participants in this study demonstrated high compliance for both experience sampling and completion of the longer, once-weekly survey, with only one participant demonstrating a compliance of <83%. There were a few participants ($n=3$) who completed the General Inner Speech Questionnaire more times than they were asked, which we handled by only accepting the responses from the time points at which the notifications were sent, but this could be seen as a limitation as these few participants did not strictly adhere to the instructions. Future work in this area should ideally limit the number of times an IP address (or similar) can fill out the surveys, therefore mitigating this concern.

Only one study, to our knowledge, has leveraged experience sampling methodology in a study sample of participants with aphasia (to study flow states; Sather et al., 2017). The individuals from Sather et al. (2017) all had mild, chronic aphasia. Overall, the Sather sample appears like the sample leveraged in the current study, which likewise involved individuals with chronic, milder aphasia and individuals within a similar age range (roughly 50-70 years). While the peer reviewed publication from Sather et al. (2017) does not discuss compliance in the nine

individuals with aphasia that participated in the experience sampling methodology to evaluate 'flow states' in their daily lives, close reading of Sather's (2015) dissertation suggests varying though largely good compliance with a higher number of notifications, albeit over a shorter period of time. The Sather (2015) thesis reports using six randomized indicators per day across a 12-hour continuous period across seven consecutive days (i.e., 42 total sampled experience attempts). Table 6 from the thesis suggests that there was a range of compliance ($M=80.42\%$, $SD=17.49\%$, $\min=42.86\%$, $\max=92.86\%$), slightly lower than is reported in our current sample ($M\sim 95\%$) but still relatively high for experience sampling.

While the best practice for dynamic sampling of inner speech appears to be mixed methodology involving experience sampling (Hurlburt et al., 2021), thought listing and interviews (Racy et al., 2020), and surveys (Alderson-Day et al., 2018; Racy et al., 2020, 2022), these were complex procedures to undertake in people with aphasia. The low number of total dropouts and the high session completion of this study is encouraging, given that we attempted to employ all three methodologies (though report only on quantitative data here). Note, though, that exclusion parameters (primarily a certain score on auditory comprehension) precluded the inclusion of individuals with more severe aphasia. The implications of this, and future directions, are discussed more below.

Experience sampling relies on self-report, which can be biased or inaccurate, especially in a population with possible comprehension and self-monitoring deficits, as discussed in Fama & Turkeltaub (2020). To lessen the concerns about self-report, participants were excluded based on poor auditory comprehension, and multiple modalities of communication (e.g., visual, written, auditory, gesture) were used throughout the study along with regular comprehension checks. Note, too, that the current samples presented with relatively little anosognosia, which likely reflects that the majority of the sample had intact self-monitoring as it related to monitoring of their own language impairments. The self-report methods used here were chosen to increase ecological validity and ability to measure contents of inner speech (Brinthaup & Morin, 2023),

while acknowledging the potential for inaccuracy. To mitigate potential biases known to occur during subjective reporting, such as the desire to please the experimenter (i.e., good-subject effect [Nichols & Maner, 2008]), experimenter JA reiterated the need for an accurate representation of what happens in daily life and encouraged honest reporting of the participant's experience in that moment. Since participants met with JA a total of four times throughout the study (Introductory session, then three interviews), this was reiterated often. Future studies should be aware that revisiting the purposes of the study, and the procedures, are likely good practice for this type of methodology when collaborating with participants who have aphasia.

Another valuable takeaway of the current study was that the virtual and remote modality of meetings (i.e., introductory meeting, interviews [not described here]), experience sampling, and questionnaires) was feasible for most participants. A small subsample elected to use a paper packet for the study, but all individuals employed some form of technology for the experience sampling procedure. Most individuals used their smartphone during the study, with only four of the 23 participants opting to use the paper packet. Those four, notably, differed in age range (50 years, 61 years, 71 years, 75 years), sex (two male, two female), education (14 years, 18 years, 20 years, 22 years), and prior occupation (art therapist, teacher, dentist, unknown/retired). It may have been a personal preference, but overall, use of technology (i.e., at least twice a day per experience sampling protocol) does not appear to be a barrier to participating.

Exploratory Question 2: Frequency of Inner Speech Use

The participants in this study reported experiencing inner speech often (nearly 80% of the sampled instances). Prior research in young adults have reported wide variation in inner speech experience, ranging from 0-75% (Brinthaup et al., 2009; Heavey et al., 2019; Heavey & Hurlburt, 2008). It should also be noted that the pseudo-random prompt intervals in the current study may have led to heightened awareness around the inner speech experience, which may

also help explain the larger proportion of study participants who experienced inner speech at the time of the notification. Although there could be multiple reasons for the differences in reported frequency, such as the varying methods (e.g., number of prompts given during experience sampling, number of total experiences sampled), this preliminary study suggests that people with chronic aphasia may experience inner speech relatively frequently in daily life.

Other studies broadly support that inner speech is available to individuals with chronic aphasia, and have used varying methods, such as asking individuals if they “can say the word in their head and it sounds correct” during object naming (Fama, Henderson, et al., 2019; Fama, Snider, et al., 2019; Fama & Turkeltaub, 2020) and if individuals with aphasia can successfully complete silent rhyme and homophone judgments (Geva, Bennett, et al., 2011; Stark et al., 2017). Empirical research in adults without brain injury supports different cognitive and neural processes for spontaneously elicited inner speech (versus non-spontaneous inner speech (what is being done in the experimental, word-level tasks) (Grandchamp et al., 2019; Hurlburt et al., 2016; Pratts et al., 2023). That is, spontaneous inner speech appears to not require as much ‘linguistic’ brain as non-spontaneous inner speech – likely because the spontaneous experience of inner speech draws upon semantic and lexical resources but perhaps fewer phonological and pre-motor resources – which means that individuals with damage to certain language regions of the brain may have heightened capacity for or awareness of spontaneous inner speech but not necessarily non-spontaneous inner speech. In the ConDialInt model (Grandchamp et al., 2019), this may reflect the fact that daily life inner speech experiences are likely more ‘condensed’ than the experimental tasks probing word-level inner speech through rhyming and silent picture naming, which both require that the person have access to detailed phonological information. Despite the potentially different underlying cognitive and neural processes, the convergence of evidence from experimental and spontaneous (or naturalistic) tasks evaluating inner speech in aphasia appears to support that inner speech can be available to a person even in the context of a language impairment due to brain injury.

Exploratory Question 3: Relationship between Inner Speech and Demographic and other Neuropsychological Data

Exploratory analyses suggested that, in our sample, there was not a significant relationship identified between sex, age, or years post-stroke with inner speech. Note that this is a preliminary study with a small sample size, and as the power values included in the results stipulate, most correlations had low power. Low statistical power can occur because of small sample size, small effects, or both. In this case, there were no statistically significant correlations even when underpowered, suggesting that a much larger sample size would be needed to measure potentially very small effects (if indeed they exist).

A study employing the Varieties of Inner Speech Questionnaire suggested a higher rate of inner speech experiences that involved evaluation / criticalness in women (Alderson-Day et al., 2018), though their sample had 75.5% females. Largely, though, the impact of sex or gender on inner speech remains under evaluated (Morin & Racy, 2022). In the current study, we did not identify a compelling relationship between sex and inner speech in daily life.

Another exploratory area of interest was the impact of age on inner speech. Beyond the appearance and awareness of inner speech later in the childhood years (Flavell et al., 1997; Geva & Fernyhough, 2019), very little research has evaluated the impact of aging (adolescence, young adult, older adult) on inner speech frequency or its themes. In the sample of adults included in our study, which ranged from age 38 to age 75, a significant relationship of age with inner speech frequency or themes was not found. A medium-strong correlation was identified for the number of themes experienced but this effect was notably underpowered. If replicated in a larger sample, this negative relationship may suggest a relationship between increasing age with decreasing inner speech themes. A study on mind-wandering, which is not quite the same thing as inner speech but may comprise aspects of inner speech, suggested that the frequency of mind-wandering decreases with age in neurotypical aging, with no difference in the decrease

due to age in individuals experiencing Alzheimer's Disease (Gyurkovics et al., 2018). The authors offer the explanation that mind-wandering is a resource-dependent process and cognitive resources decline with age. If this were applied to our study's findings, it might suggest that decreased cognitive resources result in fewer inner speech themes arising, though other factors are almost certainly also at play. For example, an increase in leisure time during retirement may decrease the need to use inner speech for certain reasons (e.g., problem solving) compared to younger adults. A much larger sample size would be needed to confirm any effect of age on inner speech.

We explored the relationship between chronicity – the number of years living with stroke consequences – and inner speech. Several anecdotal autobiographical accounts from stroke survivors who developed aphasia discuss the acute loss of inner speech, followed by a return of inner speech throughout their recovery process (Marks, 2018; Taylor, 2009). Observational evidence suggest that, in brain damaged patients who eventually recover from their trauma, self-awareness often returns in parallel with inner speech (Ojemann et al., 1996). Beyond recovery of inner speech during the acute phase, which is usually defined as the stage of spontaneous recovery during which most brain healing occurs (typically the first three months post-injury), little is known about inner speech changes in the chronic stage. Our study appears to not support a significant relationship between increased chronicity and inner speech frequency or themes in a sample of adults with a wide chronicity range (under one year to 25 years post-stroke). Therefore, years of living with chronic stroke symptoms may not be the most salient variable for describing any potential changes observed in inner speech.

Finally, we examined the extent to which anosognosia (lack of insight specific to language impairment) related to inner speech, given inner speech's strong relationship with self-monitoring (Morin, 2011a, 2011b). Documentation of aphasic anosognosia can be traced back to Wernicke but has only recently been quantified using standard assessments (Cocchini et al., 2010; Kertesz, 2010; Wernicke, 1874). Therefore, it would not be surprising to find that

experience of inner speech is negatively related to anosognosia – that is, people with probable anosognosia may report experiencing fewer instances of inner speech. Another thought could be that those with anosognosia may not reliably self-report inner speech function, and/or may over-estimate how much they use inner speech. However, to our knowledge, no study has empirically related presence of anosognosia with inner speech experience in aphasia. Our study did not find strong evidence that the amount of anosognosia, or lack of insight, related to inner speech frequency or themes. It may be because only two individuals in the current study demonstrated a discrepancy score that the original Cocchini et al. (2010) study deemed ‘anosognostic,’ i.e., $\geq$ a discrepancy of 13.1. That is, there may not have been a sufficient sample of individuals experiencing anosognosia to reliably identify any relationship with inner speech. This may once again be due to a relatively mild presentation of aphasia in the majority of the participants in this study, who are less likely to present with anosognosia (Kertesz, 2010), but it should be noted that others have not found a direct relationship between anosognosia and aphasia severity (Weinstein & Kahn, 1950, 1955).

The neural and cognitive mechanisms specific to anosognosia of language impairment in aphasia remain unexplained (Kertesz, 2010), though there is likely some relation to a self-monitoring mechanism. Some scholars have postulated a language-related self-monitoring mechanism situated in auditory comprehension (Roelofs, 2020) (which is lent some credence by the documentation of individuals with Wernicke’s aphasia as often presenting with anosognosia; Wernicke, 1874), while others have argued for a more generalized self-monitoring system less intimately tied to the language system (Nozari et al., 2011). While the current study does not directly support either claim, some level of self-monitoring seems important for understanding the experience of inner speech. This may also be why some individuals without brain damage say that they lack inner speech (Hurlburt et al., 2021), i.e., self-monitoring may be an individual difference that helps explain the inner speech phenomenon. This is a potent area of future study, in aphasia and in adults without brain damage.

It would be remiss to not comment on the research suggesting a relationship between meta-cognitive and executive function capacities with inner speech, and how that may play a role into the perception of inner speech by persons with aphasia who may have impairments in these areas. Philosopher Langland-Hassan theorized that, whilst theorists have linked inner speech with metacognition (thinking about one's thinking), it does not appear that inner speech has a clear correlation or reason for being related to metacognitive capacities given that metacognition is not thought to be particularly related to inner speech's linguistic functions, its sensory characteristics, or its consciousness (Langland-Hassan, 2014). Indeed, further work from his group suggested that inner speech impairment (as measured by a highly linguistic task, which is the covert judgment of rhyming pairs) was likely not related to metacognitive self-assessments in general (Langland-Hassan et al., 2017). Inner speech's relationship with executive function has also been explored in neurotypical adults, suggesting that disruption of inner speech (through things like articulatory suppression) results in problem solving impairments (Wallace et al., 2017). Whilst executive function impairments are not pervasive in aphasia, they can be present (Dutta et al., 2022; Fridriksson et al., 2006; Mayer et al., 2016; Purdy, 2002). No study, to our knowledge, has yet examined the relationship of executive dysfunction and inner speech in aphasia. This question is especially interesting in light of theories suggesting that speech self-monitoring relies on domain-general executive function ability (Nozari et al., 2011). This is a clear future research direction that should be explored, especially to elucidate the directionality of inner speech and executive dysfunction. For example, it may be the case that inner speech is required for higher executive function capacity and that, when executive function is impaired, inner speech is not impaired but instead simply cannot be leveraged to its full extent.

Cases of less frequent inner speech

In this exploratory pilot study, most of the people with aphasia reported frequent use of inner speech, but there were two participants who reported inner speech less often. These participants had unique clinical profiles, including moderate motor speech deficits for one participant (Participant 20) and anosognosia (lack of awareness of language deficits) and low compliance for another participant (Participant 30). These cases motivate future work on the topic, specifically investigating speech motor programming, anosognosia, and inner speech with a greater variety in the clinical profiles of the participants. Links may elucidate the nature of inner speech and its place in lexical retrieval, speech production, and self-monitoring.

Participant 30, who was classified as having anosognosia, may have reported less frequent inner speech due to a unique inner experience. During interviews, this participant reported a version of ‘condensed’ inner speech: he did not always *hear* or *say* the words in his head (Heavey & Hurlburt, 2008; Hurlburt et al., 2016), but rather *saw* the words, as if on a teleprompter. That is, phonological properties may not have been present for his experience of inner speech, but some access to visual representations of lexical-semantics appeared available. Uncovering individual differences in the phenomenon of inner speech experience is crucial for understanding how inner experience relates to behavior, as individual differences in its properties may change how and to what extent a person uses it (Lupyan et al., 2023; Roebuck & Lupyan, 2020).

Hypothesis 1: Themes of Inner Speech Use

A major purpose of leveraging the General Inner Speech Questionnaire was to evaluate the themes of inner speech use. The use of the General Inner Speech Questionnaire at the end of each week of experience sampling was intentional, given evidence that experience sampling tends to result in within-subject and across-subject patterns and differences (Hurlburt & Akhter, 2006). That is, across Hurlburt’s studies employing descriptive experience sampling, he found

that the more participants were asked to be consciously aware of inner speech (through the experience sampling notifications), the more they tended to recall about inner speech and be “conscious” of inner speech. Therefore, it was our thought to have the General Inner Speech Questionnaire come at the end of each week of experience sampling, where there would be a probability of heightened awareness as to the inner speech experiences across that week.

In this preliminary study, participants with aphasia in the current study experienced similar themes of inner speech as found across several studies in young adults without brain injury. Indeed, Racy et al. (2020) and Morin & Racy (2022) summarize that the most common inner speech experiences found in over 1000 undergraduate volunteers included content themes (negative emotions) and functions (to problem solve, plan, control emotions, and self-motivate). In the current study, functions were the most endorsed reasons for using inner speech, with solving problems, thinking critically, planning, rehearsing ahead of time, and concentrating as some of the most shared experiences. Content experiences were also prominent, with negative emotions, preferences, ‘what I want to do,’ and food/what I want to eat as experienced by many within the sample. Therefore, our findings suggest some commonality to inner speech experience in aphasia compared to individuals without brain injury.

There is a distinct lack of data about the inner speech experiences of older adults. It may be that some of the experiences found in the current study’s sample are more reflective of being an older than a younger adult, e.g., we showed that 56.52% of the sample said (at least once) that their inner speech involved health. The same could be said about using inner speech during remembering, and perhaps also with coping (with 56.52% of our participants using inner speech to cope at least once during the study), the latter of which is a much rarer use of inner speech in adults (Racy et al., 2020). A clear future direction is collection of inner speech experiences in the older adult population.

While there is a large benefit to comparing themes between adults with aphasia and neurotypical data collected previously using the restricted list of themes available on the

General Inner Speech Questionnaire, there are also drawbacks to this approach. For example, it may be that persons with aphasia endorse different themes than those on the General Inner Speech Questionnaire modified version. Our team did anticipate this, collecting thought listing data during each experience sampling session and at the end of each General Inner Speech Questionnaire administration (specifically, we asked: “What else do you talk to yourself about? Use this space to write or draw”). However, team capacity has not yet enabled us to aggregate and analyze that qualitative data. Therefore, the extent to which adults with aphasia endorse *different* themes from those in the General Inner Speech Questionnaire is the subject of future work from our group and will extend the findings presented here.

The ratings of confidence in their inner speech themes also warrant consideration here. It has been shown previously that neurotypical young adults responding to the General Inner Speech Questionnaire are generally confident in their accurate recall of inner speech experiences across the past week (Racy et al., 2020), and the present study replicated that in a small sample of persons with aphasia. However, it bears consideration that persons with aphasia may have varying judgments of confidence. There is a body of literature that suggests that persons with aphasia may not be confident in their overall communicative abilities (Babbitt & Cherney, 2010), and it bears questioning whether this extends to confidence in their perception of inner speech. Whilst it was not the goal of the current study to probe this, this is a consideration that warrants future consideration, because heterogeneity in confidence – if this study were to be extended to include a more diverse representation of persons with aphasia—may be found, and this could be used to examine if and how themes and frequency of inner speech relate to perception of each person’s confidence in reporting inner speech.

We explored the similarity of inner speech experiences across weeks, with no directional hypothesis attached to this. The rationale behind this exploratory investigation was that, because of the relationship of inner speech with context (i.e., it shifts according to scenario, mood, need, etc.) (Alderson-Day et al., 2018; Heavey et al., 2019; Racy et al., 2020), there may

be corresponding fluctuations across inner speech items across the three administrations/weeks. Indeed, it is expected that life circumstances will vary across participants. Despite this rationale, the findings from the current study did not suggest much shift in inner speech themes (total number or type) over the course of three weeks. This consistency (or lack of significant difference) may be early evidence for short-term reliability of administering the General Inner Speech Questionnaire in this sample. Longer studies (i.e., over the course of several months or even years) may be more likely to demonstrate changes in inner speech due to the higher potential of significant life or context changes occurring then. We also did not collect information about any significant life events that may have occurred during the study, and future studies should collect this to evaluate the extent to which these life events impact inner speech.

Hypothesis 2: Inner Speech's Relationship with Aphasia Severity

The hypothesis was a null hypothesis (that aphasia severity would not be related to inner speech) and statistics failed to reject the null hypothesis. This is an important finding because it suggests that future research should evaluate the extent to which individuals with more severe aphasia feel that they experience inner speech. If inner speech was only an impoverished version of overt speech, people with more severe aphasia should experience inner speech comparatively less than those with milder aphasia. Note that there were no individuals with severe aphasia in the current sample, precluding a more conclusive interpretation of the impact of more severe aphasia on inner speech. Failing to reject the null hypothesis in the current study motivates future research into the role of inner speech, such as indicating which items are most stimuable in therapy (Hayward et al., 2016) and identifying the root cause of someone's anomia (Fama, Snider, et al., 2019; Fama & Turkeltaub, 2020).

Prior work evaluating more experimentally-obtained inner speech, such as subjective inner speech experience during picture naming or rhyming, has suggested that some adults with

severe non-fluent aphasia have inner speech to a degree, and that their inner speech is often preserved more than their overt speech (Fama, Henderson, et al., 2019; Geva, Bennett, et al., 2011; Stark et al., 2017). However, spontaneous inner speech occurring in daily life in individuals with severe aphasia, inclusive of those experiencing more impairments related to auditory comprehension, is a clear future direction of research. It is likely that more accommodations need to be made to methodology and procedures to encourage participation from this group, and developing this study is a prime opportunity to employ participatory design by partnering with individuals with severe aphasia to co-design the procedures. It may also be the case that, by nature of the type of strict procedural requirements of experience sampling, that a necessary cut-off of auditory comprehension abilities must be employed to ensure study validity, and that we, as a field, may be unable to understand the daily experience of inner speech in individuals with more severe auditory comprehension or self-monitoring impairments using this type of procedure.

Conclusion

For what we believe is the first time, we dynamically evaluated self-report of inner speech experiences in the daily lives of adults with chronic aphasia. This preliminary study had low attrition and high compliance, establishing that participants could understand and execute this method, despite it being time intensive. Most participants reported experiencing inner speech, and there was a general pattern in the themes of inner speech that were common amongst the group, including using inner speech to remember, plan and motivate oneself. Demographic variables (year post-stroke, age, sex), aphasia severity, and insight into language impairment were not found to have a significant relationship with inner speech frequency or the number of themes explored using inner speech.

Broader Implications and Clinical Application

We envision several broad implications and clinical applications of assessing inner speech in daily life in adults with aphasia. By collecting inner speech data from adults with aphasia, clinicians (such as speech-language pathologists and psychologists) may choose goals that take into account that people with aphasia are doing certain tasks “in their heads” using inner speech (e.g., planning, rehearsing a conversation), meaning that (1) they are likely salient to the individual with aphasia, and (2) stimulating verbal language around those tasks during therapy may be particularly beneficial, given that inner speech has been demonstrated to relate to overt language recovery with therapy (Hayward et al., 2016). We envision that researchers, speech-language pathologists, and mental health professionals can leverage the types of procedures used in this study to better understand changes in emotional states and psychosocial health without relying heavily on verbal production. Most participants used inner speech to talk about negative emotions, in order to regulate emotions, and in order to motivate themselves.

Because of the relative strength of inner speech compared to overt speech in many people with aphasia (Fama et al., 2017; Geva, Bennett, et al., 2011; Stark et al., 2017), much of a person with aphasia’s language may be taking place inside their heads. This is especially likely in cases of comorbid dysarthria and apraxia of speech, in which overt motor speech is impacted. Specifically for apraxia of speech, which is a disorder of motor planning and programming, evidence of successful inner speech indicates that internal retrieval and modeling can be intact without successful overt word production (Fama, Henderson, et al., 2019). In the current study, less emphasis was placed on requiring inner speech to have all correct phonetic information, but in general, successful inner speech requires accurate lexical retrieval and access to semantic information. Clinicians should be aware of this possibility and recognize that skills such as “rehearsing ahead of time” may be more common in inner speech post-stroke and could be leveraged during therapy (e.g., practicing scripts using inner and overt speech; asking

if the person has inner speech for items or phrases post-therapy to gauge improvement in not only overt speech but also inner speech over time).

As discussed above, anecdotal evidence from lived experience suggests that inner speech may disappear in the acute stage of aphasia and recover over time, perhaps in tandem with or slightly ahead of the rest of language (Anbar, 2022; Marks, 2018; Taylor, 2008). This suggests that clinicians in the acute and inpatient settings may want to monitor changes in inner speech as reflective of recovery, just as they would measure expressive and receptive language changes. Along with the outcome measures that are generally collected, inner speech can be added to the unique profile of communicative abilities for the clinician's patients with aphasia.

Finally, aphasia recovery does not singularly involve language improvement. Indeed, because individuals with aphasia become more socially isolated (Code, 2003; Dalemans et al., 2010) and experience a variety of other challenges (e.g., depression, Leeds et al., 2004; Pompon et al., 2022), recovery of language and communication occurs more holistically. The Life Participation Approach to Aphasia suggests focusing on the person with aphasia's experience across a variety of domains, including personal factors, environment, and activities / participation (Chapey et al., 2018; Kagan, 2020). Measuring and leveraging inner speech in daily life may be a unique way to holistically understand aphasia recovery.

Limitations and Future Directions

We employed pseudo-random interval sampling during the study, which was chosen to increase predictability of prompts (e.g., within the same hour per day), reduce prompt fatigue, and increase compliance (Hektner et al., 2007). The pseudo-random, semi-predictable signal times carry strengths and weaknesses. For example, when people know that they will be notified for a short survey every day between 9-10AM and 4-5PM, they tend to adjust to the interruption, e.g., they may plan to not be driving or be in a quiet spot in order to accurately and quickly respond (Hektner et al., 2007). A strength of predictability is that people tend to not find the alerts

intrusive, so compliance will be high and missing data will be low – this is indeed what we found in our study. A weakness, though, is that people may rearrange their activities in order to “fit in” with the data collection regime, e.g., avoid scheduling things during those times. This may result in heightened “awareness” of inner speech, e.g., in the case of our sample, thus creating a slightly artificially high number of “in the moment” yes responses. However, given that individuals with aphasia have many unmovable commitments (e.g., support groups, therapy), we felt that the strengths of pseudo-random design outweighed the weaknesses. This design choice poses a limitation for comparison of our findings to other descriptive experience sampling studies of inner speech (e.g., Hurlburt and colleagues’ work in young adults), which involved random sampling as well as more samples per day (typically around ~six notifications per day, though the total number of notifications was similar to what was employed in our study). Future studies should replicate the findings from the current study as well as empirically explore the extent to which participants with aphasia are amenable, and reliable, in studies employing a variety of beep/prompt intervals (e.g., up to six per day) and study durations (e.g., days, weeks).

As acknowledged in the method and results, there was one option on the General Inner Speech Questionnaire which was included on the Qualtrics survey but not on the paper packet (“Replay past conversations” was listed twice rather than including “Listen to my own voice”). This was an experimenter error and results must be interpreted as such because of the missing data. However, given that replaying past conversations was reported relatively frequently across all weeks (>60% of participants) and listening to my own voice items was reported relatively infrequently across all weeks (<20% of participants), the missing data may be interpreted as largely negligible.

This study’s sample did not include any participants with severe aphasia, which may affect the relationship with aphasia severity and reports of inner speech, as their experience is not captured here. Additionally, comparisons to prior literature are limited due to the age difference between the participants, as frequency and contents of inner speech may change

throughout the lifespan regardless of brain injury. Future work involving individuals with more severe aphasia and anosognosia is the logical next step in validating the impact of each on inner speech, along with comparisons to age-matched adults without brain injury or with other neurogenic communication disorders (e.g., traumatic brain injury, right hemisphere damage, or cognitive-communication impairments). Hurlburt and colleagues' body of work also suggests that the validity and specificity of descriptive experience sampling improves with training and practice. Since this study established early evidence for feasibility of the use of descriptive experience sampling in a population of adults with aphasia, future work should evaluate the extent to which increased training (such as a longer study, or an extended introductory period) complements the results presented here.

Finally, it is of note that the General Inner Speech Questionnaire was modified based on consultations with an expert in the field of inner speech research, Alain Morin. A clear next step is to engage with persons with aphasia to further refine the scale, ensure its accessibility, and ensure its specificity to living with an acquired language impairment.

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Data Availability Statement

The data that support the findings of this study are openly available in The Open Science Framework at <https://osf.io/b6vxx/>.

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Supplementary Materials

Supplementary Table 1: Removed items from the General Inner Speech Questionnaire (middle column) to form the adapted version used in the current study (right column), organized by General Inner Speech Questionnaire section (left column). GISQ = General Inner Speech Questionnaire.

<i>Section</i>	<i>Items included in original GISQ but not included in Adapted GISQ</i>	<i>Items included in both Original and Adapted GISQ</i>
<i>I talk to myself about (content)</i>	• Behavior • Sleep • Beliefs • Music • Physical activity • Hypothetical situations • Family • Friends • Education • Appearance • Motives • Work • Ideas • Immediate environment • Intimate partner	• Positive emotions • Negative emotions • What others think about me* (modified wording) • Errands • Who I am • Bodily sensations • Preferences • People around me • Past events • What I want to do* (modified wording) • Food / what to eat • Current events • Health • Financial situation • Future • Relationships
<i>I talk to myself in order to (function)</i>	None removed	• Replay past conversations • Solve problems • Motivate myself • Control emotions • Think critically • Plan • Rehearse ahead of time • Avoid saying out loud what I want to say • Concentrate • Listen to my own voice • Cope
<i>I talk to myself when (activities)</i>	• Ruminating • Alone/bored • Studying • Performing hygiene • At school • At work • Walking • Mind wandering • Orienting myself • Taking transit	• Meeting people • Reading* (modified wording) • I want to rely on myself • Remembering • Driving

Supplementary File 1 (S1): The adapted General Inner Speech Questionnaire, complete with pictures. It includes sections on the contents (“I talk to myself about...”), functions (“I talk to myself in order to...”, and activities (“I talk to myself when...”) of inner speech use. It can be found at <https://osf.io/b6vxk/> under “Files.”

Chapter 4: Problem Solving and Inner Speech in People with and without Aphasia

Abstract

Background: The mechanisms behind connections in problem solving and language abilities are unknown (Baldo et al., 2015), and as the “little voice in the head,” inner speech is situated to be one such mechanism. People with aphasia experience varying degrees of preserved inner speech despite the severity of the language disorder (Alexander et al., 2023; Geva, Bennett, et al., 2011; Stark et al., 2017). By studying this connection in people with aphasia, there is a unique opportunity to explore the role of inner speech in cognitive processes such as problem solving. This project has two aims: 1) examine the relationship between inner speech and problem-solving abilities in adults with and without aphasia and 2) investigate the effects of interfering with inner speech during problem solving.

Method: People with aphasia and age-similar healthy adults completed assessments of cognition, inner speech, and psychosocial health. People with aphasia completed assessments of language and awareness of their language deficits as well as inner speech and problem-solving tasks. To assess the connection between inner speech and problem solving, all participants completed a non-verbal problem-solving task (Wisconsin Card Sorting Test) during a dual task paradigm with three conditions: no interference (baseline), verbal interference (articulatory suppression) and non-verbal interference (finger tapping). Articulatory suppression is known to disrupt inner speech, allowing for investigation into how inner speech may facilitate problem solving.

Results: Performance on problem-solving tasks positively related to performance on an objective task of inner speech. Adults without aphasia outperformed people with aphasia on the problem-solving task, but there were interactions between groups based on inner speech ratings, and interference conditions. There was a large learning effect from the baseline condition to the interference conditions, but the verbal interference condition showed less

improvement than the non-verbal interference condition. Aphasia severity, but not anosognosia, also predicted performance for people with aphasia.

Conclusion: This project is a comprehensive investigation into the role of inner speech in problem-solving tasks in adults with and without aphasia. Using a combination of established tasks (e.g., Wisconsin Card Sorting Test, covert rhyme judgment, successful inner speech during naming) and innovative methods (e.g., articulatory suppression, self-reported outcome measures), the project contributes valuable insights into the fields of speech-language pathology, aphasiology, and cognitive neuroscience. The results inform clinical decision-making and help to better understand the importance of inner speech in aphasia recovery and cognition.

Problem Solving, Inner Speech, and Aphasia

Although aphasia is generally considered to be a disorder of language, some aspects of cognition are affected, e.g., attention, problem solving and planning (Baldo et al., 2005, 2015; Dutta et al., 2023; Fridriksson et al., 2006; Murray, 2012). Studies have found that more severe aphasia related to more impaired reasoning and problem-solving abilities (Baldo et al., 2005, 2015), citing immense variability in both verbal and nonverbal tasks not wholly attributable to language impairment (Dutta et al., 2023; Fridriksson et al., 2006; Murray, 2017).

Inner speech, the experience of speaking to oneself in one's head, has been implicated in problem-solving abilities in neurotypical adults (Baldo et al., 2005; Wallace et al., 2017) and children (Lidstone et al., 2010), as well as other clinical populations (e.g., ADHD, autism spectrum disorder) (Corkum et al., 2008; Vissers et al., 2020). In a small number of experiments with healthy adults, interfering with inner speech led to decreases in problem-solving performance (Baldo et al., 2005; Nedergaard, Christensen, et al., 2023; Wallace et al., 2017). Problem solving is one of many cognitive abilities which may rely on executive functioning (which includes abilities such as inhibitory control, cognitive flexibility, planning, and organization). Inner speech may be a domain-general tool for problem solving and other executive functioning tasks due to its relationship with many cognitive abilities (Alderson-Day & Fernyhough, 2015).

Attention, memory, and executive function impairments may co-occur with aphasia, as people with aphasia perform worse on tasks involving these skills, although the individual cognitive profiles of people with aphasia differ (Murray, 2012). People with aphasia (PWA) may have individual variability in problem solving abilities due to inner speech, the experience of speaking to oneself in one's head, as some people may use covert verbal strategies to complete cognitive tasks. Inner speech can be remarkably intact compared with other language processes, such as overt speech, yet may likewise be affected in aphasia (Alexander et al.,

2023; Geva, Bennett, et al., 2011; Stark et al., 2017). By exploring the relationship between inner speech and problem solving, researchers gain clinical insight into how important inner speech may be in the overall recovery process. For example, PWA report using inner speech often in their daily lives (Alexander et al., 2024), but it remains unclear how the themes and uses of inner speech differ between PWA and age-similar adults without brain injury. It is possible that PWA are using inner speech in unique ways as a result of their stroke and language impairment—for example, they may be relying more heavily on inner speech for tasks that have a high cognitive load and require resources that are less efficient post-stroke, such as during problem solving.

The study of inner speech and problem solving in aphasia will inform treatment – for example, a person’s communicative deficits could stem from language impairments, which may also affect the ability to perform certain problem-solving tasks. In this case, the clinician would take the traditional approach of targeting language in therapy. If a person’s communication deficits were influenced by concomitant cognitive difficulties, the clinician could add therapy to address cognition through training and compensatory strategies (Fama et al., 2024).

Understanding the root causes of communication difficulties will help to individualize treatment and support patients with unique, tailored goals. For this reason, it is important to routinely assess cognition in people with aphasia, as aphasia severity does not adequately predict the need for cognitive assessment (Murray, 2012). With overarching knowledge of a person’s cognitive-linguistic abilities, including communication, inner speech, executive functioning, memory, and attention, clinicians may employ novel techniques such as metacognitive training (Wadams et al., 2022) to increase communicative effectiveness, life participation, and independence with activities of daily living.

Measuring Inner Speech

Due to the nature of inner speech as an internal experience, it can be difficult to measure empirically (Brinthaup & Morin, 2023). Prior literature divides the methods into two approaches: objective and subjective methods. Both have been used in aphasia research, with differing goals and assumptions about the underlying mechanisms of inner speech (Fama & Turkeltaub, 2020).

Objective measures are those that aim to empirically measure inner speech using a monitored, measurable response (Alexander et al., 2023). A popular objective measurement of inner speech is covert rhyme judgment, where subjects indicate whether the words for two pictures rhyme without saying them aloud, thereby requiring access to the word form and manipulation of it (Alexander et al., 2023; Geva, Bennett, et al., 2011; Stark et al., 2017). Validation of the involvement of the language network during inner speech comes from neuroimaging studies, where lesion-symptom mapping associated impaired covert rhyme judgment (a proxy for inner speech) with damage to dorsal brain regions often associated with language production and processing: the left inferior frontal gyrus pars opercularis and the left supramarginal gyrus, and adjacent white matter (Fama et al., 2017; Geva, Jones, et al., 2011; Pillay et al., 2014).

Subjective measures rely on the introspection of the person to indicate whether inner speech is present. Subjective report meaningfully measures a covert state, as is done when measuring pain (e.g., the Wong Baker FACES Pain Scale; Wong & Baker, 1988) or tinnitus (e.g., the Tinnitus Handicap Inventory; Newman et al., 1996). As with measures of other internal, perceptual phenomena, subjective reports are an important way of measuring inner speech because these experiences are grounded in an individual's perception. A common means of eliciting a subjective report of inner speech is to present a photo to a person (e.g., picture of an apple) and have them indicate whether or not they have the correct word with accurate phonetic

information (Fama, Henderson, et al., 2019; Fama, Snider, et al., 2019). Researchers have likewise validated this method by demonstrating that presence of subjective inner speech associates with damage to left hemisphere, dorsal brain areas related to language production (Fama et al., 2017), likely because of the emphasis on the linguistic aspects of perceiving inner speech—i.e., that they have the word form in their head. Other means of subjectively measuring inner speech rely less heavily on the linguistic components of inner speech. These subjective measures tend to be questionnaires which inform researchers about the general experience of inner speech and themes of its use in daily life, which are described in more detail in Chapters 3 and 5 (Alderson-Day et al., 2018; Alexander et al., 2024; Brinthaup et al., 2009; Racy et al., 2020).

All methods of measuring inner speech have benefits and drawbacks (Alexander et al., 2023; Brinthaup & Morin, 2023), some of which are better suited to working with PWA. For example, subjective methods require self-report, which can be difficult to confirm in any population, but especially when navigating language comprehension and production deficits. The reports are leant considerable validity because they are meaningfully related with lesion data (Fama, Snider, et al., 2019). Additionally, word-level self-report (e.g., asked about having inner speech for a picture of an object) enables investigation of lexical retrieval without need for manipulation of the information, such as during a rhyme judgment between two words, requiring fewer cognitive processes. Self-report methods additionally enable researchers to examine reports of inner speech in a variety of contexts and to understand the functions and contents of inner speech.

Inner rhyme judgment (an objective method) assumes that inner speech is required to complete the task, and because it requires confrontation naming abilities (the forced naming of some concrete referent, like a picture), may be inherently difficult for PWA and may require additional cognitive processes related to executive functioning. Successful inner speech during naming (a subjective method) and inner rhyme judgment are not perfectly correlated, however,

and they seem to be measuring different aspects of the inner speech phenomenon (Alexander et al., 2023). Namely, successful inner speech during naming assesses the access to a word and its lexical/phonological components, while inner rhyme judgment requires manipulation of that information. Further, both inner rhyme judgment and subjective reports of successful inner speech during naming narrow the scope of inner speech to a very specific linguistic component – inner speech at the word level, reflecting access to lexical-semantic and phonological information. Word-level inner speech may be particularly informative in aphasia due to the nearly universal symptom of anomia (impaired word finding), but it cannot capture the full range of inner speech as it is used throughout daily life.

Indeed, evidence from psychology, philosophy and neuroscience suggests that inner speech has various dimensions (Chapter 1; Alexander & Stark, 2024), and while the linguistic system plays a role in its function, there are other relevant cognitive systems. Philosophers hypothesize that inner speech is not equivalent to language, but rather a mental use of language which often occurs consciously and without overt articulation (Langland-Hassan, 2021). Based on fMRI, which measures brain activation indirectly through blood hemoglobin metrics, inner speech activates typical language production areas in the left hemisphere, but depending on the characteristics of the particular instance of inner speech (e.g., whether it has all phonetic components, is monologue or dialogue, and is spontaneous or intentional), it may also activate right hemisphere homologues of language production, as well as the parietal lobules and bilateral inferior frontal lobes (Grandchamp et al., 2019). Through questionnaires, young people (Racy et al., 2020) and people with aphasia reported using inner speech for tasks such as problem solving and planning (Alexander et al., 2024), further motivating research linking inner speech and problem solving in aphasia.

Interfering with Inner Speech

Given that inner speech has been highly correlated with a variety of cognitive processes as well as processes related to self-monitoring (Alderson-Day & Fernyhough, 2015; Morin, 2011; Morin & Michaud, 2007), the next plausible step is to confirm directionality: Is inner speech *necessary* for the completion of these functions?

One way to examine the effect of inner speech on other processes is to interfere with inner speech during tasks. This has most often been done through dual task paradigms (Baldo et al., 2005; Nedergaard, Christensen, et al., 2023; Wallace et al., 2017). These problem-solving tasks involve interfering with inner speech through a verbal task (i.e., repeating a word aloud or recalling words through verbal rehearsal) and are contrasted with interference using a nonverbal task (i.e., foot tapping or recalling visuospatial information). The most robust evidence for the importance of inner speech on problem solving has come through these dual task paradigms (Baldo et al., 2005; Wallace et al., 2017). However, to this author's knowledge, inner speech interference tasks have not been attempted in PWA. Dual task paradigms to evaluate the impact of aphasia on attention have been widely used, however (Hula & McNeil, 2008; McNeil et al., 2004; Murray, 2000; Salis et al., 2023), suggesting the feasibility of this paradigm in PWA. Based on results from the dual task studies in PWA, it is possible that attention will be disproportionately affected in PWA, which may add to the discrepancies between PWA and cognitively healthy adults on dual tasks.

Conducting a dual task paradigm that interferes with inner speech during problem solving will enable investigation of the underlying mechanisms of problem-solving difficulties documented in PWA (Baldo et al., 2015; Dutta et al., 2023), with inner speech abilities as a potential reason for variability in performance.

Problem Solving and Aphasia

A pervasive message about aphasia is that it is a loss of language, but intelligence is intact (Ivanova & Dronkers, 2022). Some PWA disagree with this view, reporting comprehension difficulties which make it difficult to say that intellect has not been impacted (Anbar, 2022). One cognitive issue appears to be in problem-solving abilities, which are affected in people with aphasia even when non-verbal reasoning tasks are selected (Baldo et al., 2005, 2010, 2015; Hjelmquist, 1989). These deficits in problem solving and other types of executive functioning may account for conversational difficulties in PWA (Penn et al., 2010) and co-occur with functional communication deficits (Fridriksson et al., 2006).

Lesion-symptom mapping in studies of problem solving and aphasia has associated deficits in reasoning with lesions in the left middle and superior temporal gyri as well as the left inferior parietal lobule, which are brain areas served by the middle cerebral artery (Baldo et al., 2010). Indeed, left middle cerebral artery strokes most typically cause aphasia, due to the circulation of left lateral frontal, temporal, and parietal cortices. Interestingly, the left inferior parietal lobule includes the supramarginal gyrus, which has been suggested to play a pivotal role in inner speech (Geva, Jones, et al., 2011) and, more generally, to lexical-phonological information (Hickok, 2012; Hickok & Poeppel, 2004, 2007). Therefore, inner speech may be a mechanism explaining impairments in problem solving, given overlapping brain regions involved in these processes and reports that people use inner speech to solve problems, although this may differ depending on cultural background (Kim, 2002) and how intact inner speech is after stroke (Alexander et al., 2023). With the overlapping brain regions, another possibility is that a stroke could affect both inner speech and problem-solving processes independently, depending on the precise location of the injury, but they also could be interconnected, with each process supporting the other.

Related to interconnected processes, there may be a bidirectional relationship between inner speech and some cognitive processes in PWA, specifically components of executive functioning (Fama et al., 2024). With tasks measuring response inhibition, conflict monitoring/resolution, general task-switching ability, phonological control, and semantic control, Fama and colleagues (2024) found consistent relationships between each of the executive function tasks and objective tasks of inner speech (those which require both access to and manipulation of the word, such as inner rhyme judgment). With a double dissociation between inner speech and executive function abilities (i.e., there were people who performed poorly on inner speech but adequately on executive function and vice versa), the authors concluded that the two processes do not strictly rely on one another, rather the processes support one another. Inner speech supports executive function through strategies such as self-cueing during cognitive tasks, but people may use alternative strategies, especially when inner speech may be impaired. According to Fama and colleagues (2024) executive functioning abilities likely support performance on certain inner speech tasks, especially those which require mental manipulation, as there may be components of executive functioning inherent to the task.

Gaps in the Knowledge

Although studies have used lesion symptom mapping in aphasia to link deficits in problem solving with lesions to language areas (Baldo et al., 2010), it is unknown whether those participants had deficits in inner speech. There are PWA who have relatively spared inner speech compared to overt speech based on subjective ratings, reports of successful inner speech during naming, and objective inner rhyme judgment tasks (Alexander et al., 2023; Fama et al., 2017; Fama, Snider, et al., 2019; Geva, Bennett, et al., 2011). It is possible that PWA who have stronger inner speech perform better on problem-solving tasks and are more affected by interference with inner speech. To this author's knowledge, no studies have used articulatory suppression in problem-solving tasks in PWA, and this type of dual task paradigm aids in

delineating the relationship between problem solving and inner speech. Inner speech has been related to some aspects of executive functioning (Fama et al., 2024), but more complex problem-solving abilities have not been compared to performance on inner speech tasks, which would lend credence to the links between inner speech and problem solving beyond subjective reports that people use inner speech to solve problems (Alexander et al., 2024).

Specific Aims and Objectives

This project leveraged inner speech to better understand problem solving and aphasia.

1. **Examine the relationship between inner speech and problem-solving abilities in adults with and without aphasia.**

Hypothesis: Consistent with the findings from Fama et al. (2024), better inner speech (on objective measures) will relate to better problem-solving abilities.

2. **Investigate the effects of interfering with inner speech during problem solving.**

Hypothesis: Interfering with inner speech will impact performance on problem solving for both groups, but especially for those who have higher scores on inner speech tasks, i.e., those with a stronger baseline inner speech capacity.

Exploratory sub-aim 1: Cognitive abilities, aphasia severity, and reduced insight into cognitive-linguistic and language impairment (anosognosia) will mediate inner speech's involvement in problem solving, due to reduced access to language and the known relationship of self-monitoring and inner speech.

Exploratory sub-aim 2: Self-report of inner speech during problem solving tasks as well as performance on other inner speech tasks indicate strategy use during problem solving and will relate to performance during interference conditions.

Method

Materials

Cognitive-Linguistic Assessment

Quick Aphasia Battery (Wilson et al., 2018). PWA completed this standard battery, which provides scores across core language domains and an overall aphasia severity score. The Sentence Comprehension subtest served as an exclusionary measure for the PWA group, as participants with a raw score under 30 out of 48 were excluded. This battery does not sort participants into aphasia type, but rather allows for in-depth examination of relative strengths and weaknesses, as PWA rarely fit neatly into the specific aphasia types (Wilson et al., 2018). The remote version of the battery was used, and the entire battery typically took ~20 minutes to complete.

Montreal Cognitive Assessment (MoCA) (Nasreddine et al., 2005). All participants completed this assessment as a measure of cognitive abilities such as visuospatial skills, attention, memory, language, and abstract thinking. For the cognitively healthy adult (CHA) group, this was an exclusionary measure, with a cut-off for inclusion of ≥ 23 . This cutoff was lowered from the original of 26 for a lower false positive rate. Although this assessment includes subtests which rely on language (e.g., categorical word fluency, picture naming), it is often used post-stroke (Chiti & Pantoni, 2014; Cumming et al., 2011; Yan et al., 2022) but it must be noted that, due to the reliance on language, the MoCA likely underestimates cognitive abilities of PWA (Falkauskas, 2016). As such, subtest scores which are less reliant on language (e.g., visuospatial, attention) can be reported with higher confidence in the PWA group. To allow for remote administration, participants completed the visuospatial section of the assessment on their own paper and showed it to the experimenter over the videoconferencing software (Abdolahi et al., 2016; DeYoung & Shenal, 2019; liboshi et al., 2020), with the mazes portion being said out loud rather than drawn. The MoCA typically took 5-10 minutes to complete.

Visual Analogue Test assessing Anosognosia for Language impairment (VATA-L)

(Cocchini et al., 2010). Due to the relationship between inner speech and self-awareness and monitoring (Morin, 2011), the participants with aphasia and a caregiver/loved one completed this 14-item questionnaire to test insight into language-related communication difficulties. The participant's score is subtracted from their caregiver's score, providing a discrepancy score which reflects the participant's self-awareness of their language deficits. This questionnaire typically took 15-20 minutes to complete.

Executive Function and Problem-Solving Tasks

Towers of Hanoi (Lucas, 1891). To rule out a confound of globally impaired executive functioning abilities, all participants completed the Towers of Hanoi task (Figure 1). Towers of Hanoi is a test of non-verbal planning and problem solving in which there are three poles (towers), and participants must move discs from the leftmost tower to the rightmost tower in the least moves possible. There are three rules: 1) only one disc may be moved at a time, 2) larger discs may not be placed on top of smaller discs, and 3) only the top disc of a tower can be moved. Participants completed four rounds of this task, starting with one disc (to better understand how to drag and drop from the leftmost tower to the rightmost tower) and increasing to four discs. Participants had seven minutes overall to complete the four levels. Data were collected on the number of levels complete as well as the number of moves and amount of time used to complete each level.

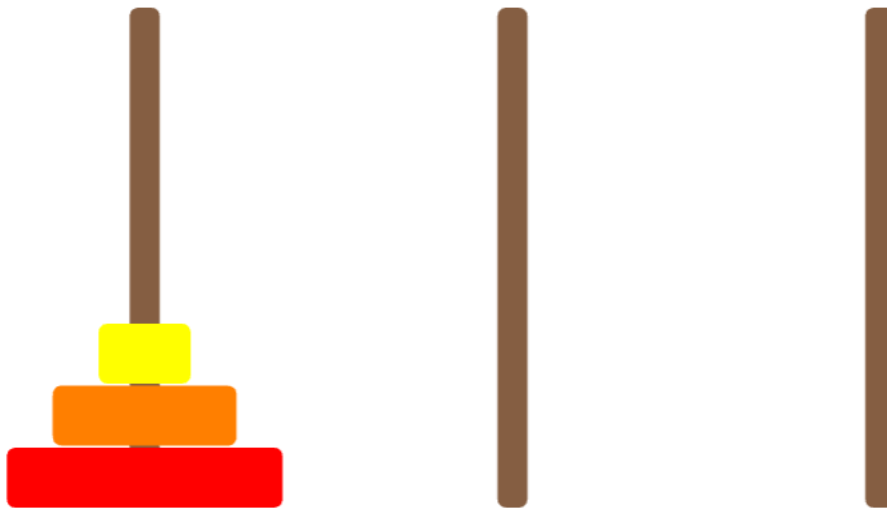


Figure 1: Example of the Towers of Hanoi, in which the participants moved the discs from the leftmost tower to the rightmost tower. This level of the task can be solved in seven moves.

Flanker Task (Eriksen & Eriksen, 1974). The Flanker Task was used to test non-verbal inhibition. After 6 practice items, participants completed 80 trials in this task: 40 congruent, 20 incongruent, and 20 neutral (Figure 2). Both accuracy and reaction time were collected. It is expected that participants will be less accurate and slower to respond in the incongruent trials, as the flanking arrows act as distractors. Participants used their keyboard to respond to this task. It took approximately 5-10 minutes.

Press **F** for
left

Press **J** for
right



Figure 2: An example of an incongruent trial in the Flanker Task, in which the arrows to either side (the “flankers”) point in the opposite direction of the arrow in the middle (the target). The correct answer in this example would be “right,” so the participant would press the J key on the keyboard.

Wisconsin Card Sorting Test (Grant & Berg, 1948). Following established procedures (Baldo et al., 2005; Nedergaard, Christensen, et al., 2023; Wallace et al., 2017), all participants completed a dual task paradigm to determine the effects of interfering with inner speech. Participants completed a computerized Wisconsin Card Sorting Test (based on Grant & Berg, 1948) (Figure 3), in which participants sorted a card into one of four groups which may match based on color, number, or shape. The participants were not told the sorting rule, but they were given feedback on their accuracy. Throughout the task, the rule changed without warning, and participants must adapt to the change. The Wisconsin Card Sorting Test has been used to assess problem solving in PWA, demonstrating feasibility, although some participants with more severe aphasia have had difficulty with the task (Baldo et al., 2005, 2015; Fridriksson et al., 2006; Purdy, 2002).

To assess the role of language in problem-solving tasks (Baldo et al., 2005, 2010, 2015; Wallace et al., 2017), there were three interference conditions during the Wisconsin Card

Sorting Test: no interference, verbal, and non-verbal. Verbal interference of inner speech used articulatory suppression, requiring the participant to repeat the sound “na” to the beat of a metronome (60 beats per minute, or one beat per second). Non-verbal interference required the participant to tap their hand on their chest to the beat of a metronome (Wallace et al., 2017). The tapping must be visible to the experimenter. All participants completed all of the conditions, for within-subjects comparison (unlike in Baldo et al. [2005], in which participants were randomly assigned to complete just one condition). The order of the two interference conditions was counterbalanced to minimize effects of learning and order, with participants randomly assigned to which interference condition (Verbal or Non-verbal) would immediately follow the baseline, and they would then complete the final interference condition. The items from the three conditions on the Wisconsin Card Sorting Test task were equivalent in difficulty. At the end of each condition, participants were asked to report how often they used inner speech on the task, with ratings of 0 (not at all), 25 (sometimes), 50 (about half of the time), 75 (most of the time), and 100 (all of the time). Due to errors in the software and connectivity issues, there were three missing data points for the inner speech rating. In each of these instances, it was only on one of the conditions for the participant, so there was available data on their individual ratings for the other conditions. For this reason, the researcher imputed the missing data points to allow for more robust data analysis. The Wisconsin Card Sorting Test took approximately 60 minutes with large time variability across participants.

You will see 4 cards near the top of the screen. Your card will be at the bottom of the screen. To match your card, click one of the cards at the top of the screen.

For example, your card could match the first card (shape), the second card (number), or the third card (color).

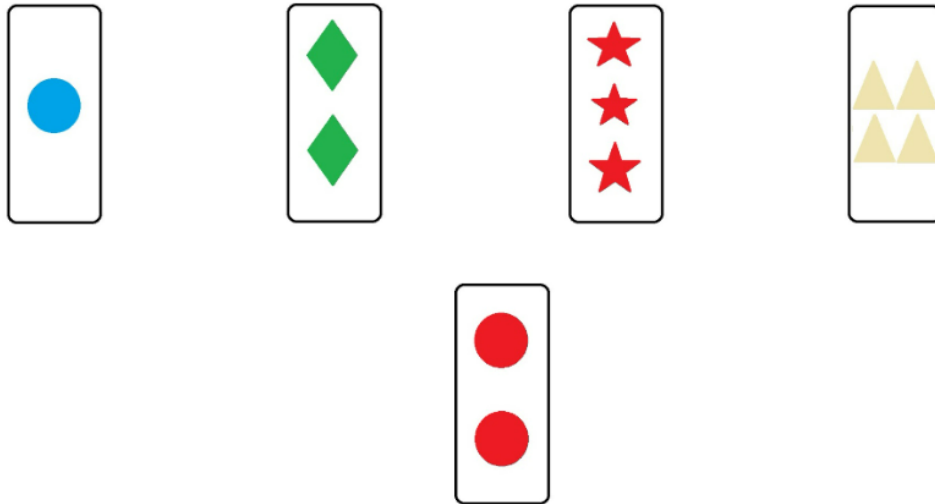


Figure 3: Example of the Wisconsin Card Sorting Test, based on Grant & Berg, 1948. This was part of the instructions given to the participants.

Inner Speech Tasks

To assess inner speech perception and abilities, a composite score from two well-established tasks of inner speech was computed (Alexander et al., 2023; Fama, Snider, et al., 2019; Geva, Bennett, et al., 2011; Stark et al., 2017). These two inner speech tasks took approximately 30 minutes.

Naming – Successful Inner Speech. Participants reported their successful inner speech during a 30-item confrontation naming task (Figure 4) – when shown a picture, participants were asked if they had the correct word for it in their heads with the right sounds in the right order (both semantic and phonetic information) (Fama, Snider, et al., 2019). They then named the picture out loud.

I can say the word for this in my head, and it sounds right.

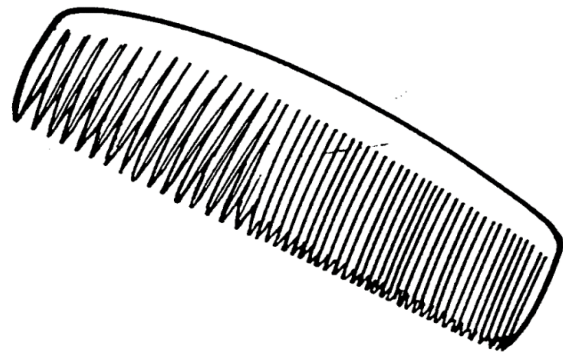


Figure 4: Example of the successful inner speech during naming task, in which the prompt on the left was shown for each picture as a reminder to indicate whether the participant can name the picture on the right (comb) silently with all the right sounds in the right order.

Inner Rhyme Judgment. Both groups completed a 30-item inner rhyme judgment task (example in Figure 5), in which they decide whether the words for two pictures rhyme and respond yes or no with a button press.

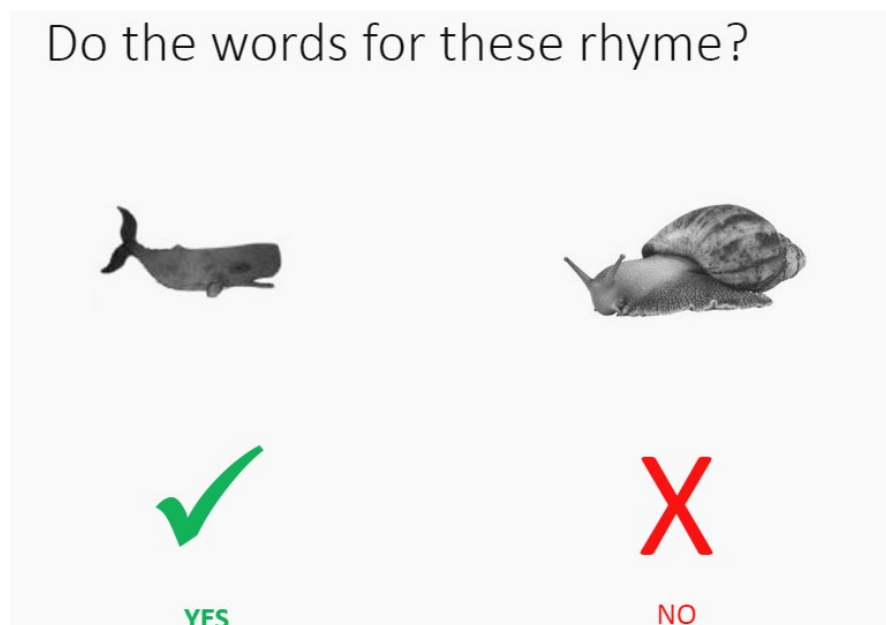


Figure 5: Example of the inner picture rhyme judgment task, in which the participants indicated whether the words for the two pictures rhyme. In this case, the correct answer would be “yes.”

Participants

Forty-seven PWA and 45 cognitively healthy adults consented to participate. To be included, PWA reported a diagnosis of aphasia from a stroke at least six months prior to testing. They also received a comprehension score of 30 or higher out of 48 on the Quick Aphasia Battery Sentence Comprehension subtest (Wilson et al., 2018) to ensure ability to understand task instructions. Participants in the CHA group were age- and sex-similar to the adults with aphasia and had normal cognition (confirmed by receiving a score of 23 or higher on the Montreal Cognitive Assessment, lowered from the original cutoff due to the virtual administration and updated guidelines [Carson et al., 2018; Elkana et al., 2020]). Both groups were aged 40-80 years and reported no other neurological disorder or disease or history of developmental language impairment. They also reported adequate vision and hearing for virtual testing.

From the original numbers, one PWA did not have the necessary equipment to complete the experimental tasks, one did not complete the full experiment in one session and did not return for a second session, six were excluded due to suspected malingering (they may have been feigning a language disorder, as the presentations of their language deficits were unusual), and six had significant right-sided hemiplegia or hemiparalysis which impacted the analysis of the WCST Non-verbal condition. This resulted in a final sample of **33 PWA** for analysis. Three of the cognitively healthy adults were also excluded from analysis, two due to cognitive concerns (scoring below a 23 on the Montreal Cognitive Assessment and exhibiting further difficulties with other cognitive tasks) and one due to connectivity issues which greatly inflated reaction times on the WCST tasks. This resulted in a final sample of **42 cognitively healthy adults** for analysis. Table 1 shows demographic factors of the two groups, with statistical comparisons where applicable. Additionally, both groups were a majority white (CHA:

97.6% White, 2.4% Black or African American; PWA: 75.8% White, 15.2% Black or African American, 6.1% Asian, 3% American Indian/Alaska Native) and none of the participants in either group were of Hispanic or Latino origin.

Table 1: Demographics of the two groups. CHA: Cognitively healthy Adults, PWA: People with Aphasia

	CHA (N=42)	PWA (N=33)	Statistics
Age			
Mean (SD)	63.8 (8.00)	62.2 (9.45)	$t(63) = -0.78$,
Median [Min, Max]	63.9 [42.1, 76.8]	61.4 [42.0, 77.4]	$p = .44$
Gender			
Female	28 (66.7%)	16 (48.5%)	$\chi^2(1) = 1.83$,
Male	14 (33.3%)	17 (51.5%)	$p = .18$
Education (in years)			
Mean (SD)	17.4 (2.22)	17.0 (2.45)	$t(65) = -0.65$,
Median [Min, Max]	17.5 [13.0, 24.0]	16.0 [12.0, 24.0]	$p = .52$
Bilingual			
No	34 (81.0%)	28 (84.8%)	Not computed:
Yes	8 (19.0%)	5 (15.2%)	Not enough cases
Quick Aphasia Battery			
Mean (SD)	NA	8.20 (1.17)	NA
Median [Min, Max]	NA	8.53 [4.16, 9.60]	
Self-Reported Depression or Anxiety			
No	34 (81.0%)	21 (63.6%)	$\chi^2(1) = 2.02$,
Yes	8 (19.0%)	12 (36.4%)	$p = .16$

Procedure

Participants completed a pre-screening form on REDCap (Harris et al., 2019), which the researchers reviewed prior to scheduling a session. There were several tasks involved in this study. Participants met with the researchers through videoconferencing on Microsoft Teams. They completed the informed consent process, with multimodal communication through visuals

on slides and verbal descriptions. After consenting, participants completed a demographic intake form to confirm their eligibility and gain information about their health, education, occupation, and medical history relevant to brain injury.

The researchers then introduced the concept of inner speech in more depth, with an interactive presentation in which the researcher and participant gave descriptions and examples of what inner speech is and is not (see Appendix Table 1 for a summary of the descriptions). Participants then completed standardized assessments described in Materials (i.e., Quick Aphasia Battery and VATA-L for PWA; Montreal Cognitive Assessment for all participants). Participants were then directed to Gorilla, an online cloud software program where the experiment was hosted (see Figure 6 for a schematic of the experimental procedure). The participants shared their screen on the video call to allow the researchers to explain each task, troubleshoot technical issues, and answer clarifying questions. Participants were randomly sorted into one of the two experiment blocks – the WCST block or the additional inner speech and problem-solving block, after which they would complete the other block. The order of tasks in the additional inner speech and problem-solving block was randomized, while in the WCST block, the baseline condition was always first, followed by the Verbal and Non-verbal conditions in random order.

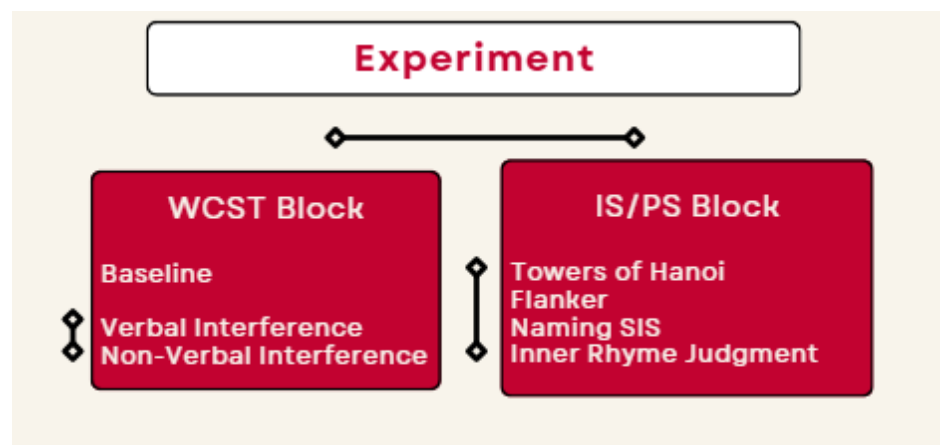


Figure 6: Diagram of the experimental procedure. WCST = Wisconsin Card Sorting Test. IS/PS = inner speech and problem solving. SIS = successful inner speech. Lines with diamonds at the end indicate randomization. After completing cognitive-linguistic assessment, participants were randomly sorted into either the WCST Block or IS/PS Block, which had further randomization.

Data Analysis

Hypothesis 1: Better inner speech will relate to better problem-solving abilities.

To examine this relationship, correlations and linear regressions were completed between inner speech tasks (objective: inner rhyme judgment, subjective: successful inner speech during naming) and problem-solving tasks (Towers of Hanoi, Flanker Test of Inhibition, and baseline WCST).

Hypothesis 2: Interfering with inner speech will impact performance on problem solving for both groups, but especially for those who have higher scores on inner speech tasks.

Prior to determining the influence of interfering with inner speech, a linear model was completed to predict participants' inner speech rating on the WCST by the interference type. This was intended to ensure that articulatory suppression did in fact interfere with inner speech on the verbal interference condition based on the perceptions of the participants.

Model Creation and Comparison

Linear modelling was used to compare the influence of verbal interference (articulatory suppression) and non-verbal interference on the WCST. Four theoretically-based models were created to attempt to explain variance in accuracy on the WCST. A Rationalized Arcsine Unit (RAU) transform was used to convert the accuracy data into a linear scale and combat any possible ceiling effects. Due to the within-subjects design, the researchers controlled for the participant and which interference condition was completed first in each model. Descriptions of the models are provided in Table 2.

Table 2: Predicting accuracy on the WCST. Descriptions of all models, including fixed and random effects as well as interactions.

Model	Descriptor	Sample	Fixed Effects	Interactions Tested	Random Effect
1	Base	All participants	- Interference Type (verbal vs. non-verbal) - Group (PWA vs. CHA) - Condition order (verbal first; non-verbal first)	Group x Interference Type	Participant ID
2	Inner Speech	All participants	- All Model 1 predictors - Inner speech assessments	- Group x Interference Type - Interference Type x Inner Speech Rating	Participant ID
3	Demographics/ Cognition	All participants	- All Model 1 predictors - Inner speech assessments - Age - Education - MoCA total score (out of 30)	- Group x Interference Type - Interference Type x Inner Speech Rating	Participant ID
4	Full	All participants	- All Model 1 predictors - Significant predictors from Model 2 - Significant predictors from Model 3 - Executive function tasks	- Group x Interference Type - Interference Type x Inner Speech Rating	Participant ID
PWA	PWA-only	PWA group	- PWA-specific assessments (aphasia severity; anosognosia) - Inner speech assessments - Interference Type - Demographics - Condition order (verbal first; non-verbal first)	Interference Type x Inner Speech Rating	Participant ID

▪ Executive function tasks = Flanker (incongruent correct count; higher scores are better), Towers of Hanoi (levels completed; higher scores are better)

▪ Other cognitive tests = Montreal Cognitive Assessment (MoCA), scored out of 30 (higher scores are better)

▪ Inner speech assessments = Picture-rhyme judgment (higher scores are better); Naming SIS (higher scores are better); Inner Speech Rating (self-reported percentage that they used inner speech on the task, 0%, 25%, 50%, 75%, 100% of the time)

▪ Demographics = Age (years), Education (years)

▪ Interference type = Comparing accuracy on baseline to Verbal and Non-verbal interference conditions

▪ Condition order = Variable describing whether Verbal or Non-verbal condition followed baseline

▪ PWA-specific assessments = Quick Aphasia Battery total score (out of 10; higher scores are better), Visual Analogue Test of Anosognosia – Language (VATA-L; a reversed score, such that smaller scores are better)

The base model tested the primary hypothesis, that interference type (verbal or non-verbal) will impact performance for both groups, but there may be an interaction (i.e., PWA may be more affected by interference). The subsequent models added to the base model to address inner speech, demographics, and cognition. These models were systematically compared with analyses of variance (ANOVA) to determine the model which best explains the participants' performance on the WCST (i.e., if model two improved on the base model, then model two became the standard of comparison for the next model iteration). The ANOVA technique compared the F-statistics and p-values of the models to see if the more complex model provided a significantly better fit to the data.

The PWA group was then examined separately to determine the influence of language abilities (overall score on the Quick Aphasia Battery) and awareness of language deficits (discrepancy score on the VATA-L) on their performance (accuracy) on all three conditions on the WCST, along with education, age, and interference type. This was also completed through linear modelling.

Results

The PWA group was significantly less accurate than the CHA group on most executive function baseline tasks: Number of levels complete on the Towers of Hanoi (but not the number of moves used to complete those levels), and in all conditions of the WCST. The two groups did not significantly differ in performance on the incongruent trials of the Flanker task, as both groups performed well on this test of inhibition. Similarly, inner speech was impaired in the PWA group when measured objectively through inner rhyme judgment, but the PWA group did not significantly differ on subjective successful inner speech from the CHA group. The similarity in successful inner speech is likely because both groups reported being able to name nearly every item in their heads, although there were a few participants in the PWA group who had less successful inner speech. See Table 2 for descriptive statistics and group comparisons.

Table 2: Descriptive Statistics and Group Comparisons on Main Outcome Measures. CHA = cognitively healthy adults, PWA = people with aphasia, WCST = Wisconsin Card Sorting Test.

	CHA (N=42)	PWA (N=33)	Statistics
WCST Baseline Accuracy (out of 64)			
Mean (SD)	51.9 (3.37)	39.3 (11.6)	$t(36) = 6.05$
Median [Min, Max]	52.5 [43.0, 58.0]	42.0 [15.0, 54.0]	$p < .001$
WCST Verbal Interference Accuracy			
Mean (SD)	52.0 (3.16)	41.8 (11.4)	$t(36) = 5.01$
Median [Min, Max]	52.0 [45.0, 57.0]	46.0 [16.0, 54.0]	$p < .001$
WCST Non-Verbal Interference Accuracy			
Mean (SD)	52.4 (3.30)	41.5 (11.4)	$t(36) = 5.27$
Median [Min, Max]	53.0 [40.0, 58.0]	43.0 [12.0, 58.0]	$p < .001$
Successful Inner Speech during Naming – Percent Correct			
Mean (SD)	100 (0)	96.7 (10.5)	$t(32) = 1.82$
Median [Min, Max]	100 [100, 100]	100 [40.0, 100]	$p = .08$
Inner Rhyme Judgment – Percent Correct			
Mean (SD)	92.3 (4.98)	76.3 (15.5)	$t(37) = 5.72$
Median [Min, Max]	92.7 [78.1, 100]	78.1 [41.5, 95.1]	$p < .001$
Flanker Incongruent Trials Accuracy (out of 20)			
Mean (SD)	20.0 (0.216)	19.2 (2.33)	$t(32) = 1.74$
Median [Min, Max]	20.0 [19.0, 20.0]	20.0 [7.00, 20.0]	$p = .09$
Towers of Hanoi – Levels Complete			
Mean (SD)	3.86 (0.354)	3.58 (0.561)	$t(51) = 2.52$
Median [Min, Max]	4.00 [3.00, 4.00]	4.00 [2.00, 4.00]	$p = .02$
Towers of Hanoi – Total Number of Moves on Levels 1 through 3			
Mean (SD)	13.7 (4.48)	14.8 (4.09)	$t(71) = -1.16$
Median [Min, Max]	11.0 [11.0, 30.0]	13.0 [11.0, 29.0]	$p = .25$

Hypothesis 1: Relationship between Inner Speech and Problem-Solving Variables

As anticipated, the CHA group neared ceiling on both inner speech tasks. For the PWA group, performance on the inner rhyme judgment task was significantly correlated with the cognitive variables of number correct on the baseline condition of the WCST $r(37) = .35$, $p = .03$, FDR-adjusted $p = .059$, levels complete on the Towers of Hanoi $r(37) = .51$, $p < .001$, FDR-adjusted p

$< .001$, and the number correct on the incongruent items of the Flanker task $r(37) = .43$, $p = .006$, FDR-adjusted $p = .02$. Successful inner speech during naming was not significantly correlated with performance on any of the three baseline cognitive measures (all p 's greater than .4). Figure 7 shows the distribution of the scores along with a line representing the correlation between the two inner speech tasks and each cognitive variable.

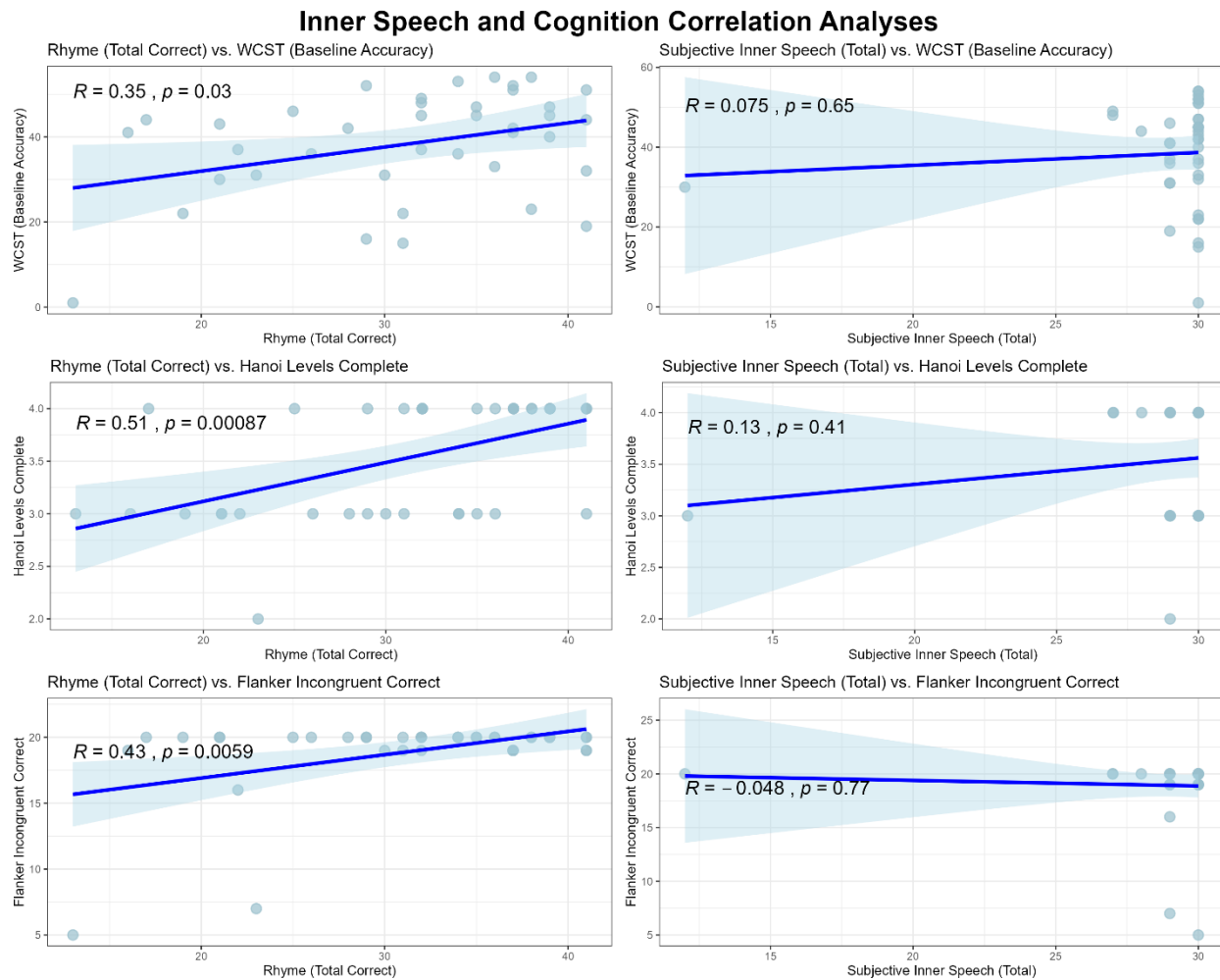


Figure 7: Correlation Analyses of Inner Speech and Cognitive Measures for the PWA Group.

The statistics reported in these figures are Pearson's R , along with the uncorrected p -values.

Hypothesis 2: Influence of Interfering with Inner Speech

Predicting Inner Speech

As depicted in Figure 8, the verbal interference condition significantly impacted the perception of inner speech. The verbal interference condition significantly predicted a lower subjective rating of the amount of speech experienced during the task (a reminder: a rating of 0, 25, 50, 75, or 100 could be selected immediately after the task): ($\beta = -15.48$, $SE = 4.56$, $t(146) = -3.40$, $p < .001$). Perhaps surprisingly, the non-verbal interference condition also negatively impacted perceived frequency of inner speech during the WCST, although to a lesser degree ($\beta = -9.52$, $SE = 4.56$, $t(146) = -2.09$, $p = .04$). Group membership did not significantly predict inner speech rating ($\beta = -10.44$, $SE = 7.23$, $t(136) = -1.45$, $p = .152$), and there was no significant interaction between group and interference type (all p 's $> .2$).

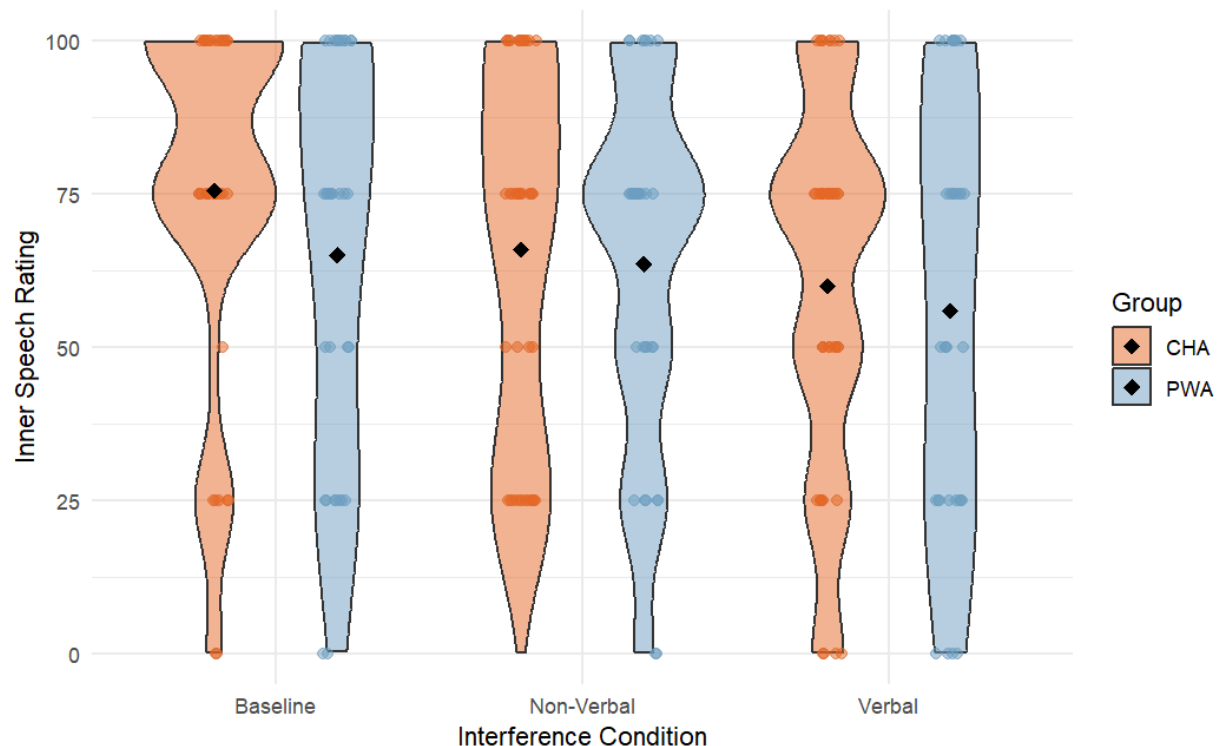


Figure 8: Inner Speech Ratings on each condition by Group. Participants answered the question: “How often did you use inner speech during this task?” and had a scale of five rating options from 0-100% of the time.

Predicting Performance on the WCST

As described in the data analysis section, four linear models were used to predict accuracy on the WCST.

Base Model: In Model 1, which used the interaction between group and interference type to predict accuracy, only group membership was a significant predictor, with the PWA group performing worse than controls. Neither interference type nor its interaction with group were significant predictors of accuracy. Additionally, the order of the interference conditions did not significantly influence performance.

This model was fit using a linear mixed-effects approach with Restricted Maximum Likelihood (REML, criterion = 1765.3) and included a random intercept for participant. See Table 3 for the statistics on all predictors in Model 1.

Table 3: Model 1 – Predicting WCST Accuracy with Group Membership and Interference Type

Predictor	B	SE	df	t	p	95% CI
Fixed Effects						
(Intercept)	139.74	2.84	107	49.25	< .001	[134.24, 145.25]
Group (PWA vs. CHA)	-25.18	3.64	125	-6.90	< .001	[-32.25, -18.11]
InterferenceType (Nonverbal vs. Baseline)	0.94	2.15	146	0.44	.661	[-3.24, 5.12]
Interference Type (Verbal vs. Baseline)	0.17	2.15	146	0.08	.938	[-4.01, 4.35]
Group × InterferenceType (Nonverbal)	3.44	3.24	146	1.06	.290	[-2.86, 9.74]
Group × InterferenceType (Verbal)	4.57	3.24	146	1.41	.160	[-1.73, 10.87]
ConditionOrder1st (Motor First)	-1.73	3.12	72	-0.55	.580	[-7.81, 4.34]
Random Effects						
Variance	SD					
Record ID (Intercept)	148.45	12.18				
Residual	96.82	9.84				

Model fit: AIC = 1783.30; BIC = 1814.05; logLik = -882.65 (df = 9)

PWA = people with aphasia; CHA = cognitively healthy adults. Bolded values in the p column indicate significant predictors at $p < .05$, while italicized values would indicate significant predictors at $p < .1$.

Inner Speech Model: Model 2 predicted accuracy from inner speech ratings (amount of inner speech used during each condition), successful inner speech (subjective measure of naming), inner rhyme judgment, the interaction between group and interference type, the interaction between inner speech ratings and interference type, and the order in which the participant completed each condition. Table 4 shows statistics on the predictors of Model 2.

Table 4: Model 2 – Predicting WCST Accuracy with Inner Speech

Predictor	B	SE	df	t	p	95% CI
Fixed Effects						
(Intercept)	65.73	22.82	75	2.88	.005	[22.11, 109.40]
Group (PWA vs. CHA)	-18.76	4.10	107	-4.58	< .001	[-26.59, -10.93]
InterferenceType (Nonverbal vs. Baseline)	14.33	4.49	151	3.19	.002	[5.65, 22.96]
Interference Type (Verbal vs. Baseline)	5.77	4.15	150	1.39	.167	[-2.22, 13.79]
Group × InterferenceType (Nonverbal)	1.78	3.15	144	0.57	.572	[-4.28, 7.85]
Group × InterferenceType (Verbal)	3.48	3.15	144	1.11	.271	[-2.59, 9.54]
Condition Order 1st (Motor First)	-1.69	3.06	71	-0.55	.581	[-7.54, 4.16]
Inner Speech Rating	0.15	0.04	190	3.32	.001	[0.06, 0.23]
Interference Type (Nonverbal) × Inner Speech	-0.18	0.06	153	-3.23	.002	[-0.29, -0.07]
Interference Type (Verbal) × Inner Speech	-0.06	0.05	151	-1.06	.291	[-0.16, 0.05]
SIS Total	1.41	0.77	71	1.84	.070	[-0.06, 2.88]
Rhyme Correct	0.54	0.34	70	1.56	.122	[-0.12, 1.19]
Random Effects						
	Variance	SD				
Record ID (Intercept)	133.73	11.56				
Residual	89.55	9.46				

Model fit: AIC = 1782.20; BIC = 1830.03; logLik = -877.10 (df = 14)

PWA = people with aphasia; CHA = cognitively healthy adults; SIS = successful inner speech during naming. Bolded values in the p column indicate significant predictors at $p < .05$, while italicized values indicate significant predictors at $p < .1$.

In this model, group membership remained a significant predictor, with the PWA group performing worse than controls. Non-verbal interference was also a significant predictor of better performance, whereas verbal interference was not. Inner speech ratings significantly predicted accuracy, while successful inner speech and inner rhyme judgment showed marginal or non-significant effects.

Additionally, the interaction between non-verbal interference and inner speech rating was significant, suggesting that higher inner speech ratings were associated with reduced accuracy under non-verbal interference conditions. The corresponding interaction for verbal interference was not significant. Neither the interactions between group and interference type nor the order of condition presentation were significant.

This model was fit using a linear mixed-effects approach with REML (REML criterion = 1754.2) and included a random intercept for participant. Model 2 significantly improved upon Model 1 ($\chi^2(5) = 24.42, p < .001$) based on an ANOVA comparison, suggesting that inner speech measures contributed meaningful explanatory power. Figure 9 shows the relationship between inner speech ratings and accuracy on each of the conditions of the WCST, showing the interaction between interference condition and inner speech ratings. Especially on the baseline and verbal interference conditions, the perception of more frequent inner speech during the task predicted better accuracy.

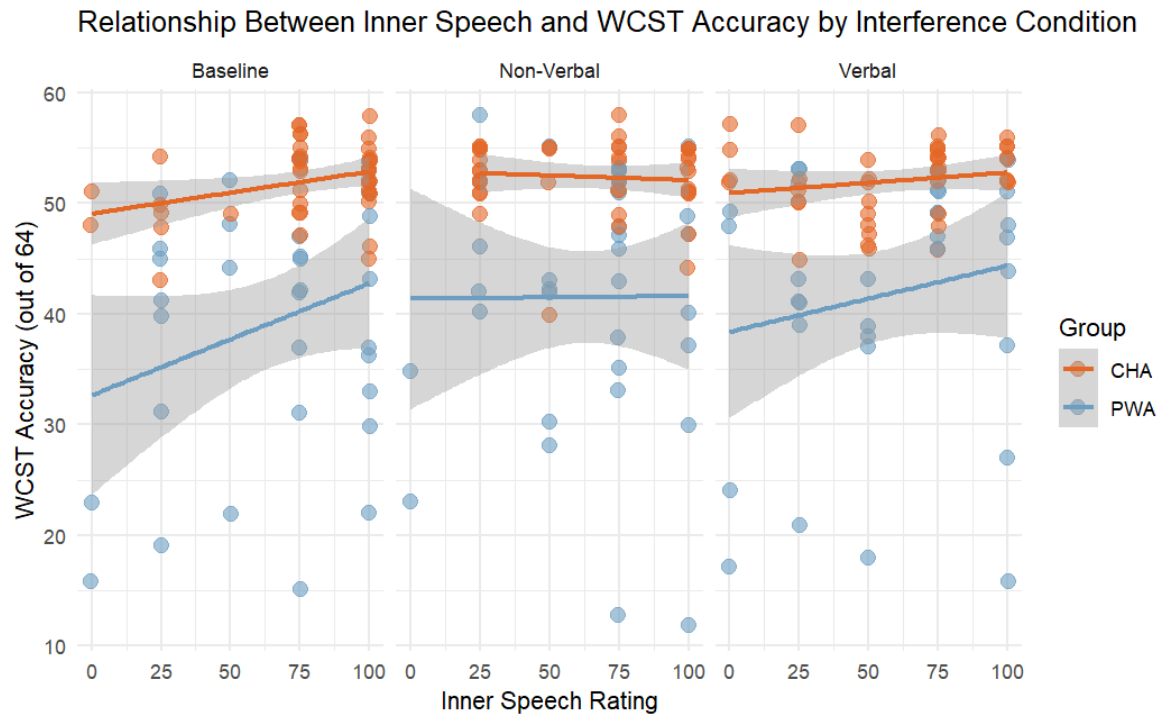


Figure 9: Inner Speech Rating and Wisconsin Card Sorting Test accuracy, separated by interference condition. Participants were asked how often they used inner speech during the task, from 0-100%.

Inner Speech and Demographics Model: Model 3 extended Model 2 by including the interaction between group and interference type, the order in which the participant completed each condition, successful inner speech, the interaction between interference type and inner speech ratings, inner rhyme judgment, age, education, and the participant's score on the MoCA. Belonging to the PWA group significantly predicted worse accuracy. The non-verbal interference condition continued to predict better accuracy than the baseline condition, while the verbal interference condition did not significantly differ from baseline. Table 5 shows the statistics on the predictors of Model 3.

Table 5: Model 3 – Predicting WCST Accuracy with Inner Speech, Demographics, and Basic Cognition

Predictor	B	SE	df	t	p	95% CI
Fixed Effects						
(Intercept)	71.70	25.85	70	2.77	.007	[23.31, 120.06]
Group (PWA vs. CHA)	-13.39	4.39	96	-3.05	.003	[-21.62, -5.15]
Interference Type (Nonverbal vs. Baseline)	13.88	4.49	151	3.09	.012	[5.12, 22.44]
Interference Type (Verbal vs. Baseline)	5.50	4.15	151	1.33	.187	[-2.48, 13.50]
Group × InterferenceType (Nonverbal)	1.85	3.15	144	0.59	.557	[-4.21, 7.92]
Group × InterferenceType (Verbal)	3.53	3.15	144	1.12	.263	[-2.53, 9.60]
Condition Order 1st (Motor First)	-1.70	2.96	68	-0.58	.565	[-7.24, 3.82]
Inner Speech Ratings	0.14	0.04	191	3.17	.002	[0.06, 0.22]
InterferenceType (Nonverbal) × Inner Speech	-0.18	0.06	153	-3.13	.002	[-0.28, -0.07]
InterferenceType (Verbal) × Inner Speech	-0.05	0.05	152	-1.01	.313	[-0.15, 0.05]
SIS Total	0.54	0.80	68	0.68	.499	[-0.95, 2.03]
Rhyme Correct	0.03	0.37	67	0.09	.928	[-0.66, 0.73]
Age	-0.05	0.17	67	-0.27	.786	[-0.36, 0.27]
Education (in years)	-0.14	0.65	67	-0.21	.833	[-1.36, 1.08]
MoCA	1.61	0.55	67	2.94	.004	[0.59, 2.64]
Random Effects						
	Variance	SD				
Record ID (Intercept)	120.90	10.99				
Residual	89.53	9.46				

Model fit: AIC = 1779.66; BIC = 1837.73; logLik = -872.83 (df = 17)

PWA = people with aphasia; CHA = cognitively healthy adults; SIS = successful inner speech during naming. Bolded values in the p column indicate significant predictors at $p < .05$, while italicized values would indicate significant predictors at $p < .1$.

Inner speech ratings remained a significant predictor of accuracy, suggesting that higher inner speech ratings were associated with better performance. Additionally, higher MoCA scores

were associated with better accuracy. The interaction between non-verbal interference and inner speech rating was significant, indicating that under non-verbal interference conditions, higher inner speech ratings were associated with lower accuracy. The corresponding interaction for verbal interference was not significant. Other demographic factors, including age and years of education, were not significant predictors of accuracy. Similarly, inner rhyme judgment and condition order were not significant predictors. The interaction between group and interference type remained non-significant for both non-verbal and verbal interference.

This model was fit using a linear mixed-effects approach with REML (REML criterion = 1745.7) and included a random intercept for participant. Model 3 significantly improved upon Model 2 ($\chi^2(3) = 9.48, p = .024$) based on an ANOVA comparison, indicating that the addition of demographic factors (including the MoCA, a cognitive test) contributed meaningful explanatory power.

Baseline Cognition Model: Finally, Model 4 incorporated the interaction between group and interference type, the order in which the participant completed each condition, the interaction between interference type and inner speech ratings, the participant's MoCA score, the number of incongruent items correct on the Flanker task, and the number of levels completed on the Towers of Hanoi. Table 6 shows the statistics on the predictors in Model 4.

Table 6: Model 4 – Predicting WCST Accuracy with Inner Speech and Cognition/Executive Function

Predictor	B	SE	df	t	p	95% CI
Fixed Effects						
(Intercept)	77.34	19.67	74	3.93	< .001	[39.99, 114.70]
Group (PWA vs. CHA)	-13.19	4.16	103	-3.17	.002	[-21.10, -5.28]
InterferenceType (Nonverbal vs. Baseline)	13.92	4.48	152	3.11	.002	[5.21, 22.51]
Interference Type (Verbal vs. Baseline)	5.54	4.15	151	1.34	.183	[-2.44, 13.54]
Group × InterferenceType (Nonverbal)	1.85	3.14	144	0.59	.557	[-4.21, 7.92]
Group × InterferenceType (Verbal)	3.53	3.14	144	1.12	.264	[-2.54, 9.59]
ConditionOrder1st (Motor First)	-1.34	2.82	70	-0.47	.637	[-6.70, 4.02]
Inner Speech	0.14	0.04	193	3.19	.002	[0.06, 0.22]
InterferenceType (Nonverbal) × Inner Speech	-0.18	0.06	154	-3.15	.002	[-0.28, -0.07]
InterferenceType (Verbal) × Inner Speech	-0.05	0.05	152	-1.03	.306	[-0.15, 0.05]
MoCA sum	1.61	0.47	69	3.45	.001	[0.72, 2.49]
Flanker Incongruent Correct	-0.49	1.03	69	-0.48	.631	[-2.45, 1.45]
Hanoi Levels Complete	4.25	3.55	69	1.20	.235	[-2.49, 10.99]
Random Effects						
	Variance	SD				
Record ID (Intercept)	114.94	10.72				
Residual	89.51	9.46				

Model fit: AIC = 1769.05; BIC = 1820.29; logLik = -869.53 (df = 15)

PWA = people with aphasia; CHA = cognitively healthy adults. Bolded values in the p column indicate significant predictors at $p < .05$, while italicized values would indicate significant predictors at $p < .1$.

The significant predictors remained the same as in Model 3. Belonging to the PWA group significantly predicted worse accuracy. The non-verbal interference condition predicted better accuracy than the baseline condition, while the verbal interference condition was not significantly different from baseline. Performance on the WCST separated by group and condition is depicted in the violin plot in Figure 10, showing the distribution of performance.

Higher inner speech ratings continued to predict better accuracy, as did higher MoCA scores. The interaction between non-verbal interference and inner speech rating remained significant, indicating that under non-verbal interference conditions, higher inner speech ratings were associated with lower accuracy. The interaction between verbal interference and inner speech was not significant. Neither the number of incongruent items correct on the Flanker task nor the number of levels completed on the Towers of Hanoi significantly predicted accuracy.

This model was fit using a linear mixed-effects approach with REML (REML criterion = 1739.1) and included a random intercept for participant. Model 4 did not significantly improve upon Model 3 ($\chi^2(2) = 0.00$, $p = 1.00$), indicating that the additional cognitive measures did not contribute to explaining accuracy beyond the predictors already included in Model 3. Since the inclusion of cognitive control measures (Flanker and Towers of Hanoi) did not significantly improve the model, **Model 3**, the inner speech and demographics model, remains the best-fitting model for predicting accuracy. Model 3 does include the cognitive measure of the MoCA, which appears to capture the participants' baseline cognitive abilities without the need for additional baseline cognitive measures.

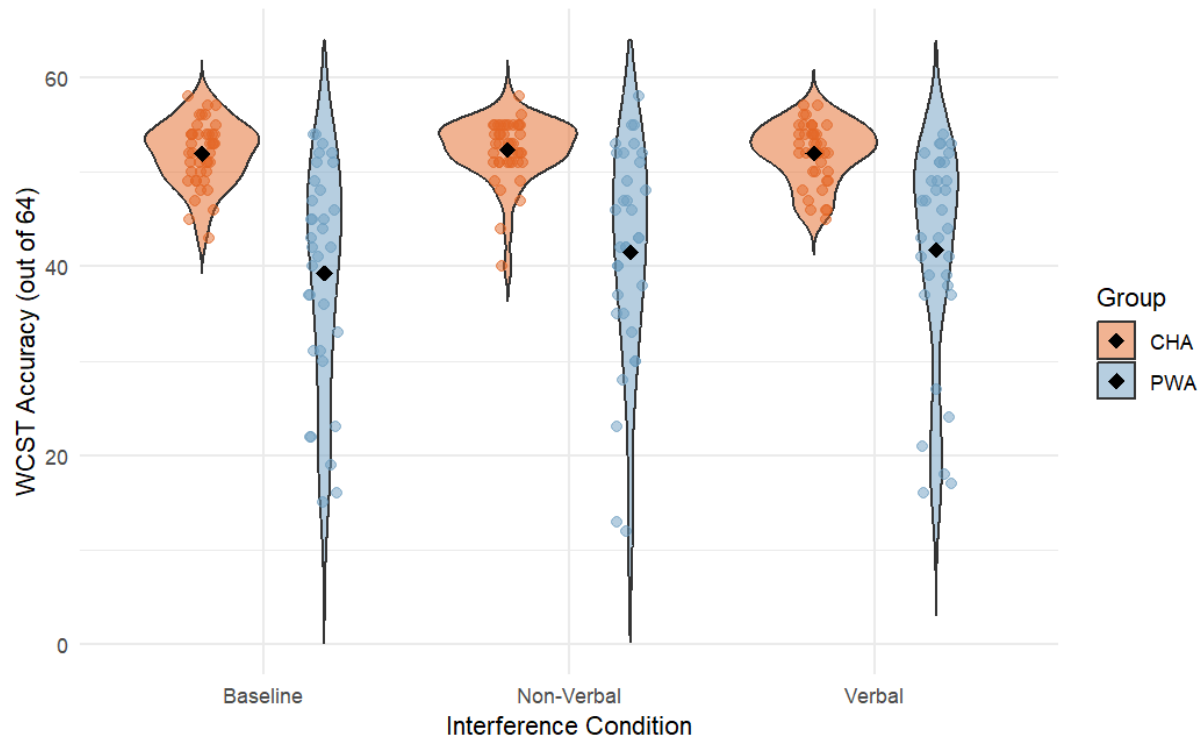


Figure 10: Performance on the Wisconsin Card Sorting Test based on group and interference conditions. CHA: cognitively healthy adults, PWA: people with aphasia, Non-verbal: hand tapping during the task, Verbal: saying “na” during the task. Black diamonds represent the means.

Role of aphasia severity and anosognosia

A separate linear mixed-effects model was conducted for PWA to examine the influence of interference type, inner speech rating, aphasia severity, anosognosia (VATA-L discrepancy score), and the order of interference conditions on accuracy in the WCST. The model, fit using REML (REML criterion = 681.9), included a random intercept for participant. Table 7 shows the statistics on the predictors in the PWA model.

Table 7: PWA Model – Predicting WCST Accuracy with Language Abilities and task-based Inner Speech

Predictor	B	SE	df	t	p	95% CI
Fixed Effects						
(Intercept)	48.61	24.02	31	2.02	.052	[3.57, 93.76]
InterferenceType (Nonverbal vs. Baseline)	23.43	7.01	61	3.34	.001	[9.82, 36.73]
Interference Type (Verbal vs. Baseline)	12.61	6.24	61	2.02	.048	[0.59, 24.51]
Inner Speech Rating	0.24	0.08	84	2.81	.006	[0.08, 0.40]
QAB Overall	6.38	2.79	28	2.29	.030	[1.13, 11.61]
VATA-L Discrepancy	-0.68	0.41	28	-1.67	.106	[-1.45, 0.09]
Condition Order 1st (Motor First)	-3.46	6.47	28	-0.54	.597	[-15.59, 8.72]
Interference Type Nonverbal x Inner Speech	-0.28	0.10	61	-2.88	.005	[-0.47, -0.09]
Interference Type Verbal x Inner Speech	-0.09	0.09	61	-1.04	.302	[-0.26, 0.08]
Random Effects						
	Variance	SD				
Record ID (Intercept)	285.9	16.91				
Residual	135.1	11.62				

Model fit: AIC = 817.50; BIC = 845.82; logLik = -397.75 (df = 11)

PWA = people with aphasia; QAB = Quick Aphasia Battery; VATA-L = Visual-Analogue Test assessing Anosognosia for Language Impairment. Bolded values in the p column indicate significant predictors at $p < .05$, while italicized values would indicate significant predictors at $p < .1$.

For both verbal and non-verbal interference conditions, there was increased accuracy after completing the baseline condition, demonstrating a significant learning effect. Inner speech rating was a significant predictor, with higher reported inner speech associated with improved accuracy. Importantly, an interaction between non-verbal interference and inner speech was observed, such that higher inner speech ratings were associated with a smaller benefit from the

learning effect during the non-verbal interference condition. Aphasia severity, as measured by the QAB Total Score, was also a significant predictor of accuracy, indicating that individuals with more severe aphasia performed worse on the WCST. However, anosognosia (VATA-L discrepancy score) was not a significant predictor of performance, nor was the order of the interference conditions.

These findings suggest that while greater aphasia severity is associated with worse WCST accuracy, both non-verbal and verbal interference conditions were linked to improved performance from the baseline. Additionally, inner speech appears to support task accuracy, although its role may differ depending on interference type.

Discussion

Hypothesis 1: Relationship between Inner Speech and Problem Solving

The correlations between all the baseline cognitive measures and one of the two inner speech measures (objective inner rhyme judgment but not subjective successful inner speech) highlight the relationship between language and cognition that has been investigated in a variety of ways (Baldo et al., 2005; Fama et al., 2024; Wallace et al., 2017). These tests all measure some related aspects of executive functioning (flexibility, problem solving, planning, and inhibition), and the correlations between inner rhyme judgment and these measures show that inner speech, and especially the ability to manipulate information in inner speech, may be a tool for other cognitive tasks, even if those tasks do not inherently involve language. As demonstrated by double dissociations in Fama et al. (2024), the relationship between performance on cognitive tasks and the objective task of inner speech for PWA is not straightforward. Despite the correlation between these measures, there are people who do well on one but not the other, adding to the evidence that inner speech may support domain-general cognition, but it is not necessary for it, and vice versa. Whether a person uses inner speech to support their cognitive

abilities may depend on several factors including the specific task, the environment, and personal abilities/preferences.

Due to the heterogeneity in the phenomenon of inner speech in general for people with and without aphasia (Alexander et al., 2024; Lupyan et al., 2023), differences in its use for problem solving are to be expected. This aligns with Vygotsky's (1934) foundational theories on inner speech, proposing that inner speech emerges from the internalization of social dialogue and plays a crucial role in the development of higher-order cognitive abilities such as planning, self-regulation, and reasoning. While inner speech may scaffold problem solving for some, its role is not uniform across individuals, particularly in the context of acquired language disorders.

Hypothesis 2: Interfering with Inner Speech

The articulatory suppression (verbal interference) condition did interfere with inner speech, according to the participant's reports of how often they used inner speech during the task. Interestingly, though, despite the suppression of inner speech, people did not necessarily perform worse on the verbal interference condition (or the non-verbal interference condition) compared to baseline. This lack of decline in performance is at least partially due to a robust learning effect – because the baseline condition always came first, participants performed worse (lower accuracy) on that condition even though there was no interference, as the task was novel to them. The within-subjects design of the current study was a methodological difference from Baldo et al. (2005), which randomly assigned participants to complete a single version of the WCST. With this learning effect, one may have expected *less improvement* on the verbal interference condition than on the non-verbal interference condition, rather than *worse performance*, as verbal interference would limit the benefits of task familiarity. Indeed, people performed significantly better on the non-verbal interference condition compared to baseline, but this was not true for the verbal interference condition. This finding suggests an impact of

articulatory suppression (which limits access to inner speech) on accuracy, thereby supporting the hypothesis that interfering with inner speech will have a negative impact on problem solving. The finding that verbal interference dampened gains in problem solving that would have been expected due to learning effects aligns with prior work demonstrating that articulatory suppression can impair performance on tasks that may rely on inner speech, such as reasoning, verbal memory, impulse control, and planning (Emerson & Miyake, 2003; Tullett & Inzlicht, 2010). Although the WCST is a nonverbal task, it demands cognitive flexibility, working memory, and reasoning, all processes that may benefit from verbal mediation. Thus, verbal interference may impair these executive processes by removing a cognitive tool or strategy.

Although inner speech was disrupted during the verbal interference condition, some people still performed well. This finding suggests that not all individuals use or require inner speech for problem-solving tasks like the WCST. Indeed, a subset of participants reported not using inner speech on any condition, while others reported consistent use across all conditions. Interestingly, all individuals in the cognitively healthy adult (CHA) group reported using inner speech during the non-verbal interference condition, suggesting a more habitual reliance on this strategy. It could be that there were a variety of strategies employed to complete the WCST, and inner speech supported performance but was not necessary. These individual differences echo findings by Nedergaard and colleagues (Nedergaard, Wallentin, et al., 2023) who demonstrated that while inner speech may support performance on certain executive tasks, it is not universally required. Rather, individuals may flexibly engage different strategies depending on personal preferences, task demands, and cognitive strengths. In this study, while higher overall ratings of inner speech were associated with better performance on the WCST, an interaction emerged in the non-verbal interference condition: greater inner speech use in that condition was paradoxically linked to poorer performance. One interpretation is that participants who typically relied on inner speech attempted to do so even under non-optimal conditions, resulting in cognitive conflict or divided attention. Alternatively, those who reported more inner speech

during interference may have experienced greater difficulty adapting to the constraints of that condition.

Taken together, these findings highlight the complexity of inner speech as a cognitive resource, one that can facilitate performance under typical conditions but may be vulnerable to interference, especially when it is a primary strategy for navigating cognitive tasks. They also underscore the importance of considering individual variability in inner speech use and cognitive strategy selection when interpreting task performance, particularly in clinical populations such as individuals with aphasia.

Aphasia Severity and Anosognosia

Aphasia severity (as measured by the total score on the Quick Aphasia Battery) predicted accuracy on the Wisconsin Card Sorting Test (WCST), indicating that more severe language difficulties are associated with greater challenges in problem solving—even on a nonverbal task that does not explicitly require language. This aligns with previous research suggesting that executive function deficits are common in aphasia and may be partially independent of language impairment, though the two often interact (Murray, 2012, 2017). Specifically, individuals with more severe aphasia tend to exhibit greater impairments in domains such as cognitive flexibility, working memory, and inhibitory control, all of which are essential for successful performance on tasks like the WCST. Dutta and colleagues (Dutta et al., 2023) have highlighted the variability in executive function performance among people with aphasia (PWA), emphasizing the need to consider individual cognitive profiles rather than assume uniform deficits across the population.

One potential mechanism underlying the relationship between language impairment and problem-solving difficulties is the availability and quality of inner speech. In the current study, higher inner speech ratings were associated with better WCST accuracy for the PWA group, suggesting that retained or partially intact inner speech may serve as a compensatory scaffold

for executive functioning. This is consistent with theoretical perspectives indicating that inner speech supports self-regulation, rule maintenance, and planning—core components of executive function (Barkley, 2001; Vygotsky, 1934). However, the benefits of inner speech appeared to diminish under conditions of non-verbal interference, potentially due to increased cognitive load or competition for shared cognitive resources.

Taken together with prior work, the findings from the current study reinforce the notion that executive function deficits in aphasia are not merely secondary to language impairments but instead represent a distinct yet interacting domain of difficulty. The role of inner speech may vary across individuals based on both linguistic and cognitive capacities, underscoring the importance of personalized approaches in assessment and intervention for cognitive-linguistic rehabilitation.

Similar to prior work (Alexander et al., 2024), the lack of awareness of language deficits (anosognosia) did not predict the influence of interfering with inner speech in people with aphasia. This could be because there were only three participants who had a VATA-L discrepancy score with an absolute value of 12 or greater, which is the cutoff for anosognosia (Cocchini et al., 2010), so there was not a large representation of lack of awareness of deficits. The fact that nearly all the participants had intact awareness of their linguistic deficits is a strength to this study, however, as the researchers can be more confident in their subjective ratings. Because of the connections between self-monitoring, evaluation, and inner speech, and its relation to performance on other cognitive tasks (Gade & Paelecke, 2019; Perrone-Bertolotti et al., 2014), there still could be some connection between awareness and problem-solving abilities with a sample of people with aphasia and co-occurring anosognosia.

Clinical Implications

For people with aphasia who may also have concomitant cognitive difficulties, or who use inner speech as a strategy for problem solving, there may be benefits to addressing both cognition

and language in therapy sessions. Incorporating inner speech into metacognitive strategy training (Wadams et al., 2022) may improve performance on problem-solving tasks for people with aphasia. Speech-language pathologists can introduce inner speech as one possible strategy among several options, including visual supports and tools, working backward, and guess-and-check techniques. They may then evaluate their client's strengths and weaknesses with these strategies and work together to facilitate problem solving in everyday life situations.

Limitations

Because of the within-subjects design, there was a large learning effect which made the results more difficult to interpret. To combat this, the models controlled for the presentation order of the conditions and the participants themselves, but performance still improved as participants adapted to the WCST.

Virtual administration allowed the researchers to reach people in remote locations with varied backgrounds and led to great interest in the study, but there were drawbacks to being remote. While adequate internet connection was a necessity for inclusion in the study, there were some connectivity issues throughout the study for various participants. For some participants, this was limited to longer waiting times for loading, but for others, there were issues with the presentation of certain items. Due to the virtual administration of the tasks, hand tapping was chosen as the non-verbal interference task so that the researchers could see the action and monitor performance. After the tasks, several participants commented that the non-verbal condition felt like "trying to rub my belly and pat my head," so it is possible that having a two-handed dual task was more difficult than other iterations of the motor dual task, such as foot tapping instead of hand tapping. One control participant even reported that the motor task also interfered with his inner speech, as he found himself losing track of the current "rule" due to his focus on hand tapping. This corresponds with the significant decrease in inner speech ratings from the baseline to the non-verbal interference condition. The two-handed nature of the task

also led to exclusion of several of the participants with aphasia who had lingering right-sided weakness or paralysis, lowering the number of observations and the representativeness of the study.

To ensure ability to understand task instructions and to minimize frustration with the tasks, there was a language comprehension cutoff, which may also impact the relationships between aphasia severity and other measures due to having fewer people with more severe aphasia. Additionally, only one person with aphasia scored below 50% on the successful inner speech during naming task, meaning either that this group had intact access to their inner speech, or that they misunderstood the concept of successful inner speech even with clear instructions. The administration of the successful inner speech task in this study differed slightly from prior work in our lab (Stark et al., in revision), in that participants were in control of the advancement from one item to the next, meaning that the experimenter had less opportunity to stop the task and clarify the concept.

Future Directions

MoCA scores were a significant predictor of performance on the WCST, and upon further investigation, it was the delayed recall subtest which drove this relationship, as none of the other subtests significantly predicted performance. The theory behind this is that verbal working memory and inner speech are related (Marvel & Desmond, 2012; Potagas et al., 2011), and both may relate to executive function and naming abilities (Fama et al., 2024; Sayers et al., 2023). Future studies could work to tease apart more aspects of verbal working memory to further understand the connections between language and cognition.

From these findings, inner speech may be a useful strategy for problem solving for some people with aphasia. Future work may address inner speech as part of a metacognitive strategy training approach to post-stroke cognitive-linguistic therapy.

The WCST was chosen due to its ability to be adapted for virtual administration and its nonverbal nature, but people may not rely on inner speech in order to complete the WCST at baseline, and if they do, it may be limited to the repetition of single words, more like “rehearsal” in verbal working memory rather than complex problem solving. For example, a person may rehearse the current rule in their inner speech “shape, shape, shape” and when the rule changes, they may have a brief moment of longer inner speech, “it must be color or number now,” then they return to the thread of the single repeated word. Other problem-solving tasks may be more likely to elicit the use of more complex inner speech, and future studies could attempt to use more verbal tasks to see the impact of interfering with inner speech. These tasks could also be tasks that people may encounter in their everyday lives, such as scheduling appointments or managing finances, to increase the ecological validity of the research. Future work may use methods unique to aphasia, such as voxel-based lesion symptom mapping, to determine areas of brain injury which may be related to both inner speech and problem-solving difficulties. Such a study would further uncover the neurolinguistic mechanisms underlying both inner speech and problem solving, helping to explain the nature of post-stroke cognitive-linguistic impairments.

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Appendix

Appendix Table 1: Inner Speech Summary. This was a part of the introduction to inner speech, which was an interactive presentation with each participant.

Inner speech is:	Inner speech is NOT:
Verbal (thinking)	Non-verbal thinking
A commentary	Silent thinking
In words	Sensations
Self-talk	Images
Inner speaking	Pure emotions
Talking to yourself	Wordless thinking
In your own voice	
Words or sentences	
Mostly silent	
Monologue or dialogue	
Meaningful or nonsensical	
More or less frequent	
Spontaneous or voluntary	

Chapter 5: The General Experience of Inner Speech, Emphasizing Psychosocial Health in Aphasia and Aging

Abstract

Background: Anecdotally, people with aphasia have reported changes in their inner speech after a brain injury. In the chronic phase of recovery, there may still be differences in the ways that people with aphasia use inner speech in comparison to cognitively healthy peers, and this may relate to differences in psychosocial health.

Method: 39 participants with aphasia and 43 age-similar cognitively healthy adults completed tasks and questionnaires of inner speech, problem solving, and quality of life. Participants were compared on responses to an adapted version of the General Inner Speech Questionnaire (GISQ, Racy et al., 2020), which was further validated for use in aphasia and with cognitively healthy adults. Selected questions on the GISQ were analyzed for relationships to emotions, problem solving, health, and communication.

Results: People with and without aphasia reported similar frequency and themes of inner speech, but cognitively healthy peers had more variety of inner speech, as they selected more responses on average on the GISQ. The two groups overlapped on the most commonly reported contents, functions, and activities of inner speech. Certain questions on the GISQ related to scores on measures of health-related quality of life, including using inner speech to replay past conversations and control emotions, to talk about positive emotions, and when meeting people. The GISQ shows some construct validity when compared to the Varieties of Inner Speech Questionnaire - Revised (Alderson-Day et al., 2018), although each appears to measure distinct parts of the inner speech phenomenon.

Conclusion: The findings here support that inner speech may change post-stroke, as people with chronic aphasia experience less variety in their inner speech than people without aphasia. The ways in which people use inner speech may relate to their psychosocial health.

Inner speech, also known as internal monologue, self-talk, and inner voice, is silent, verbal thought, or the “little voice in the head.” Inner speech is used by adults to talk about a variety of topics (contents), for a variety of reasons (functions), and in a variety of situations (activities) (Morin & Racy, 2022; Racy et al., 2020). Inner speech is an individual, perceptual experience, as are most inner phenomena (e.g., pain, tinnitus), and people do not use inner speech in the same ways, so it is by nature heterogeneous. With changes in language after a stroke, the ways in which inner speech is used may be altered. Some people with aphasia (an acquired language disorder which affects over two million people in the United States) report that in acute recovery from their brain injury, there was a “silence” or “quiet” in place of inner speech (Marks, 2018; Taylor, 2008), and people with aphasia use metaphors to describe the experience of inner speech (or lack thereof) in their autobiographical accounts, including conceptualizing aphasia as silence or as a spiritual experience (Tichborne et al., 2024a, 2024b). However, some people with aphasia report using inner speech much more after than before their stroke (Alexander et al., 2023), and they often report using inner speech to talk about health, to solve problems, and to rehearse ahead of time (Alexander et al., 2024). These findings indicate that inner speech may be a useful tool for people with aphasia as they navigate their environments with the additional challenges presented by language impairment. For example, tasks that may have been automatic prior to the stroke, such as calling the doctor’s office, now require advanced planning and rehearsal.

A stroke can affect multiple aspects of a person’s life, including physical, communicative, and social/emotional domains (Hilari et al., 2003; Manning et al., 2021; Moss et al., 2021). Models and frameworks such as the World Health Organization’s International Classification of Functioning, Disability, and Health (ICF) (World Health Organization, 2002) and the Living with Aphasia: Framework for Outcome Measurement (A-FROM) (Kagan et al., 2008) help situate the study of inner speech in aging and aphasia to better understand the implications on quality of life. The ICF model integrates the medical and social models of disability to address

biopsychosocial factors of health. In this model, a person's functioning is seen as the interaction between health conditions and contextual factors. A diagram of the model reprinted below shows the interconnectedness of each factor, where there are levels of human functioning: the body (or body parts), the whole person, and the person in their social environment (Figure 1).

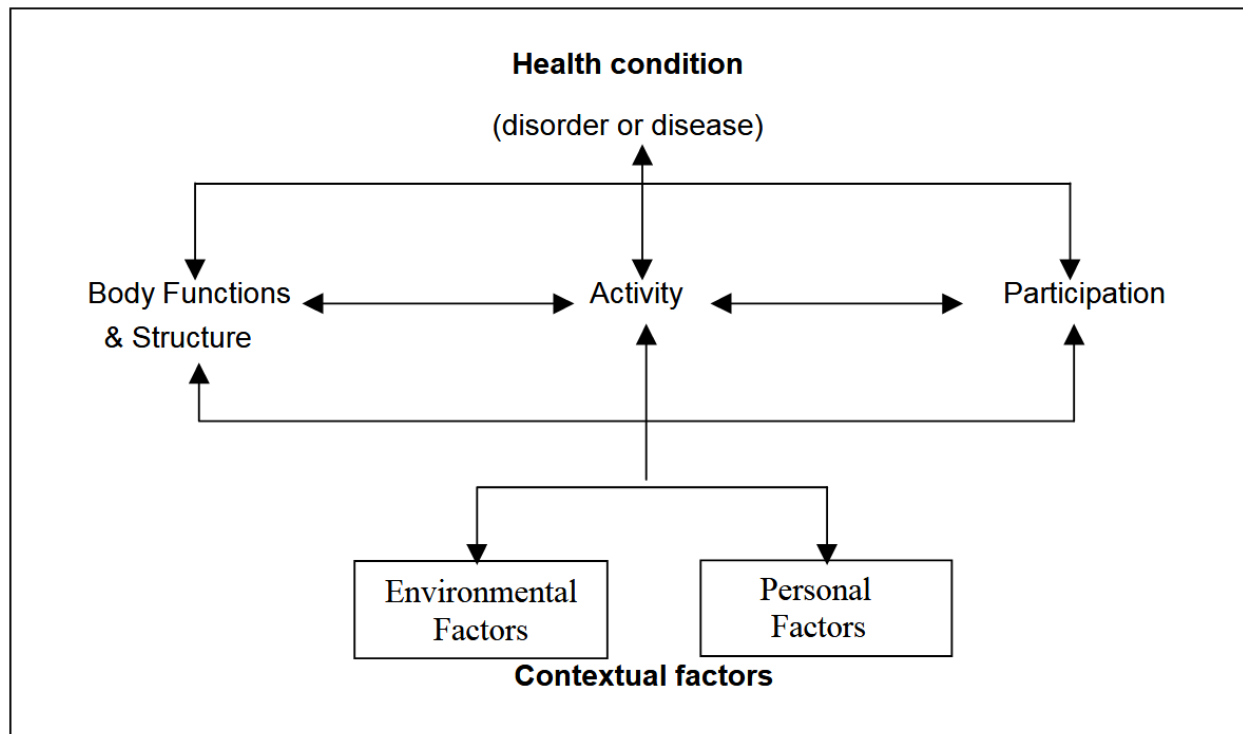


Figure 1: Representation of the World Health Organization's International Classification of Functioning, Disability, and Health model of disability, depicting the interaction between contextual factors (i.e., environmental and personal) and health conditions (body functions and structure, activity, and participation) (image from World Health Organization, 2002).

Inner speech and aphasia fit into the ICF's model as follows:

1. Health Condition (specific to aphasia, although people may have additional concomitant health conditions)
 - a. Body functions and structure: People with aphasia have a brain injury, generally affecting cortical structures associated with language in the left hemisphere including the frontal, temporal and/or parietal lobes. Damage to these areas can

affect speech and language production and comprehension (aphasia severity) and may impact other cognitive and psychological processes depending on the location and size of the injury.

- b. Activity: People with aphasia may have limitations in activities related to communication, such as word-finding, speaking, understanding others, reading, and writing. These limitations may also extend to language in the head, or “inner speech,” which could in turn limit activities such as solving complex problems or planning.
 - c. Participation: Due to impairments in body functions and activity limitations, people with aphasia may have participation restrictions related to their communication and cognitive difficulties. For example, they may have difficulty returning to work if their work involves language-based tasks, such as making presentations, having conversations, or scheduling.
2. Contextual Factors (specific to the individual, although there may be commonalities in people with aphasia)
- a. Environmental factors: This includes factors such as legal and social structures, access to healthcare, and architectural concerns related to mobility and accessibility.
 - b. Personal factors: Demographic factors (e.g., age, gender, race, and education), social background, profession, personal experiences, behavior and character.

Similarly, the A-FROM builds upon the ICF model to cater to stakeholders in the aphasia community (people with aphasia, loved ones, and clinicians). This framework emphasizes that quality of life when living with aphasia is impacted by several overlapping factors (Figure 2).

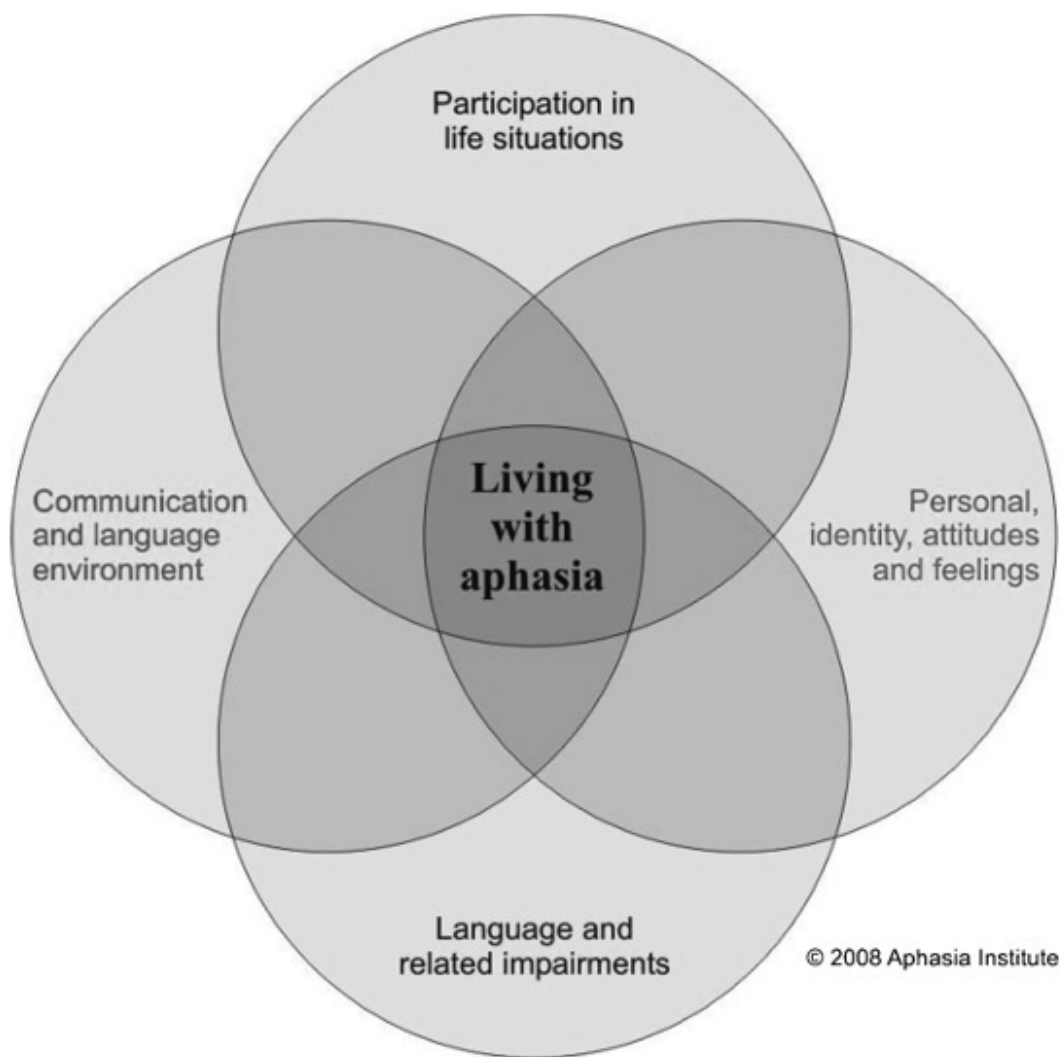


Figure 2: A diagram of the Living with Aphasia: Framework for Outcome Measurement, demonstrating the overlapping relationships between communication and language environment, language and related impairments, personal identity, attitudes, and feelings, and participation in life situations (image from the Aphasia Institute, Kagan et al., 2008).

Related to the A-FROM life domain of personal, identity, attitudes and feelings (expanded from ICF's personal factors), a person's identity is the way they perceive and define themselves, including beliefs, values, roles, and personality. Identity can develop and change over time due to life experiences. People with aphasia may have a difference in their self-concept due to changes in societal and familial roles (Fotiadou et al., 2014). The stories that

people tell affect their self-concept and identity, and these stories are likely formulated – at least in part – in inner speech (Boksay Pap, 2016; Fanti Rovetta, 2023; Shadden & Hagstrom, 2007). For people going through a large shift in identity (Shadden, 2005; Shadden & Hagstrom, 2007), such as recovery post-stroke or reaching retirement, inner speech may be a mechanism of identity reformation.

The use of inner speech could have positive or negative effects on quality of life dependent on the contents, functions, and activities of inner speech for each person. For example, if a person uses inner speech for rumination and often dwells on what is difficult for them now, people who have lost contact, and the burden they may present to their family, their inner speech may harm their mental and emotional health. Conversely, if a person uses inner speech to motivate themselves, control emotions (as is done in cognitive behavioral therapy), and to talk about resilience or progress, inner speech may benefit their mental and emotional health.

A pilot study from our lab with 23 participants suggested that individuals with aphasia use inner speech to self-motivate and plan (Alexander et al., 2024), which is complementary to what younger, cognitively healthy populations report (Racy et al., 2020). Most of the people with aphasia reported that they spoke to themselves about health, with people further clarifying that they talk about living with aphasia in their inner speech. Over three observations, an average of 77% of people with aphasia reported using inner speech for problem solving, 82% for motivation, 61% for emotional control, and 65% to rehearse ahead of time, with 42% reporting talking about identity, and 65% about health. Some of these uses may be more common for people with aphasia than for adults without a history of brain injury due to concomitant cognitive difficulties (Mayer et al., 2017) and physical, mental, social, or emotional health issues (Gainotti, 1997; Kelleher et al., 2024; Kong, 2021; Moss et al., 2021; Strong & Randolph, 2021). Indeed, depression is prevalent in people with aphasia, with one study reporting that 62% of people with aphasia met the criteria for depression a year after their stroke (Kauhanen et al., 2000). Many

people with aphasia report that mental and emotional health are some of their greatest concerns (Moss et al., 2021; Wallace et al., 2017). With the possibility of relatively spared inner speech compared to overt speech (Alexander et al., 2023; Geva et al., 2011; Stark et al., 2017), interventions which leverage this relative strength may be optimal for mental health services. Because of the relationships between inner speech, identity/self-reflection (Morin, 2018; Morin & Michaud, 2007; Racy & Morin, 2024) and cognitive processes (Alderson-Day & Fernyhough, 2015; Fernyhough & Borghi, 2023; Nedergaard et al., 2023), understanding its use in everyday life may illuminate ways to address quality-of-life concerns in older adults with and without aphasia. Both groups may be navigating substantial life changes, including retirement, shifts in familial roles, and social isolation. The ways in which people use inner speech may highlight the ways that they approach these changes.

Clinically, post-stroke changes in inner speech may affect mental health. These changes in emotional, psychological, and psychosocial factors have long been recognized as essential to understanding and improving quality of life for stroke survivors (Gainotti, 1997; Moss et al., 2021), and inner speech is often used for self-regulation of emotions and mental states in adults without brain injury (Kross et al., 2014; Morin & Racy, 2022). Investigation into the experience of inner speech and differences from people with aphasia to age-similar cognitively healthy adults could lead to a better understanding of the difficulties surrounding adaptation to life with aphasia. People with aphasia (PWA) are likely reconstructing their own identity after stroke (Shadden, 2005; Taubner et al., 2020), and inner speech may be one way to do so. Using questionnaires to explore the general experience of inner speech in adults with and without aphasia is a promising next step to tackling mental health concerns in these populations, as PWA may report more use of inner speech for factors related to psychosocial health, such as positive and negative emotions, identity, motivation, and emotional control.

Although our lab has examined the feasibility of using questionnaires to assess inner speech in PWA (Alexander et al., 2024), there was no age-similar comparison group. Therefore,

it is difficult to ascertain what themes of inner speech are unique to PWA and which intersect with themes commonly experienced during aging. The current study was planned to systematically compare the phenomenology of subjectively experienced inner speech in PWA and age-similar adults, and in particular, compare the use of inner speech for problem solving and mental health in these two populations. Additionally, this study explores the connections between performance on inner speech and problem-solving tasks with answers on general questionnaires of inner speech and quality of life.

Specific Aims and Objectives

This project used the adapted General Inner Speech Questionnaire to investigate the general experience of inner speech in people with aphasia and neurologically cognitively healthy adults (CHA), further validating its use.

1. **Identify the differences in the general inner speech experience between PWA and age-similar adults without brain injury.**

Hypothesis: Based on pilot data, PWA will report using inner speech to talk about health and to rehearse ahead of time, and this may be less common in age-similar adults without brain injury.

Exploratory sub-aim: Identify how PWA differ from adults without brain injury on use of inner speech to address psychosocial health (e.g., controlling emotions, motivation, identity), and how inner speech relates to measures of psychosocial health.

2. **Validate the adapted General Inner Speech Questionnaire (GISQ) with the Varieties of Inner Speech Questionnaire – Revised (VISQ-R).**

Hypothesis: Evaluative, Other People, and Positive/Regulatory means on the VISQ-R will predict whether people select certain responses on the GISQ (think critically, people around me/replay past conversations, and positive emotions/control emotions, respectively).

Method

Materials

Cognitive-Linguistic Assessment

Quick Aphasia Battery (Wilson et al., 2018). PWA completed this standard battery, which provides scores across core language domains and an overall aphasia severity score. The Sentence Comprehension subtest served as an exclusionary measure for the PWA group. This battery does not sort participants into aphasia type, but rather allows for in-depth examination of relative strengths and weaknesses, as PWA rarely fit neatly into the specific aphasia types (Wilson et al., 2018). The remote version of the battery was used, and the entire battery typically took approximately 20 minutes to complete.

Montreal Cognitive Assessment (MoCA) (Nasreddine et al., 2005). All participants completed this assessment as a measure of cognitive abilities such as visuospatial skills, attention, memory, language, and abstract thinking. For the cognitively healthy adult (CHA) group, this was an exclusionary measure. Although this assessment does have subtests which rely on language, it is often used post-stroke, and aphasia is often not a barrier to completing the MoCA (Chiti & Pantoni, 2014; Cumming et al., 2011; Yan et al., 2022). Due to the reliance on language, the MoCA underestimates cognitive abilities of people with aphasia (Falkauskas, 2016), so the experimenter proceeded with that caveat, and scores which are less reliant on language (e.g., visuospatial, attention) can be reported with higher confidence. To allow for remote administration, participants completed the visuospatial section of the assessment on their own paper and showed it to the experimenter over the videoconferencing software (Abdolahi et al., 2016; DeYoung & Shenal, 2019; liboshi et al., 2020), with the mazes portion being said out loud rather than drawn. The MoCA typically took 5-10 minutes to complete.

Inner Speech Questionnaires

General Inner Speech Questionnaire (Racy et al., 2020). All participants completed the shortened, adapted General Inner Speech Questionnaire developed in our lab, which has three sections: contents (“I talk to myself about...” with 16 items), functions (“I talk to myself in order to...” with 11 items), and activities (“I talk to myself when...” with 6 items). This questionnaire provides information about participants’ perception of daily inner speech, including whether they are currently experiencing inner speech at the start of the survey (yes, no, or not sure) and how often they experience inner speech during their daily lives (Never 0%, Sometimes 25%, About half of the time 50%, Most of the time 75%, and Always 100%). The adaptation includes visual supports for each item (Figure 3) and a confidence rating scale at the end of each section (Figure 4), along with the opportunity to provide additional examples of inner speech beyond the selected items. This questionnaire usually took approximately 10-15 minutes to complete.

For the rest of the survey, think about your inner speech in the **past 6 months**. Please **select** the response(s) that you agree with. I talk to myself about:

 Positive Emotions
 Negative emotions
 What others think about me

Figure 3: Example of the adapted General Inner Speech Questionnaire web version, which participants completed through Qualtrics.

How confident are you in the accuracy of your responses? Please **select** the number that corresponds to your **confidence level**.

1 - Not confident at all	
2	
3	
4	
5 - Very confident	

Figure 4: Example of the confidence rating scale after each section of the General Inner Speech Questionnaire, with visual supports to aid understanding of the 5-point scale.

Varieties of Inner Speech Questionnaire-Revised (Alderson-Day et al., 2018). The CHA group completed the Varieties of Inner Speech Questionnaire-Revised, administered to gain further construct validity of the adapted General Inner Speech Questionnaire. This questionnaire has 26 questions and evaluates five factors of inner speech: dialogic (whether the inner speech has a back-and-forth quality), evaluative/critical, other people (whether inner speech includes voices of others), condensed (whether the inner speech includes all phonetic components), and positive/regulatory. Participants select how often they have certain experiences related to inner speech, such as, “I evaluate my behavior using my inner speech. For example, I say to myself, ‘that was good’ or ‘that was stupid.’” They responded with ratings on a scale from 0 (never) to 7 (all of the time). This questionnaire uses more complex language than the General Inner Speech Questionnaire and has not before been validated for use in

PWA, and thus only the CHA group completed it. This questionnaire took approximately 10 minutes to complete.

Health-related Quality of Life Questionnaires

Stroke and Aphasia Quality of Life Scale (SAQOL-39) (Hilari et al., 2003). PWA completed the SAQOL-39 to address health-related quality of life, including psychosocial health. This scale was developed for and normed on PWA, with 39 questions related to physical, communicative, and social abilities and experiences. The SAQOL-39 has aphasia-friendly features, such as anchors on the rating scales, bolded/emphasized words, and simple language. Patient-reported outcome measures are becoming more prominent in the field of aphasiology and healthcare in general (Cohen & Hula, 2020). Governing bodies such as the National Institute of Health and American Speech-Language Hearing Association promote them (Irwin, 2012), recognizing that patient-reported outcomes in patients with language disorders pose unique challenges, but these challenges can be addressed with proper communication facilitation and/or reports from a trusted other. In this study, PWA had the opportunity to give reports with multiple modalities for both the experimenter administering the assessment and the participant answering the questions, and if needed, a trusted other provided additional information. This scale took approximately 10 minutes to complete.

Patient-Reported Outcomes Measurement Information System (PROMIS) (Cella et al., 2007). The CHA group completed two forms of the PROMIS: Positive Affect and Ability to Participate in Social Roles and Activities. These forms are relevant for measuring psychosocial quality of life and normed on the general population. They use a 5-point scale from “Never” to “Always” in answer to prompts such as, “In the past 7 days I can keep up with my social commitments.” These are adaptive measures, so the questions that each participant received were based on their responses to previous questions. These questionnaires took approximately 10 minutes to complete in total, and sometimes less due to the adaptive nature.

Participants

Thirty-nine people with aphasia and 43 cognitively healthy adults participated in this study. Demographic and neuropsychological information and comparisons can be found in Table 1 below, showing that the groups were similar on age and education. Additionally, both groups were a majority white (CHA: 97.7% White, 2.3% Black or African American; PWA: 76.9% White, 15.4% Black or African American, 5.1% Asian, and 2.6% American Indian/Alaska Native). None of the participants in the CHA group and only 2.6% of the PWA group were of Hispanic or Latino origin. Participants in the aphasia group were required to have chronic aphasia (at least six months) from a left hemisphere stroke.

Table 1: Demographic Information and Statistical Comparisons between Groups

	CHA (N=43)	PWA (N=39)	Statistics
Age			
Mean (SD)	63.7 (7.93)	61.4 (9.35)	$t(75) = -1.21$,
Median [Min, Max]	63.7 [42.1, 76.8]	60.4 [42.0, 77.4]	$p = .23$
Gender			
Female	29 (67.4%)	19 (48.7%)	$\chi^2(1) = 2.23$,
Male	14 (32.6%)	20 (51.3%)	$p = .14$
Education (in years)			
Mean (SD)	17.3 (2.29)	17.1 (2.60)	$t(76) = -0.33$,
Median [Min, Max]	17.0 [13.0, 24.0]	16.0 [12.0, 24.0]	$p = .74$
Bilingual			
No	34 (79.1%)	32 (82.1%)	Not computed;
Yes	9 (20.9%)	7 (17.9%)	Not enough cases
Quick Aphasia Battery Overall Score			
Mean (SD)	NA	7.77 (1.57)	NA
Median [Min, Max]	NA	8.33 [3.19, 9.60]	
Self-Reported Depression or Anxiety			
No	35 (81.4%)	26 (66.7%)	$\chi^2(1) = 1.62$,
Yes	8 (18.6%)	13 (33.3%)	$p = .20$

Procedure

These data were collected as part of a larger project, with the procedure described in Chapter 4. Participants completed the assessments and questionnaires from the “Materials” section of this manuscript, generally within one session, but sometimes spread into two depending on the needs of the participant. Participants met with a researcher on videoconferencing software (Microsoft Teams). The cognitive-linguistic assessments were completed directly in the call, as was the PROMIS for the CHA group. Participants were then directed to an experiment software

called Gorilla, in which they completed the remaining tasks, with the inner speech questionnaires being a redirect to surveys hosted on Qualtrics. The participants shared their screens so that the researcher could troubleshoot technical difficulties, explain the tasks, and read and explain each question as needed, depending on the preferences and abilities of each participant.

Data Analysis

Hypothesis 1: PWA will report using inner speech to talk about health and to rehearse ahead of time, and this may be less common in age-similar adults without brain injury.

Due to the binary nature of the data, Pearson's chi-squared tests were used to compare the groups on how frequently inner speech was used in relation to health and rehearsing ahead of time.

Exploratory sub-aim: Identify how PWA differ from adults without brain injury on use of inner speech to address psychosocial health (e.g., controlling emotions, motivation, identity), and how inner speech relates to measures of psychosocial health.

Similar to the main hypothesis, Pearson's Chi-squared tests were used to compare the groups on controlling emotions, motivation, and identity. To compare inner speech and health-related quality of life, correlations were used between responses on the GISQ and scores on the PROMIS (for the CHA group) and the SAQOL (for the PWA group), as well as self-reported depression and anxiety.

Hypothesis 2: Evaluative, Other People, and Positive/Regulatory means on the VISQ-R will predict whether people select certain responses on the GISQ (think critically/avoid saying out loud what I want to say, people around me/replay past conversations, and positive emotions/negative emotions/control emotions, respectively).

For the CHA group, linear modelling was used to predict VISQ-R Evaluative, Other People, and Positive/Regulatory means based on the responses outlined on the GISQ outlined above.

Results

Before addressing the main hypotheses, differences in the reports of frequency of inner speech, current inner speech, and total number of responses selected in people with and without aphasia were evaluated. Based on Table 2 below, there are no group differences in the frequency of inner speech or whether people were currently experiencing inner speech at the beginning of the questionnaire. Cognitively healthy adults, however, selected significantly more responses on the GISQ overall, indicating using inner speech to talk about more things (contents), for more reasons (functions), and in more situations (activities). Table 3 shows the means for the three domains of health-related quality of life on the SAQOL-39 for the PWA group. Participants rated their physical health as slightly higher than their psychosocial or communication health. None of the SAQOL-39 domains were significantly related to aphasia severity (overall score on the QAB), with all p -values greater than .23.

Table 2: Descriptive Statistics of Basic Comparisons of Current Inner Speech, Inner Speech Frequency, and the Variety of Inner Speech. CHA: cognitively healthy adults, PWA: people with aphasia, GISQ: General Inner Speech Questionnaire

	CHA (N = 43)	PWA (N = 39)	Statistics
Current (in-the-moment) Inner Speech			
No	9 (20.9%)	11 (28.2%)	$\chi^2(2) = 2.11,$ $p = .35$
Not sure	1 (2.3%)	3 (7.7%)	
Yes	33 (76.7%)	25 (64.1%)	
Frequency of Inner Speech (percent of time in daily life)			
Mean (SD)	57.56 (20.80)	56.41 (30.74)	$t(65) = 0.06,$
Median [Min, Max]	50.00 [25.00, 100.0]	50.00 [0, 100.0]	$p = .95$
Number of Responses on the GISQ (out of 33 possible)			
Mean (SD)	24.72 (4.589)	21.00 (8.775)	$t(65) = -2.37,$
Median [Min, Max]	25.00 [15.00, 33.00]	23.00 [0, 32.00]	$p = .02$

Table 3: Psychosocial Health – Stroke and Aphasia Quality of Life Means – PWA group. Ratings are from 1-5, with 1 being more negative experiences/difficulty in the domain, and 5 being more positive experiences/independence.

	Overall (N = 39)
SAQOL-39 Physical Mean (rating from 1-5)	
Mean (SD)	4.354 (0.6337)
Median [Min, Max]	4.625 [2.750, 5.000]
SAQOL-39 Communication Mean	
Mean (SD)	3.150 (0.6934)
Median [Min, Max]	3.286 [1.714, 4.286]
SAQOL-39 Psychosocial Mean	
Mean (SD)	3.688 (0.7457)
Median [Min, Max]	3.875 [1.875, 4.938]

Participants in the CHA group completed the VISQ-R and two questionnaires from the PROMIS related to their ability to participate in social roles and activities and their positive affect. Descriptive statistics are shown in Table 4, with means near the center of the scale on the Condensed, Dialogic, Evaluative, and Positive/Regulatory domains (from 1-7, means near 3.5 would be the center) demonstrating that this group had a mix of “shorthand” inner speech (condensed) and full sentences with all phonological and grammatical characteristics, monologic and dialogic inner speech, they sometimes evaluated themselves in inner speech, and they sometimes encourage or calm themselves using inner speech. This group did not experience the voices of other people in their inner speech very often, which is consistent with the absence of psychopathology in the group.

Table 4: Inner Speech and Psychosocial Health-related Quality of Life – CHA Group. VISQ-R = Varieties of Inner Speech Questionnaire – Revised. PROMIS = Patient-Reported Outcome Measurement Information System.

	Overall (N = 43)
VISQ-R Condensed Mean (possible range of 1-7)	
Mean (SD)	3.121 (0.9342)
Median [Min, Max]	3.000 [1.800, 5.400]
VISQ-R Dialogic Mean	
Mean (SD)	3.940 (1.069)
Median [Min, Max]	3.600 [2.000, 6.400]
VISQ-R Evaluative Mean	
Mean (SD)	3.259 (0.6213)
Median [Min, Max]	3.143 [1.714, 4.714]
VISQ-R Other People Mean	
Mean (SD)	1.749 (0.8084)
Median [Min, Max]	1.600 [1.000, 4.400]
VISQ-R Positive Regulatory Mean	
Mean (SD)	3.628 (0.6712)
Median [Min, Max]	3.750 [2.500, 4.750]
PROMIS Participate T-score (range of 0-100)	
Mean (SD)	53.82 (6.013)
Median [Min, Max]	53.50 [41.10, 67.50]
PROMIS Affect T-score	
Mean (SD)	49.07 (6.803)
Median [Min, Max]	48.40 [35.30, 71.60]

Aim 1: Differences in Inner Speech Use for Adults with and without Aphasia

Hypothesis: Based on pilot data, PWA will report using inner speech to talk about health and to rehearse ahead of time, and this may be less common in age-similar adults without brain injury. This hypothesis was not supported: with the specific, pre-selected content item of “Health,” there was no significant difference between the groups, $\chi^2 (1, N = 82) = 0.27, p = .60$. With the

function to “Rehearse ahead of Time,” there was also no significant difference between the groups, $X^2 (1, N = 82) = 1.0285, p = .31$.

In relation to exploratory sub-aim, for selected psychosocial responses, chi-square tests revealed that the CHA group used inner speech to “motivate myself” significantly more than the PWA group, $X^2 (1, N = 82) = 4.06, p = .04$, but there was no significant difference in using inner speech to “control emotions,” $X^2 (1, N = 82) = 1.07, p = .30$ or to talk about identity, $X^2 (1, N = 82) = 0.18, p = .67$.

With a more exploratory, data-driven approach, it is clear that most of the items on the GISQ were endorsed by most of the participants, regardless of group. Supported by the findings of a significant difference in number of items selected, more items are endorsed by a higher percentage of the CHA group than the aphasia group.

Figure 5 shows the most frequently reported contents: Most people with aphasia used inner speech to talk about what they want to do (85%), food/what to eat (77%), and errands (74%). Most people without aphasia talked about the same things, although the order was different (errands 100%, what they want to do 98%, food/what to eat 93%).

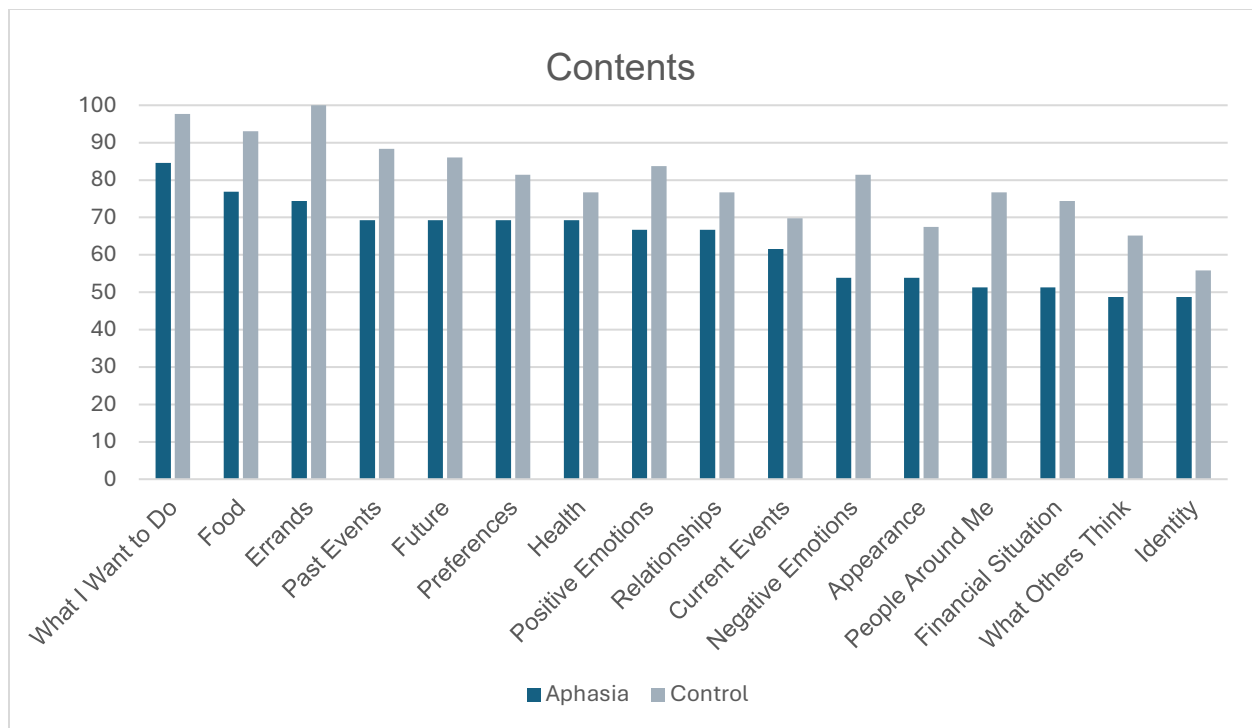


Figure 5: Most Frequently Reported Contents on the GISQ. The percentage of respondents in each group is represented on the y-axis, with the PWA group in darker blue and the CHA group in light blue.

Figure 6 shows the most frequently reported functions: Most people with aphasia used inner speech in order to solve problems (85%), plan (79%), and think critically (74%). As with the contents, people without aphasia had the same top 3 reported functions, but with higher percentages of people reporting each function (plan 98%, solve problems 93%, think critically 88%).

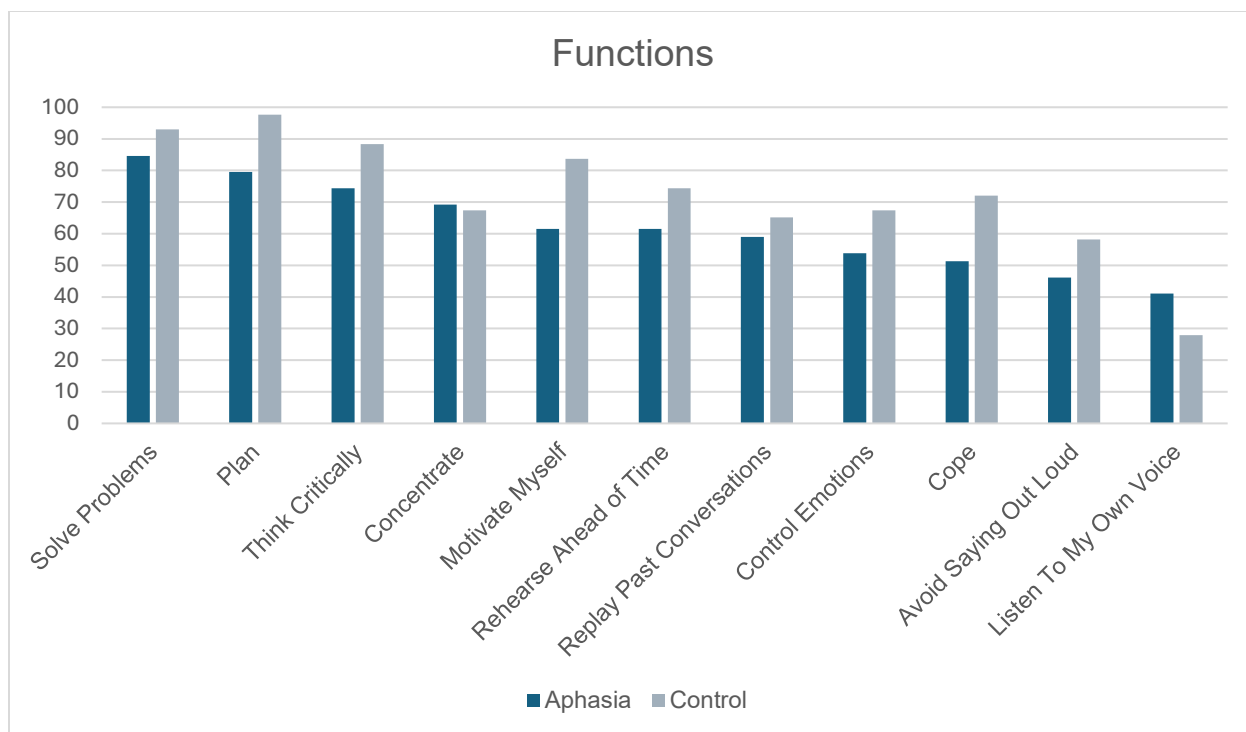


Figure 6: Most Frequently Reported Functions on the GISQ. The percentage of respondents in each group is represented on the y-axis, with the PWA group in darker blue and the CHA group in light blue.

Figure 7 shows the most frequently reported activities: Most people with aphasia reported using inner speech when remembering (87%), alone (74%), and reading (67%). People without aphasia reported using inner speech when remembering (98%), alone (88%), and driving (77%). Only 53% of people with aphasia reported using inner speech while driving, likely due to some people no longer driving after their stroke.

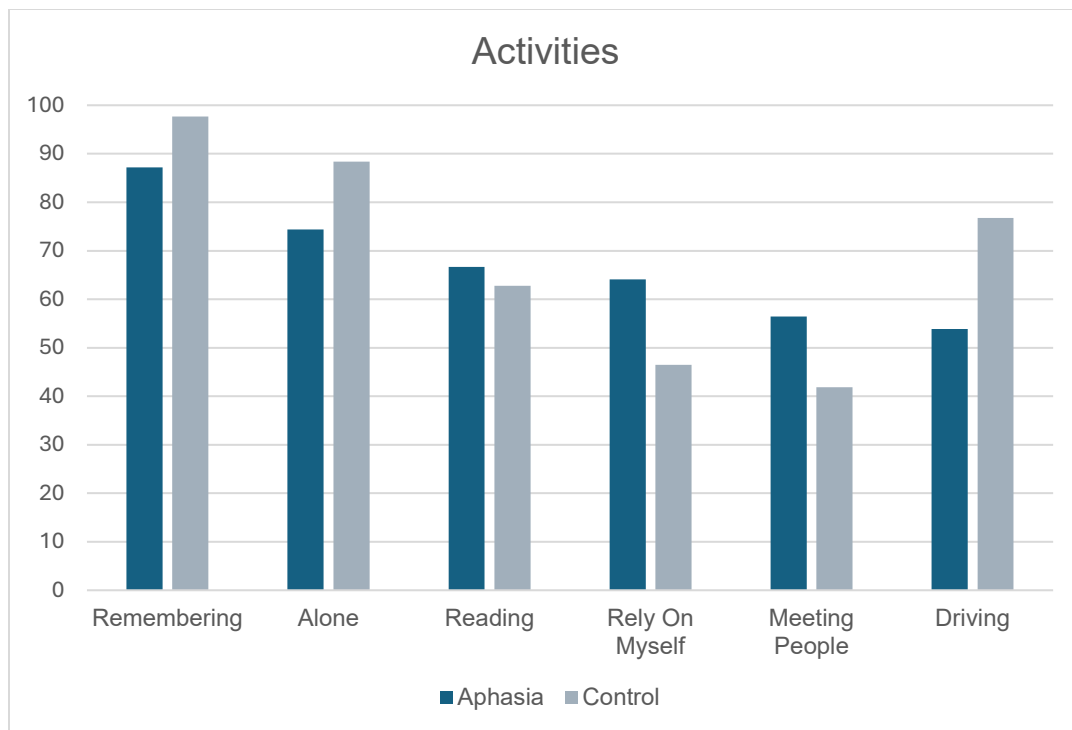


Figure 7: Most Frequently Reported Functions on the GISQ. The percentage of respondents in each group is represented on the y-axis, with the PWA group in darker blue and the CHA group in light blue.

Overall, there seems to be overlap in the most commonly reported contents, functions, and activities of inner speech use for people aged 40-80 with and without aphasia. Differences on the GISQ may be due in part to physical/participatory capabilities.

GISQ and Measures of Health-Related Quality of Life

Specific questions on the GISQ, selected due to their relevance, were correlated to the three domains on the SAQOL-39: physical, psychosocial, and communication. The correlation heat map in Figure 8 shows that the SAQOL-39 significantly correlated with two of the selected questions on the GISQ. Using *p*-values uncorrected for multiple comparisons, higher scores on the psychosocial domain are negatively correlated with using inner speech to replay past conversations ($r[37] = -2.97, p = .005$) and to control emotions ($r[37] = -2.11, p = .04$). Inner speech frequency was not correlated with depression and anxiety ($r[37] = 1.74, p = .09$), but

using inner speech to control emotions ($r[37] = 2.95, p = .005$) and for self-motivation ($r[37] = 2.17, p = .04$) were positively correlated with depression and anxiety. Expectedly, many of the GISQ questions correlate with one another, and the SAQOL-39 domains correlate with one another, as shown in Figure 8.

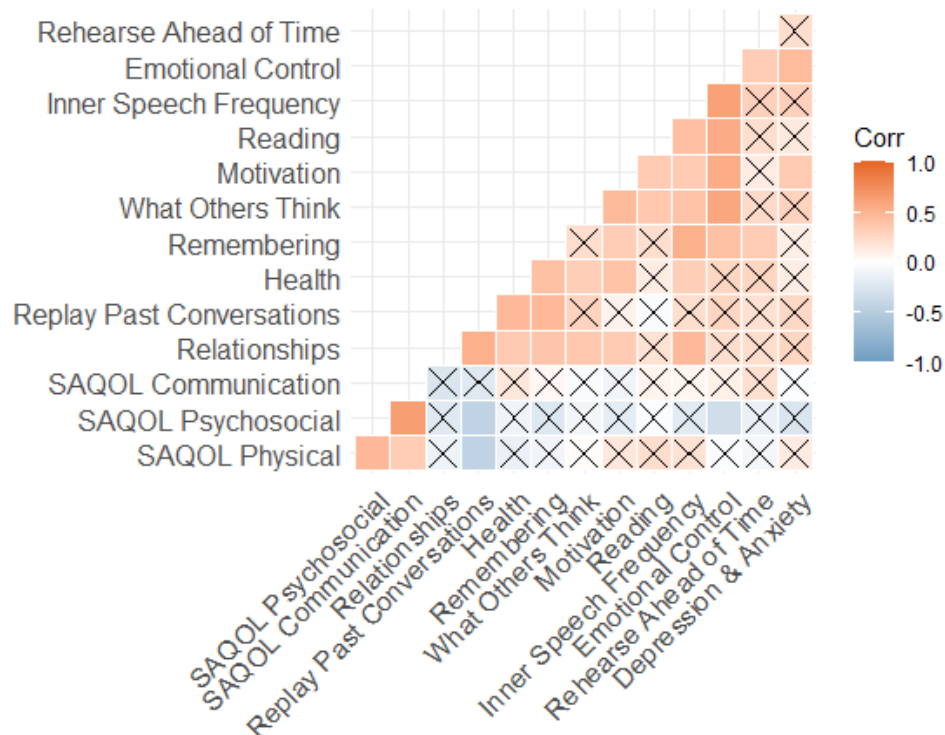


Figure 8: Correlation Matrix of SAQOL and GISQ Variables for PWA. Darker colors represent stronger correlations, with red indicating positive correlations and blue indicating negative. An x appears in each box in which the correlation was not statistically significant, based on an uncorrected p -value of .05.

For the cognitively healthy adults, other questions from the GISQ were chosen based on their relevance to test correlation with the PROMIS Affect and Participation scores. As shown in Figure 9, the PROMIS Affect score was positively correlated with using inner speech to talk about positive emotions ($r[41] = 2.52, p = .01$) and when meeting people ($r[41] = 2.23, p = .03$). It was negatively correlated with self-reported depression and anxiety ($r[41] = -2.28, p = .03$). The PROMIS Participation score was also negatively correlated with depression and anxiety

($r[41] = -2.63, p = .01$). Inner speech frequency was positively correlated with depression and anxiety ($r[41] = 2.26, p = .03$), while using inner speech to talk about positive emotions was negatively correlated with depression and anxiety ($r[41] = -3.11, p = .003$). The PROMIS scores correlated with each other, as did several questions on the GISQ, as shown in Figure 9.

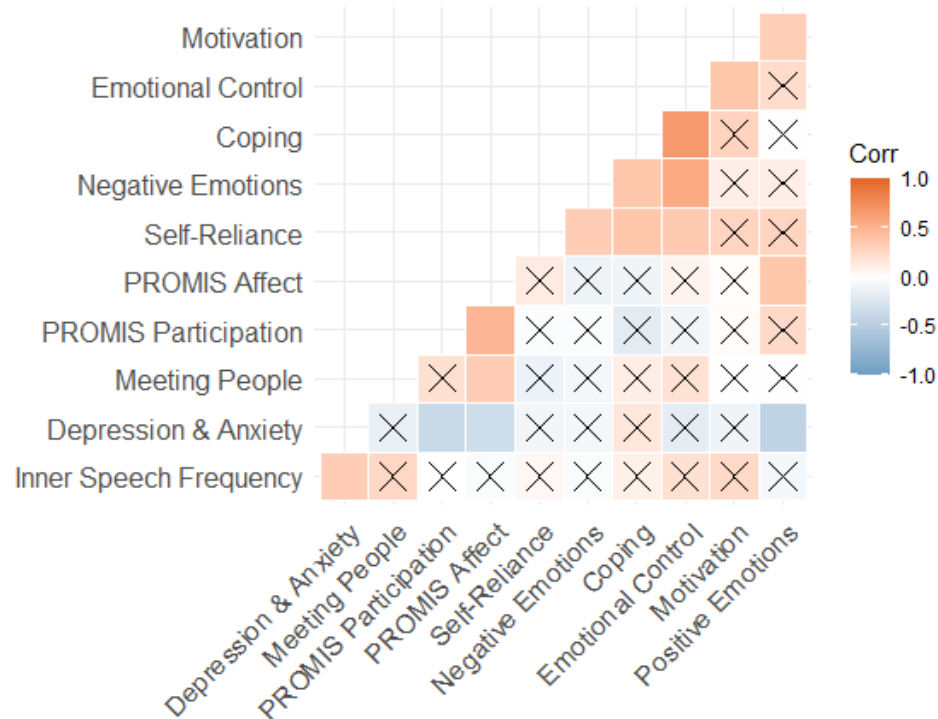


Figure 9: Correlation Matrix of PROMIS and VISQ Variables for CHA. Darker colors represent stronger correlations, with red indicating positive correlations and blue indicating negative. An x appears in each box in which the correlation was not statistically significant, based on an uncorrected p -value of .05.

Aim 2: Construct Validity of the GISQ, as compared to the VISQ-R

Based on the specific questions asked in each of the surveys, the Evaluative, Positive/Regulatory, and Other People means on the VISQ-R were the most relevant to evaluate in comparison to the GISQ. No models were made for the Condensed and Dialogic Means from the VISQ-R, as the GISQ does not have questions that address whether inner

speech is in full words/sentences or just fragments (condensed) or if inner speech is perceived in monologue or has a back-and-forth, conversational quality (dialogic).

Evaluative Model: A multiple regression was conducted to predict VISQ-R Evaluative scores based on using answers to the GISQ's "think critically" and "avoid saying out loud what I want to say", and their interaction. The overall model was significant, $R^2 = 0.23$, $F(3,39) = 3.99$, $p = 0.014$. "Think Critically" was a significant predictor, $\beta = -8.50$, $t(39) = -2.87$, $p = 0.007$. "Avoid Saying Out Loud" was not a significant predictor on its own, $\beta = -5.00$, $t(39) = -1.39$, $p = 0.173$, but the interaction term was significant, $\beta = 8.27$, $t(39) = 2.16$, $p = 0.037$. The model explained 23.5% of the variance (Adjusted $R^2 = 17.6\%$) and was statistically significant ($F[3,39] = 3.992$, $p = 0.0143$).

Other People Model: Here, using responses from the GISQ on using inner speech to "replay past conversation" and to talk about "people around me" were used to predict the phenomenon of "other people" from the VISQ-R. The model explained only 10.3% of the variance (Adjusted $R^2 = 5.8\%$) and did not reach overall significance ($F[2,40] = 2.287$, $p = 0.1147$). "Replaying past conversations" from the GISQ significantly predicted the perception of "other people" on the VISQ-R ($\beta = 2.71$, $p = 0.0397$). Using inner speech about "people in the environment" (GISQ) was not a significant predictor of "other people" (VISQ-R) ($\beta = -0.87$, $p = 0.5500$).

Positive/Regulatory Model: This model explored how the GISQ's answers about using inner speech to "control emotions", and to talk about "negative emotions" and "positive emotions" influenced "positive/regulatory" inner speech from the VISQ-R. The model was statistically significant ($F[3,39] = 2.927$, $p = 0.0456$), explaining 18.4% of the variance (Adjusted $R^2 = 12.1\%$). Negative emotions significantly predicted lower positive regulatory inner speech ($\beta = -3.30$, $p = 0.0286$). Control over emotions and positive emotions were not significant predictors ($p = 0.1387$ and $p = 0.0957$, respectively).

Discussion

Differences in Inner Speech Frequency

Overall, adults with and without aphasia reported similar current (in-the-moment) inner speech at the beginning of the GISQ, i.e., “*Are you experiencing inner speech right now?*”, with 64.1% of the PWA group and 76.7% of the CHA group answering “Yes.” These results are similar to those in Alexander et al. (2024), where PWA answered “Yes” 78% of the time over 30 different occasions prompting them to examine their inner speech. This compares with a wide range of inner speech frequency for college students, with a range from 0-75% and an average of 26% of the time during descriptive experience sampling (Heavey & Hurlburt, 2008). This indicates that the specific methodology of asking a person to examine their current inner experience may matter, but adults aged 40-80, regardless of whether they have aphasia, may use inner speech more often than college-aged students.

The two groups also had similar frequencies of inner speech in their daily lives when selected from a 5-point scale including 0, 25, 50, 75, and 100% of the time (mean: 56.41%, range: 0-100% of the time for PWA; mean: 57.56%, range: 25-100% for CHA). There were two people with aphasia who reported never experiencing inner speech, which they stated was different from their experiences prior to their strokes. Everyone in the CHA group reported at least some inner speech, although there were some people in this group who used it less frequently. According to Brinthaup et al. (2009), most college students engaged in self-talk (which included inner speech and “private speech,” or self-directed speech out loud) frequently, with a majority of respondents choosing sometimes, often, or very often on a 5-point Likert scale, which was also seen in a version of the self-talk scale adapted to target just inner speech (Ren et al., 2016). Both of those scales ask about the frequency of certain types of inner speech or self-talk: self-criticism, self-reinforcement, self-management, and social assessment, so frequency is related only to those domains and not inner speech in general. Similarly, college

students in Heavey et al. (2019) reported using inner speech 68.3% of the time, on average. Rather than using a Likert scale, Heavey et al. (2019) had participants place a mark on a line, allowing for more fine-grained differences in inner speech frequency. Together, these findings suggest that inner speech is a common experience across individuals, including people with aphasia, although its frequency may vary considerably.

Differences in Inner Speech Use

There was a difference in the number of GISQ themes (contents, functions, and activities) that were endorsed by each group, indicating less variety in the inner speech of people with aphasia. The version of the adapted GISQ used in the current study had one more item (using inner speech when “alone”) than the version used in Alexander et al. (2024), but the average total number of responses for PWA was similar to the PWA group from this study, (20.12, or 62.88%, of the 32 possible in the previous study versus 21, or 63.64%, of the 33 possible in this study). The CHA group from this study, in contrast, selected more responses overall on average (24.72, or 74.91% of the 33 possible). This means that, on average, people with aphasia may be using inner speech just as often as their cognitively healthy age-similar peers, but they may talk in their heads about fewer topics/for fewer reasons/during fewer situations, for example This restricted range of inner speech experiences could reflect underlying limitations in linguistic flexibility, vocabulary, executive function, or metacognitive awareness following stroke, and may contribute to differences in how inner speech supports cognitive tasks.

The only statistical difference between groups on the pre-selected questions from the GISQ was using inner speech for self-motivation, which was reported more by the cognitively healthy adults than the people with aphasia. This could be due to a reliance on multimodal, external cues for people with aphasia, such as visual reminders or routines rather than internal, speech-based cues. Using inner speech for self-motivation was one of the most-commonly reported themes of inner speech in Alexander et al. (2024), with over 80% of PWA selecting that

answer across three administrations of the adapted GISQ. This compares to 83.72% of the CHA group and only 61.54% of the PWA group in this study, further highlighting the individual, heterogeneous nature of the phenomenon of inner speech. Certain participants with aphasia in this study may demonstrate less inner speech for motivation, but not all people with aphasia lack self-motivational inner speech.

When examining the most-selected contents, functions, and activities of inner speech for each group, there were strong similarities between groups, although almost all items were endorsed by fewer people with aphasia than cognitively healthy adults. The only exceptions were in order to concentrate and “listen to my own voice,” and when reading, “I want to rely on myself,” and meeting people. These items highlight the underlying cognitive-linguistic components of inner speech, as people with aphasia may have more difficulty with attention (Murray, 2000, 2012), may use inner speech during other linguistic tasks to monitor correctness or to support reading abilities, for self-reliance in tasks which the CHA group may complete without a second thought, and when attempting to recall the name and personal details of a new acquaintance, which again may be simpler for adults without aphasia. Some of these differences were very slight, but it is interesting to note the few items that defied the usual pattern of more endorsement by the CHA group.

This research builds on prior research using forms of the GISQ in young adults (Morin & Racy, 2022; Racy et al., 2020) and in people with aphasia (Alexander et al., 2024). Studies in young adults reported the most common contents and functions of inner speech use being problem solving, negative emotions, emotion control, planning, and self-motivation (Morin & Racy, 2022). For the adults without aphasia aged 40-80 in the current study, the most common contents and functions of inner speech use were errands, what I want to do, planning, problem solving, and food/what to eat, which shares some of the same themes as young adults, but it seems that older adults may use inner speech less for emotional regulation and more for day-to-day planning of everyday tasks. The most reported contents and functions for people with

aphasia (solve problems, what I want to do, plan, food, errands, and think critically), mesh more with their peers of a similar age than with the young adults from Morin and Racy's work (2022), although we must recognize that the GISQ from the current study has been adapted from its original form. Both groups reported using inner speech when remembering and alone, but the group without aphasia reported using inner speech while driving more than the people with aphasia, which may relate to abilities and independence after stroke.

When comparing the participants from the current study to the PWA from Alexander et al. (2024), the most common contents and functions of inner speech also overlap – participants in that study reported using inner speech for planning, self-motivation, critical thinking, about food/what to eat, and for problem solving (in that order). Due to the nature of the current study being part of a larger study on problem solving and inner speech, participants may have been more aware of their inner speech for problem solving in the current study than in prior studies.

Psychosocial Health and Inner Speech

In general, the adapted GISQ did not map well onto the measures of health-related quality of life, although not all 33 items were examined due to the theoretically driven approach. Worse psychosocial health was related to the frequency with which people with aphasia used inner speech to replay past conversations and control emotions. This finding may relate to the idea presented earlier that inner speech may be used differently by different people, and those who engage in rumination will have worse mental health (Zawadzki, 2015). For cognitively healthy adults, scores on the measure of positive affect unsurprisingly correlated with using inner speech to talk about positive emotions, but higher frequency of inner speech was positively correlated with reporting depression or anxiety. This finding further supports that using inner speech is not inherently good or bad for mental health – rather, the contents, functions, and activities (i.e., the contexts) of inner speech matter.

This is consistent with prior research suggesting that inner speech can both support emotional regulation and exacerbate psychological distress, depending on its tone, purpose, and repetition. For example, McCarthy-Jones and Fernyhough (2011) propose that the dialogic nature of inner speech can sometimes become critical or intrusive, particularly in individuals with anxiety or depressive symptoms. They related the original Varieties of Inner Speech Questionnaire to anxiety, especially the evaluative and motivational domains. Brinthaup and colleagues (Brinthaup, 2019; Brinthaup et al., 2009) also distinguish between constructive self-talk (e.g., problem-solving, encouragement) and dysfunctional self-talk (e.g., rumination, self-criticism), which have divergent effects on emotional well-being. Therefore, in both people with and without aphasia, the quality and intentionality of inner speech may influence whether it acts as a tool for self-regulation or a source of psychological strain.

Validating the Adapted GISQ with the VISQ-R

The GISQ and VISQ-R evaluate some complementary, distinct, and overlapping aspects of inner speech. The GISQ focuses on the contents, functions, and activities of inner speech use – what it is about, why it is used, and when it is used. The particular questions on the GISQ correspond with some of the VISQ-R domains, but it was expected prior to testing that the GISQ would not have questions that related to the dialogic or condensed domains of the VISQ-R. When predicting scores on VISQ-R Evaluative, which includes ratings such as, “I think in inner speech about what I have done, and whether it was right or not,” there was an interaction between using the GISQ’s inner speech for critical thinking and suppression/inhibition of verbal expression. This interaction suggests that self-evaluative inner speech is not simply a function of critical thinking or suppression alone but rather emerges from their combined influence. Individuals who think critically yet refrain from verbalizing their thoughts may instead turn inward, engaging in heightened self-evaluative inner dialogue.

When predicting scores on VISQ-R Other People, which includes ratings such as, “I experience the voices of other people asking me questions in my head,” using inner speech to replay past conversations (from the GISQ) was a significant predictor. That suggests that for this group of cognitively healthy adults, the experience of other people’s voices in their minds was linked to imagined interactions, including reflecting on real conversations, and less on thinking about people in general. These ideas have been discussed in prior research in philosophy (Morin, 2020) showing that imagined interactions and inner speech can overlap, but each includes topics or components that are unique to it.

When predicting scores on VISQ-R Positive/Regulatory, which includes ratings such as, “I calm myself down by talking silently to myself,” using inner speech to talk about negative emotions significantly predicted lower scores, while both using inner speech to control emotions and to talk about positive emotions were not significant (from the GISQ). The predictive relationship of talking about negative emotions may indicate that, related to the findings on psychosocial health measures, people who experience more negative emotions and talk about them in their inner speech may be less likely to use inner speech for self-regulation. It is surprising, however, that the other emotion-related questions on the GISQ were not predictive of positive/regulatory scores on the VISQ-R. This could be related to differences in the format of the questionnaires (1-7 frequency rating scale vs. binary reports). Overall, this analysis indicates that the GISQ and VISQ-R may have some shared constructs, but they seem to be capturing different dimensions of inner speech.

Previous research has validated the VISQ and VISQ-R against both qualitative interviews and other self-report measures to confirm their factor structures and external validity. For example, (Alderson-Day et al., 2018) confirmed the structure of the VISQ with factors including evaluative, other people, positive/regulatory and condensed domains of inner speech, noting correlations with measures of auditory verbal hallucinations. Similarly, McCarthy-Jones and Fernyhough (2011) validated early iterations of the VISQ using experience sampling and

found consistent patterns of dialogic and evaluative inner speech across individuals. These studies have established that the VISQ captures distinct experiential qualities of inner speech that relate to both normal and atypical cognitive phenomena.

Importantly, the GISQ has undergone validation by Morin and colleagues (Morin & Racy, 2022; Racy et al., 2020) who established its internal reliability and construct validity in samples of cognitively healthy adults. Their work showed that the GISQ effectively captures a broad range of inner speech content and contexts, and correlates meaningfully with other self-report and behavioral measures of metacognition, self-awareness, and introspective thinking, such as the Self-Talk Scale, Mind-Wandering Questionnaire, and the Rumination-Reflection Questionnaire (Brinthaup et al., 2009; Mrazek et al., 2013; Trapnell & Campbell, 2011). These studies support the GISQ's utility as a flexible and theoretically grounded measure of inner speech, particularly for exploring individual differences in how, when, and why inner speech is used.

Together, the GISQ and VISQ-R offer distinct yet complementary perspectives on inner speech. While the VISQ-R emphasizes phenomenological features like dialogality and condensation, the GISQ targets practical uses and situational triggers for inner speech. This makes the GISQ especially valuable for research in applied settings (e.g., clinical populations, educational contexts), where the goals are often to understand how inner speech functions in everyday tasks rather than its qualitative experience alone.

Clinical Implications

In this study and as already established (Curran et al., 2020), experiencing depression or anxiety significantly impacts psychosocial health, including affect and the ability to participate in social activities for cognitively healthy adults as well as post-stroke quality of life. These impacts relate to the ICF domain of personal factors and the A-FROM domains of personal, identity, attitudes, and feelings. Some aspects of psychosocial health and quality of life were related to

our measures of inner speech, so speech-language pathologists and mental health professionals may consider the examination of inner speech for their clients, especially when related to rumination. This would play into existing therapeutic approaches such as cognitive behavioral therapy (Baker et al., 2024), but using “inner speech” may be a useful way of reframing the idea, as it can be seen as a relative strength for some people with aphasia, and it can be approached as a tool rather than a hindrance. The GISQ and VISQ-R can be used to evaluate distinct aspects of the phenomenon of inner speech and may be complementary to one another in clinical situations.

Limitations

Our group adapted the GISQ from its original form to have fewer questions and less complicated answer types, better suited to working with people with aphasia. The original form of the GISQ went beyond the binary question of “do you experience inner speech about this topic/for this reason/during this situation?” and used a Likert scale to determine the frequency of certain types of inner speech (e.g., 0 = never; 1 = rarely; 2 = sometimes; 3 = often; 4 = very often; 5 = all the time). This would have allowed for more variation in the responses and could have made more robust statistical analysis possible. Additionally, there was a tendency (especially in the CHA group) to select the responses rather than leave them blank, but there was no indication of frequency or importance of that item. For example, if a person selects “Health,” that does not indicate whether they talked about it more or less than “Negative Emotions,” just that sometime in the past six months, they had used inner speech about that topic. For people with very mild aphasia or cognitively healthy adults, the original form of the GISQ may be a more informative tool, but for the population of people with aphasia, including those with moderate-severe language deficits, the adapted form is more feasible.

The frequency of inner speech question may have benefited from more possible options or a sliding scale to select the percentage of the time that the participant experiences inner speech, as people may have been hesitant to choose the extremes (never or always).

Future Directions

Given our lab's past work and others in this area, clear extensions include further capturing inner speech in the daily environment or through interviews across the lifespan to see where changes in inner speech may happen along with the life course. Additionally, specific questions on the use of inner speech to reframe identity after significant life changes such as retirement or stroke may highlight this potential avenue of study of psychosocial health in the aging population with and without aphasia.

This study was done with people with chronic aphasia, but some anecdotal evidence reports that inner speech was most different in the acute stages after the stroke (Marks, 2018; Taylor, 2008). Future work may investigate inner speech throughout the recovery process to see its relationship with overt speech and psychosocial health outcomes.

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Chapter 6: Inner Speech, Aphasia, Problem Solving, and Aging – Discussion and Next Steps

Each of the previous chapters addressed questions related to inner speech, aphasia, and aging. Overall, there are multiple facets of the phenomenon of inner speech to address – at the word level, in daily life, during problem solving, and as a connection to psychosocial health. These multiple facets were investigated with a variety of methodologies – inner rhyme judgments, rating scales, questionnaires, dual-task interference, and subjective reports of successful inner speech during naming. The variety of perspectives and methods provided a broad view of the questions related to inner speech and raised several questions for future investigations. This chapter will summarize and interrelate the findings from each chapter and provide takeaways and future directions for the study of inner speech in the fields of neuroscience and speech, language and hearing sciences.

Chapter 1: Why study inner speech in aphasia, and why use interdisciplinary perspectives?

Studying inner speech in aphasia helps researchers, clinicians, and other stakeholders understand the cognitive and neural processes underlying language use and recovery after brain injury. Although inner speech is studied outside of aphasiology, the perspective of people with an acquired brain injury allows for additional investigation into the relationship between language and thought as well as the specific areas of the brain which contribute to cognitive processes including inner speech. Using interdisciplinary perspectives provides context from multiple fields which have independently studied inner speech – psychology, philosophy, linguistics, neuroscience, and aphasiology. These disciplines use different methods, define inner speech in different ways, and approach their investigations from different theoretical backgrounds. Integrating insights from various disciplines allows aphasiologists and neuroscientists to better understand the phenomenon of inner speech and develop more effective investigations and interventions.

Chapter 2: How can you measure a phenomenon that takes place in someone's head, and what information do different types of measurements give?

As with other internal, perceptual phenomena such as pain or tinnitus, one of the main ways to measure inner speech is through subjective reports, either in the moment or through questionnaires. The fact that these methods are subjective does not devalue them, as they are the only way to directly measure certain aspects of inner speech, such as perceived access to words, differences in inner speech before and after the stroke, and the contents, functions, and activities of inner speech use in daily life. These self-reports capture the phenomenological and contextual richness of inner speech, providing insight into how individuals personally experience and utilize their inner voice – information that would be otherwise inaccessible through objective means alone. Additionally, measuring inner speech subjectively is: reliable, as supported by prior test-retest studies; valid, as self-report of inner speech aligns with other measures such as rhyme judgment and phonological awareness, although they may measure slightly different things; meaningful, as subjective reports vary by individual; and clinically informative, as it gives a sense of where in the process of word access and output there may be disruptions, and subjective reports of inner speech in daily life may aid health professionals to better understand the lived experiences of people with aphasia.

In this dissertation, inner speech was also measured and manipulated in more objective ways, including inner rhyme judgments and articulatory suppression (in Chapter 4), which can give information concerning the ability to access and manipulate information in inner speech and the role of inner speech in a task, respectively. These tasks provide performance-based indices that are less influenced by self-perception or memory bias, offering a valuable complement to subjective accounts. Notably, the objective measure of inner rhyme judgment was significantly related to aphasia severity, while the subjective measures were not. This divergence is important. The relationship between objective inner speech performance and aphasia severity suggests that certain components of inner speech, particularly those involving

phonological access and manipulation, may be sensitive indicators of underlying language impairments. In contrast, the lack of correlation between subjective reports and aphasia severity could reflect the heterogeneity of personal strategies, compensations, or metacognitive insight following stroke. Therefore, a more holistic approach that incorporates both subjective and objective methods is crucial for a comprehensive understanding of inner speech in aphasia. It allows researchers and clinicians to distinguish between the experience of inner speech and the functional capacity to use it, thereby deepening our understanding of the nature of the phenomenon of inner speech overall.

Chapter 3: How do people with aphasia use inner speech in their daily lives, beyond the word level?

People with aphasia reported using inner speech often in their daily lives, in 78% of instances during experience sampling. They used inner speech to talk **about** food and what to eat, what they wanted to do, and their preferences; **in order to** solve problems, think critically, and motivate themselves; and **when** remembering things, wanting to rely on themselves, and reading. These findings illustrate that inner speech remains a meaningful and frequent part of daily mental life for many individuals with aphasia, even after stroke. Importantly, the experience of inner speech as evaluated here did not relate to aphasia severity or anosognosia, with the caveat that the sample of PWA tended to be milder in presentation and that participants with more severe auditory comprehension deficits were excluded from the study due to concerns about their ability to understand and follow the protocol.

This work motivates a deeper investigation into how inner speech may differ in adults with and without aphasia – not only in frequency, but also in terms of the *contents*, *functions*, and *activities* in which it occurs. Given that inner speech is a major medium for self-reflection, planning, emotional regulation, and identity maintenance, differences in how inner speech is used could have implications beyond language function alone. Specifically, this work motivated the aim in Chapter 5 to investigate inner speech in relation to *psychosocial health*, with goals of

explaining how people adapt to post-stroke changes, including how they cope emotionally, stay motivated, and maintain their sense of self. Understanding the phenomenology of inner speech in aphasia could therefore offer novel insights into quality of life, mental health, and resilience in this population.

To support this line of inquiry, the General Inner Speech Questionnaire (GISQ) (Racy et al., 2020) was adapted for use with people with aphasia in this study. The adapted version was designed to reduce linguistic complexity while preserving the theoretical integrity of the original measure, allowing individuals with aphasia to report on a wide range of inner speech experiences. The GISQ focuses on what people talk to themselves about (contents), why they use inner speech (functions), and when they tend to use it (activities or contexts). This adaptation allowed for a broader, more accessible evaluation of inner speech that extended beyond traditional word-level tasks.

The adapted GISQ was administered again in Chapter 5, allowing for comparisons between the aphasia group and a group of cognitively healthy adults. This enabled the exploration of both group differences and individual variability in inner speech use, as well as its relationship to other cognitive and emotional factors. Together, the findings from Chapters 3 and 5 contribute to a richer understanding of inner speech as a complex, context-sensitive phenomenon, one that has meaningful links to both language and psychosocial functioning.

Chapter 4: What happens to people with and without aphasia when we disrupt the use of inner speech while they solve problems (i.e., how is language related to reasoning skills) and how are different ways of measuring inner speech related to different types of cognitive tasks?

Participants in this study completed the Wisconsin Card Sorting Test (WCST) (Grant & Berg, 1948) under different conditions, including a verbal interference condition in which they repeatedly said “na” out loud. This articulatory suppression task effectively disrupted inner speech, as confirmed by participants’ ratings of how frequently they used inner speech during the task. While there was a strong learning effect across conditions, likely due to the novel

nature of the task, participants demonstrated less improvement in performance from baseline during the verbal interference condition than during a comparable non-verbal motor interference condition. These findings suggest that inner speech contributes meaningfully to reasoning and problem-solving performance, and disrupting it can impair those processes, while higher ratings of inner speech use during the task related to better performance.

Importantly, the study helps to clarify the complex relationship between verbal working memory and inner speech, especially in the context of executive functioning. While baseline verbal working memory, an individual's general ability to hold and manipulate verbal information, likely explains a substantial portion of the variance in problem-solving performance across participants, it does not fully account for what happens during cognitive tasks. Instead, what appears to matter most for executive functioning in these tasks is how individuals call on inner speech in context, during the task itself.

In other words, someone may have strong verbal working memory capacities (e.g., performing well on digit span or sentence repetition tasks) but still fail to use inner speech effectively during reasoning tasks. Conversely, someone else may have limited verbal working memory but strategically and effectively use inner speech as a scaffolding tool in specific contexts. In this study, ratings of inner speech use during each WCST condition (i.e., context-specific inner speech) were more strongly associated with executive functioning performance than general trait-like measures of successful inner speech use or objective rhyme judgment. This distinction is echoed in the divergent findings for different inner speech measures. The *objective inner rhyme judgment task* (from Chapter 2), which requires recognizing and manipulating the sounds of words without vocalizing them, was related to cognitive abilities such as inhibition, reasoning, planning, and problem solving. This suggests a link to verbal working memory, which is required for holding phonological information online and manipulating it. On the other hand, the *subjective measure of successful inner speech*, self-report on how well people feel they can name words silently, was not predictive of executive functioning

performance. This supports the idea that trait-based inner speech ratings and task-embedded inner speech behaviors reflect different cognitive operations.

Ultimately, these findings highlight the importance of measuring inner speech dynamically, as a process that can shift across tasks and contexts, rather than as a static trait. This has important implications for understanding reasoning skills in both neurotypical adults and those with aphasia.

Chapter 5: How are people with and without aphasia using inner speech – for problem solving, emotional control, motivation, rehearsal? Do people with aphasia talk about health and identity in their inner speech more than adults without aphasia? How does inner speech relate to psychosocial health?

People with and without aphasia reported using inner speech at approximately the same frequency throughout their daily lives, echoing findings from Chapter 3. However, the cognitively healthy adults endorsed more varied inner speech across the dimensions of content (what inner speech is about), function (why it is used), and activity (when or in what context it is used), suggesting that aphasia may impact not the quantity of inner speech, but the *diversity* of its use. This is consistent with previous research suggesting that inner speech becomes more limited or constrained following brain injury, potentially reflecting disruptions in language access or executive control mechanisms needed to engage in more flexible self-directed cognition. Despite these differences in variety, the most reported themes of inner speech were similar across groups. These included solving problems, thinking about what one wants to do, planning, food-related thoughts, remembering, errands, and critical thinking. This similarity supports the idea that certain topics or cognitive routines, like making plans or organizing the day, may be part of a core set of inner speech experiences for middle-aged and older adults, regardless of neurological or cognitive status.

Interestingly, not every participant reported using inner speech to solve problems, reinforcing Chapter 4's finding that inner speech is one of several available strategies for non-

verbal problem solving. This variability may reflect individual cognitive styles or differences in metacognitive awareness; some individuals may favor visual or action-based strategies over verbal mediation.

In terms of content more closely tied to identity and wellbeing, there were no significant group differences in the frequency of using inner speech to talk about health, identity, or rehearsing for future conversations. This finding extends the experience sampling data from Chapter 3 and suggests that inner speech about these topics may be common across adulthood, rather than specific to living with aphasia or recovering from a neurological injury. Prior work by Morin and colleagues (Morin, 2009, 2011a, 2011b; Racy & Morin, 2024) has emphasized the role of inner speech in self-reflection and self-knowledge, and this study provides preliminary evidence that themes related to identity, health, and planning for the future may be common, though potentially experienced or interpreted differently depending on life context.

Importantly, inner speech use was not uniformly associated with positive or negative psychosocial outcomes. For people with aphasia, more frequent inner speech about replaying past conversations and controlling emotions was negatively correlated with psychosocial quality of life. These patterns may reflect rumination, where inner speech becomes repetitive and focused on emotionally salient or socially fraught content. This interpretation aligns with prior research by Zawadzki (2015) and others, suggesting that inner speech can contribute to maladaptive self-focus or emotional dysregulation in the context of anxiety, depression, or social stress.

Consistent with this, self-reported depression or anxiety also correlated with several measures of inner speech across both groups. In particular, individuals who reported more frequent or emotionally charged inner speech tended to report poorer emotional wellbeing. These findings underscore the idea that inner speech is not inherently helpful or harmful – rather, its impact on mental health depends on how and why it is used. Supportive or

motivational inner speech may bolster self-regulation and resilience, whereas repetitive or self-critical inner speech may exacerbate emotional difficulties.

Together, the findings from Chapter 5 deepen our understanding of how inner speech supports daily cognition, reflects identity-related processing, and interacts with emotional wellbeing in both clinical and non-clinical populations.

Clinical Implications and Future Directions

Aligning with current initiatives in aphasia therapy, such as person-centered care and individualized treatment, people with aphasia have heterogeneity in their experience with inner speech, and this suggests that rehabilitation should be tailored to a person's specific experience. For example, some people may benefit from therapies that recognize and leverage their relative strengths in inner speech compared to overt speech and use this as a strategy for complex cognitive tasks or emotional regulation. Integrating inner speech measures into assessment protocols for people with aphasia may provide a more comprehensive picture of cognitive-linguistic function and aid in understanding where there may be breakdowns in the speech and language production process. Additionally, future studies may further tease apart the *static* skills of verbal working memory or inner rhyme judgment and the *dynamic* use of strategies such as inner speech to support higher-order cognitive abilities.

The links found between inner speech, emotional control, and psychosocial health highlight the need for mental health interventions that are tailored to people with aphasia. Overall, regardless of language abilities, encouraging positive self-talk and emotional regulation strategies may improve a person's psychosocial quality of life, which has been supported in young adults (Boukraa, 2016; Grzybowski & Brinthaup, 2022; Guerin et al., 2010). Future work may investigate inner speech with a variety of measures in a longitudinal way, following people with aphasia as they recover and adults without aphasia as they navigate the natural seasons of their life course. Because inner speech is such an individualized experience,

it may be more meaningful to see changes in the same person. For people with aphasia, this also includes having measures of inner speech which could be reasonably simple for administration in the acute phases of post-stroke recovery, as inner speech may be an indicator of improvement in access to language. Additionally, researchers could explore more methods of investigating the neural underpinnings of inner speech and problem solving, such as via voxel-based lesion-symptom mapping (finding areas of brain injury which relate to performance) and mapping of neural networks (functional and structural).

Conclusion

This dissertation explored the multifaceted nature of inner speech in healthy aging and aphasia, shedding light on its role in cognitive processes, daily life, and psychosocial well-being. Across the chapters, the findings demonstrated that inner speech is a dynamic and complex phenomenon that extends beyond language production. It influences processes such as problem solving, emotional regulation, and self-perception. By integrating a variety of measures from interdisciplinary fields, this research provided a comprehensive view of inner speech in aphasia.

A key takeaway from this work is that many people with aphasia continue to use inner speech frequently in their lives, but their inner speech appears to be less varied in content and function compared to their age-similar neurologically healthy counterparts. The findings also suggest that inner speech is one of many possible strategies for problem solving, with some individuals relying more on verbalized reasoning while others may use nonverbal or visual approaches. The use of inner speech may also depend on the task at hand, environmental factors, and language abilities.

The findings from this dissertation also highlight the relationship between inner speech and psychosocial health. While inner speech may be useful for solving problems or planning, certain uses – such as replaying past conversations – were associated with lower psychosocial

quality of life. This suggests that while inner speech may serve as a tool for self-reflection and control, it can also reinforce negative thought patterns in some individuals, warranting further investigation into its psychological implications.

From a clinical perspective, these findings emphasize the possible benefits of assessing inner speech as part of aphasia rehabilitation. Traditional language assessments may overlook the cognitive and emotional dimensions of inner speech, yet these aspects could influence an individual's recovery and quality of life. Future interventions could incorporate strategies to support inner speech use in problem solving and emotional regulation, tailored to the specific needs of individuals with aphasia.

Looking ahead, future research should further investigate the neural mechanisms underlying inner speech, explore its role in other neurological populations, and examine longitudinal changes in inner speech post-stroke. By continuing to integrate interdisciplinary perspectives, we can deepen our understanding of inner speech and its broader implications for language, cognition, and well-being.

In sum, this dissertation underscores that inner speech is not merely an internal dialogue; it is a crucial cognitive tool that shapes thought, guides behavior, and influences emotional resilience. Recognizing its role in both healthy and disordered populations can enhance our approaches to assessment, intervention, and our broader understanding of the mind's inner voice.

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- Racy, F., & Morin, A. (2024). Relationships between Self-Talk, Inner Speech, Mind Wandering, Mindfulness, Self-Concept Clarity, and Self-Regulation in University Students. *Behavioral Sciences*, 14(1), Article 1. <https://doi.org/10.3390/bs14010055>

- Racy, F., Morin, A., & Duhnych, C. (2020). Using a Thought Listing Procedure to Construct the General Inner Speech Questionnaire: An Ecological Approach. *Journal of Constructivist Psychology*, 33(4), 385–405. <https://doi.org/10.1080/10720537.2019.1633572>
- Zawadzki, M. J. (2015). Rumination is independently associated with poor psychological health: Comparing emotion regulation strategies. *Psychology & Health*, 30(10), 1146–1163. <https://doi.org/10.1080/08870446.2015.1026904>

Julianne M. Alexander

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EDUCATION

Dual PhD: Neuroscience and Speech, Language and Hearing Sciences

Doctoral advisor: Brielle C. Stark, PhD

Indiana University, Bloomington, IN

Advanced to Candidacy: July 2023

Completed: June 2025

Master of Arts: Speech-Language Pathology

University of Iowa, Iowa City, IA

May 2018

Bachelor of Science: Major – Speech and Hearing Sciences, Minor – Psychology

Honors Thesis: Perceptual adaptation to nonnative speech by school-aged children

Thesis advisor: Tessa Bent, PhD

Bachelor of Arts: Major – Spanish: Hispanic Linguistics track, Minor – Linguistics

Semester abroad in Buenos Aires, Argentina

Indiana University, Bloomington, IN

May 2016

LICENSES AND CERTIFICATIONS

Certification of Clinical Competence, American Speech-Language and Hearing Association

Licensed Speech-Language Pathologist, State of Indiana

PUBLICATIONS

Published Manuscripts

Alexander, J. M. and Stark, B.C. (2024). Interdisciplinary approaches to understanding the inner speech, with emphasis on the role of incorporating clinical data. *European Journal of Neuroscience*. <https://doi.org/10.1111/ejn.16470>

Alexander, J. M., Hedrick, T., & Stark, B. C. (2024). Inner speech in the daily lives of people with aphasia. *Frontiers in Psychology*. <https://doi.org/10.3389/fpsyg.2024.1335425> Preprint and data: <https://osf.io/gbw3t/>

Stark, B.C., **Alexander, J.M.**, Hittson, A., Doub, A., Igleheart, M., Streander, T., and Jewell, E. (2023). Test-retest reliability of microlinguistic information derived from spoken discourse in persons with chronic aphasia, *Journal of Speech, Language, and Hearing Research*. https://pubs.asha.org/doi/10.1044/2023_JSLHR-22-00266 Preprint and data: <https://osf.io/4zcpn/>

Alexander, J.M., Langland-Hassan, P., and Stark, B.C. (2023). Measuring inner speech objectively and subjectively in aphasia. *Aphasiology*. 1–26. <https://doi.org/10.1080/02687038.2023.2235761> Preprint and data: <https://osf.io/8sjny/>

Manuscripts in Review

Alexander, J. M. and Anderson, R. A. (in review). Inner speech in bilingual adults: A scoping review with clinical applications. *International Journal of Bilingualism*.

Stark, B. C., **Alexander, J. M.**, Stockrahm, E., Taylor, R., Nielsen, P., Barron, B., & Kaur, P. (in revision). Inner speech as a mechanism of naming improvement after lexical-semantic therapy in chronic aphasia. <https://doi.org/10.31234/osf.io/n2eqj>

PRESENTATIONS AND POSTERS

Alexander, J.M., Hedrick, T., Stark, B.C. (2023). Inner Speech Frequency, Consistency, and Themes in the Daily Lives of People with Post-Stroke Aphasia. *World Stroke Congress*, Toronto, Canada.

Stark, B.C., **Alexander, J.M.**, Stockrahm, E., Taylor, R., Nielsen, P., Hedrick, T., Kaur, P., Barron, B., Yagoda, B., Thompson, O., Perry, S. (2023). Inner speech's role in treated naming recovery in aphasia. *Clinical Aphasiology Conference*, Atlantic City, NJ.

Stark, B.C., Hedrick, T., **Alexander, J.M.** (2023). Relationship of Aphasia Severity and Anosognosia to Inner Speech Perception in Individuals with Aphasia. *Clinical Aphasiology Conference*, Atlantic City, NJ.

Alexander, J.M., Hedrick, T., Stark, B.C. (2023). Inner Speech in the Daily Lives of People with Aphasia. *Clinical Aphasiology Conference*, Atlantic City, NJ.

Alexander, J.M., Langland-Hassan, P., and Stark, B.C. (2022). Characterizing Inner Speech in Aphasia: Subjective and Objective Measures. *American Speech, Language and Hearing Association Convention*, New Orleans, LA.

Alexander, J.M., Langland-Hassan, P., and Stark, B.C. (2022). Measuring inner speech in aphasia and neurotypical controls. *Academy of Aphasia*, Philadelphia, PA.

Alexander, J.M., Hittson, A., Doub, A., Igleheart, M., Jewell, E., Streander, T., and Stark, B.C. (2021). Virtually Assessing Test-Retest Reliability of Spoken Discourse Informativeness Measures in Persons with Aphasia. *American Speech, Language and Hearing Association Convention* (virtual).

Hittson, A., Doub, A., **Alexander, J.M.**, Igleheart, M., Jewell, E., Streander, T., and Stark, B.C. (2021). Test-retest stability of informativeness during discourse in persons with aphasia. *8th Meeting of the Nordic Aphasia Conference* (virtual).

PROFESSIONAL EXPERIENCE

Research Assistant, Indiana University, Fall 2020 - present

NEURAL Research Lab, Brielle C. Stark, PhD

Instructor of Record, Indiana University, Summer 2021 - Spring 2023

Department of Speech, Language, and Hearing Sciences

Speech-Language Pathologist, September 2019 - June 2020

American Senior Communities, IN

Speech-Language Pathologist, October 2018 - October 2019

St. Joseph Health System, Mishawaka, IN

Speech-Language Pathologist, June 2018 - September 2019

Select Rehabilitation, South Bend, IN

Graduate Teaching Assistant, University of Iowa, Fall 2016 - Fall 2017

Department of Spanish and Portuguese

Research Assistant, University of Iowa, Summer - Fall 2017

Language in Aging and Aphasia Laboratory, Jean K. Gordon, PhD

Research Assistant, Indiana University, 2013-2016

Speech Perception Lab, Tessa Bent, PhD

TEACHING

Courses Taught

SLHS-S111 Phonetics (Undergraduate course, in-person)

Indiana University, Bloomington – Fall 2022 and Spring 2023

Enrollment ~ 65 each semester

SLHS-S333 Childhood Language (Undergraduate course, online)

Indiana University, Bloomington – Summer 2021

Enrollment ~ 25

SPAN:1002 Elementary Spanish II (Undergraduate course, in-person)

University of Iowa, Iowa City – Fall 2016 (2 sec.), Spring 2017 (2 sec.), Fall 2017

Enrollment ~ 20 each section,

Additional Course Development

Aphasia (Graduate course)

Developed as part of the Transforming Your Research into Teaching workshop series from the Center for the Integration of Research, Teaching and Learning. Implemented backwards course design, prioritizing the topics for which students will need enduring understanding, while integrating my own research expertise in psychosocial health.

Training

Indiana University's Center for Innovative Teaching and Learning's Graduate Teaching Apprenticeship Program in Higher Education: Associate Level (completed on 9/25/24) and Practitioner Level (completed on 1/16/25)

SERVICE

Mentor, Groups Scholars STEM Summer Research Experience Program (2023 and 2024)

Student Representative, Academy of Neurologic Communication Disorders and Sciences Membership Committee (2023-present)

Ad-Hoc Journal Reviewer, Journal of Communication Disorders, Cognitive Neuropsychology

Student Representative, Program in Neuroscience Diversity, Equity, and Inclusion Committee (2021-2023)

Coordinator, Speech, Language and Hearing Sciences PhD Student Organization (2021-2022)

Camp Leader, Girls with Nerve Summer Camp, Indiana University (2021)

Student Representative, Department of Speech, Language and Hearing Sciences Diversity, Equity, and Inclusion Committee (2020-2022)

GRANTS, FELLOWSHIPS, & HONORS

- William Orr Dingwall Dissertation Fellowship in the Foundation of Language, 2024
- American Speech-Language-Hearing Foundation New Century Scholars Doctoral Scholarship, 2023
- Groups Scholars STEM Summer Research Experience Program Mentor, Summer 2023 and 2024
- Larry and Marty Humes Student Research Travel Fund Award, Spring 2023
- College of Arts and Sciences Travel Award, Spring 2023
- National Institute on Deafness and Other Communication Disorders (NIDCD)-sponsored Academy of Aphasia Young Investigator Travel Fellowship, 2022
- Academy of Neurologic Communication Disorders and Sciences Conference Student Fellowship, 2022
- Program in Neuroscience Travel Award, Fall 2021, Fall 2022, Spring 2023
- Indiana University Bloomington Provost's Travel Award for Women in Science, Fall 2021
- Shirley J. & Thomas L. Salmon Scholarship at the University of Iowa, 2017
- General Honors Notation from Hutton Honors College at Indiana University, 2016
- Founder's Scholar, Indiana University, 2016
- Hutton Honors College Research Grant, 2015
- Hutton International Experiences Program Grant, 2014
- Council on International Educational Exchange Ambassador Grant, 2014
- Oliver W. Storer Scholarship, 2012

Memberships

- American Speech-Language and Hearing Association (ASHA)
- Academy of Neurologic Communication Disorders and Sciences
- Academy of Aphasia
- Indiana Speech-Language-Hearing Association
- ASHA Special Interest Groups 2: Neurogenic Communication Disorders, 15: Gerontology, and 20: Counseling
- FOQUS Aphasia Working Group